

Econophysics — Lecture Notes

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September 9, 2026

Preface

These notes are intended to serve as a supplementary resource for the Econophysics course within the master's program in Physics at the University of Pavia.

The complete source code of these lecture notes is freely available on GitHub at the following repository:

<https://github.com/AlessandroCasti/econophysics-lecture-notes.git>

Readers who notice typographical errors or encounter any other issues are kindly invited to submit a pull request describing the problem. Contributions of this kind are greatly appreciated and will be addressed as promptly as possible.

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Notation

Stochastic processes, mathematical finance, and statistical physics each bring their own notational traditions, and this book inevitably inherits all three. Within each chapter every symbol is defined at first use and its meaning is unambiguous; across chapters, however, a few letters are unavoidably overloaded (S , B , σ , α , r being the main offenders). The tables below collect the recurring symbols, chapter by chapter, and flag the collisions explicitly.

Probability and stochastic processes (Chapters 2–4)

$P_n(x_n, t_n; \dots; x_1, t_1)$	joint probability density of order n
$P(x, t \mid x', t')$	conditional (transition) probability density
$\langle \cdot \rangle$, $\mathbb{E}[\cdot]$	ensemble average (the two notations are used interchangeably)
$\text{Var}[\cdot]$, $\text{Cov}[\cdot]$	variance and covariance
$\mathcal{N}(m, \sigma^2)$	Gaussian distribution of mean m and variance σ^2
$a_1(x, t)$, $a_2(x, t)$	Kramers–Moyal coefficients: drift and diffusion
v , D	drift and diffusion when constant
L_{FP}	Fokker–Planck operator
$j(x, t)$, $J(r)$	probability current (flux)
$\mathcal{L}_{\text{OM}}$	Onsager–Machlup Lagrangian
$W(t)$	Wiener process; $dW(t)$ its Itô differential, $[dW]^2 = dt$
$\xi(t)$	white noise, $\xi = dW/dt$ in the distributional sense
$a(x, t)$, $b(x, t)$	drift and noise amplitude of a generic Itô SDE
κ , $\tau = 1/\kappa$	restoring rate and relaxation time of the OU process

Derivatives and pricing (Chapter 5)

$S(t), S_t$	spot price of the underlying stock
μ, σ	historical drift (return rate) and volatility of the stock
r	risk-free interest rate (constant in this chapter)
X, T	strike price and maturity of an option
C, P, O	call, put, and generic option price
$\{x\}_+$	positive part, $\max\{x, 0\}$
$\Pi(S, t)$	cash position of the writer's hedged portfolio
$\Delta, \Theta, \rho, \nu$	the Greeks (hedge ratio, time decay, rate and vega sensitivities)
$N(x)$	cumulative standard normal distribution (<i>not</i> the Gaussian density $\mathcal{N}$)
d_1, d_2	the two Black–Scholes arguments, $d_1 = d_2 + \sigma\sqrt{T-t}$
P, Q	historical and risk-neutral measures in the Björk supplement
L_t, φ_t, W_t^Q	likelihood process, Girsanov kernel and Q-Wiener process
$B_t, \Pi_t[X]$	bank account and claim price in that supplement (not the bond B or cash Π elsewhere)
u, d, p	up/down factors and risk-neutral probability of the CRR tree
$S_{i,j}, C_{i,j}$	stock and option values at tree node (time i, j up-moves)

Interest rates and bonds (Chapter 6)

$B(r, t; T)$	bond price; normalized so that $B(T) = 1$
$R(t, T)$	yield to maturity; its plot against T is the yield curve
$r(t)$	spot rate, $r(t) = \lim_{T \rightarrow t} R(t, T)$
$f(r, t), S(r, t)$	drift and diffusion of the rate SDE (<i>warning</i> : this S is not the stock price of Chapter 5)
$\mu(t, T), \sigma(t, T)$	return and volatility of the bond (<i>warning</i> : not the stock parameters of Chapter 5)
$q(r, t)$	market price of risk
α, θ, σ	speed of mean reversion, long-run mean, and noise amplitude of the Vasicek/OU rate
$\theta^* = \theta + \sigma q / \alpha$	risk-adjusted long-run mean
$A(t, T), B_0(t, T)$	coefficient functions of the exponential-affine bond price (B_0 is <i>not</i> a bond price)

Advanced models (Chapter 7)

α	Lévy index of a stable distribution, $0 < \alpha \leq 2$ (<i>warning</i> : unrelated to the Vasiček α of Chapter 6)
δ	scale parameter of a stable law
$\tilde{\phi}(k)$	characteristic function (Fourier transform of the PDF)
a_N	rescaling factor of a sum of N i.i.d. variables, $a_N = N^{1/\alpha}$
$P_{>}(x)$	complementary cumulative distribution (CCDF)
z	normalized (standardized) log-return
$f[Y], g[Y]$	volatility function and vol-of-vol in stochastic volatility models
ρ	correlation of the two Wiener processes (leverage effect if $\rho < 0$)
ν, σ	variance and vol-of-vol in the Heston pricing section (instantaneous stock volatility is $\sqrt{\nu}$)
λ	price of volatility risk

Econophysics (Chapter 8)

r	individual gross income (<i>warning</i> : not the interest rate of Chapters 5–6)
$A(r), B(r)$	drift and diffusion of the income Fokker–Planck equation (sign convention: $E[dr r] = -A(r) dt$ away from a reflecting boundary)
$T_r = B_0/A_0$	effective temperature of income (Boltzmann regime)
$\alpha = 1 + a/b$	Pareto exponent of the income CCDF tail

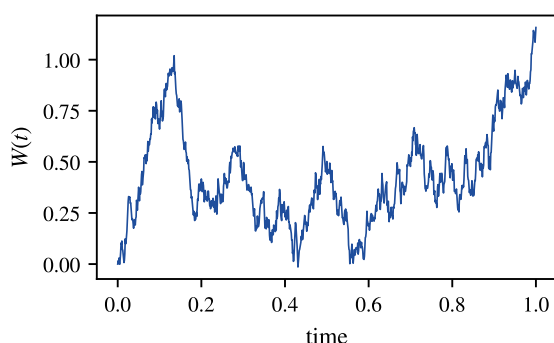
Introduction

The interest of natural scientists in social phenomena has been present for a long time. In the late twentieth century it became clear that some social and economic systems can be studied effectively using tools from theoretical physics, especially statistical physics. This observation helped give rise to the field known as *econophysics*. The term was introduced by H. Eugene Stanley at a conference in Kolkata in 1995.

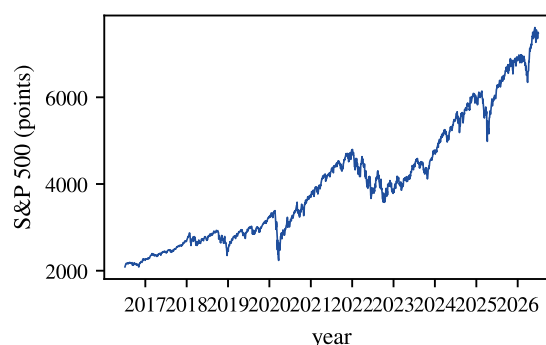
By the end of the 1990s many physicists and mathematicians were working on economic and financial problems. For example, a 1998 Nature report stated that 48% of the surveyed 1997 UK EPSRC-funded doctoral graduates in mathematical and statistical physics were working in finance one year after graduation [17].

Econophysics applies methods from theoretical physics, especially statistical physics, to the study of financial markets and related economic phenomena, which are treated as complex systems. The aim is to build simple, testable models that capture collective effects emerging from the interactions of many agents, following a *data-driven* approach.

A simple illustration of the analogy between a physical and a financial process is the resemblance between Brownian motion and the time series of asset prices. Compare (a) a one-dimensional projection of a Brownian particle's trajectory with (b) an example spot-price time series.



(a) One-dimensional Brownian motion (sample path)



(b) S&P 500 daily closes, 5 July 2016–2 July 2026. Source: cached FRED series SP500.

This visual similarity is striking given that the two processes originate from very different mechanisms: one is a physical diffusion, the other results from the collective behaviour of many traders responding to supply and demand.

Brownian Motion

2.1 Historical observations

The history of Brownian motion dates back to the early observations of the botanist Robert Brown. While observing pollen grains suspended in a liquid, he noted their highly erratic and jagged trajectory. These particles were subsequently named *Brownian* particles.

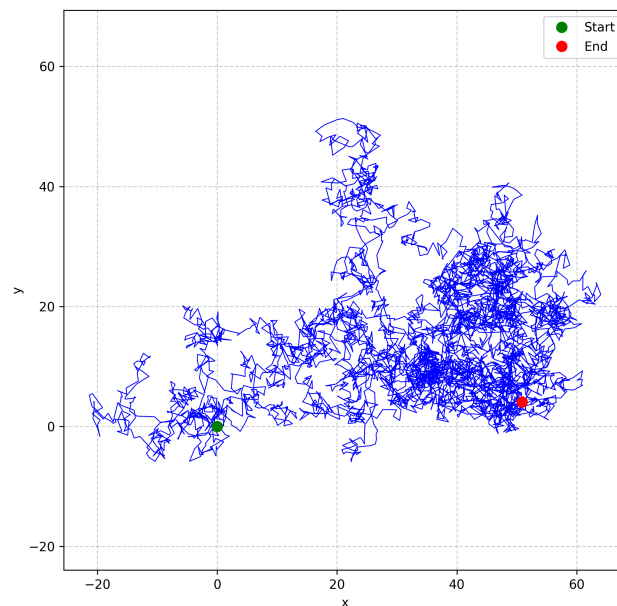


Figure 2.1: Simulation of a Brownian trajectory in two dimensions.

Specifically, Brown made the following qualitative observations:

- i) The motion never seems to cease.
- ii) Different particles always move independently of one another.
- iii) The particle's material composition (organic or inorganic) or its mass density do not seem to matter.
- iv) The motion becomes faster and more agitated as the fluid's temperature increases or as its viscosity decreases.

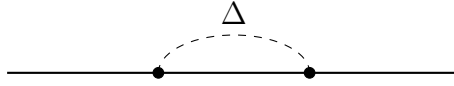
- v) The motion exhibits scale invariance (it looks similarly jagged at different magnifications).
- vi) The trajectory appears to have no tangent at any point.

The first theoretical characterization of this phenomenon was provided by Albert Einstein in two revolutionary papers published in 1905 and 1906 [10, 11]. We will now illustrate the key steps of his reasoning.

2.2 Einstein's papers

2.2.1 From random walk to the diffusion equation

He considered a one-dimensional, infinitely extended fluid composed of an ensemble of particles. The Brownian particle undergoes a series of discrete displacements, or "jumps", over very small, successive time intervals, denoted by τ , much smaller than the observation time T .



To describe the statistics of these jumps, Einstein introduced an auxiliary function, $\phi(\Delta)$, which is the probability density function (PDF) for a jump of magnitude Δ . The term $\phi(\Delta)d\Delta$ represents the probability that a single jump has a magnitude between Δ and $\Delta + d\Delta$. This function is defined by two fundamental properties based on physical reasoning:

- **Normalization:** The particle must perform a jump of some magnitude, so the total probability over all possible jumps must be equal to 1. This is expressed by the normalization condition:

$$\int_{-\infty}^{\infty} \phi(\Delta) d\Delta = 1. \quad (2.1)$$

- **Symmetry:** In the absence of any external force or drift, the fluid is isotropic. Therefore, a jump to the right ($+\Delta$) is as probable as a jump of the same magnitude to the left ($-\Delta$). This implies that the function ϕ must be an even function:

$$\phi(\Delta) = \phi(-\Delta). \quad (2.2)$$

The main goal, however, is to find $p(x, t)$, which represents the number of particles per unit length at position x and time t . If we consider $p(x, t)$ as the probability density for a single particle, it must be normalized such that its integral over all space is 1:

$$\int_{-\infty}^{\infty} p(x, t) dx = 1. \quad (2.3)$$

The probability of finding the particle in the infinitesimal interval $[x, x + dx]$ is therefore $p(x, t)dx$.

The core idea is to relate the particle density at time $t_f = t + \tau$ to the density at time $t_i = t$. The density of particles at position x at the future time $t + \tau$, denoted $p(x, t + \tau)$, is the sum of probabilities of particles arriving at x from all possible previous positions. A particle that ends up at x must have been at a position $x + \Delta$ at time t and then performed a jump of size Δ . Summing over all possible jumps Δ :

$$p(x, t + \tau) = \int_{-\infty}^{\infty} p(x + \Delta, t) \phi(\Delta) d\Delta. \quad (2.4)$$

Now, we perform a Taylor expansion for small τ and small Δ . The left side is expanded in time:

$$p(x, t + \tau) \approx p(x, t) + \tau \frac{\partial p(x, t)}{\partial t} + \dots; \quad (2.5)$$

the term $p(x + \Delta, t)$ inside the integral is expanded in space around the position x :

$$p(x + \Delta, t) \approx p(x, t) + \Delta \frac{\partial p(x, t)}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 p(x, t)}{\partial x^2} + \dots; \quad (2.6)$$

substituting these expansions back into Equation (2.4):

$$p(x, t) + \tau \frac{\partial p}{\partial t} \approx \int_{-\infty}^{\infty} \left[p(x, t) + \Delta \frac{\partial p}{\partial x} + \frac{\Delta^2}{2} \frac{\partial^2 p}{\partial x^2} \right] \phi(\Delta) d\Delta. \quad (2.7)$$

We can split the integral by linearity:

$$p(x, t) + \tau \frac{\partial p}{\partial t} \approx p(x, t) \underbrace{\int \phi(\Delta) d\Delta}_{=1} + \frac{\partial p}{\partial x} \underbrace{\int \Delta \phi(\Delta) d\Delta}_{=0} + \frac{1}{2} \frac{\partial^2 p}{\partial x^2} \int \Delta^2 \phi(\Delta) d\Delta. \quad (2.8)$$

The terms simplify based on the properties of ϕ .

So we are left with:

$$\tau \frac{\partial p(x, t)}{\partial t} \approx \frac{1}{2} \frac{\partial^2 p(x, t)}{\partial x^2} \int_{-\infty}^{\infty} \Delta^2 \phi(\Delta) d\Delta. \quad (2.9)$$

Dividing by τ , we get:

$$\frac{\partial p(x, t)}{\partial t} = D \frac{\partial^2 p(x, t)}{\partial x^2}. \quad (2.10)$$

This is the *diffusion equation* (with $D = 1$ it becomes the well known heat equation), where D is the *diffusion coefficient*, defined as:

$$D := \frac{1}{2\tau} \int_{-\infty}^{\infty} \Delta^2 \phi(\Delta) d\Delta. \quad (2.11)$$

Since $\Delta^2 \geq 0$ and $\phi(\Delta) \geq 0$, the integral is positive, thus $D > 0$. The dimensions of D are length squared per unit time, $[D] = L^2 T^{-1}$.

2.2.2 Solution of the diffusion equation

We now solve the diffusion equation with a specific initial condition. Physically, it is natural to assume that the process starts with the particle (or all particles) located exactly at the origin at time $t = 0$. This condition is represented mathematically by the Dirac delta function, $\delta(x)$. It represents an instantaneous, infinitely concentrated point source. The solution $p(x, t)$ will then describe how this initial certainty "diffuses" into a probability distribution over time.

The problem is thus defined by the partial differential equation and its initial condition:

$$\begin{aligned} \frac{\partial p(x, t)}{\partial t} &= D \frac{\partial^2 p(x, t)}{\partial x^2}; \\ p(x, 0) &= \delta(x). \end{aligned} \quad (2.12)$$

We solve this using the Fourier transform. Let $p(k, t)$ be the Fourier transform of $p(x, t)$ with respect to the spatial variable x . The transform pair is defined as:

$$\begin{aligned} \text{Inverse Transform: } p(x, t) &= \int_{-\infty}^{\infty} e^{ikx} p(k, t) dk; \\ \text{Transform: } p(k, t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-ikx} p(x, t) dx. \end{aligned} \quad (2.13)$$

First, we transform the initial condition. Using the integral representation of the Dirac delta, $\delta(x) = \frac{1}{2\pi} \int e^{ikx} dk$,

$$p(k, 0) = \frac{1}{2\pi}. \quad (2.14)$$

Next, we transform the diffusion equation itself. The time derivative remains, while the spatial derivative becomes a multiplication by $(ik)^2 = -k^2$:

$$\mathcal{F} \left[\frac{\partial p}{\partial t} \right] = D \cdot \mathcal{F} \left[\frac{\partial^2 p}{\partial x^2} \right] \implies \frac{\partial p(k, t)}{\partial t} = D(-k^2)p(k, t). \quad (2.15)$$

This transforms the partial differential equation in (x, t) into a simpler ordinary differential equation in the variable t for each value of k :

$$\begin{aligned} \frac{\partial p(k, t)}{\partial t} &= -Dk^2 p(k, t); \\ p(k, 0) &= \frac{1}{2\pi}. \end{aligned} \quad (2.16)$$

This is a first-order linear ODE with constant coefficients, which can be solved by separation of variables.

Let's treat k as a constant parameter and solve the ODE for the function $f(t) = p(k, t)$:

$$\frac{df(t)}{dt} = (-Dk^2)f(t). \quad (2.17)$$

Separating the variables f and t :

$$\frac{df}{f} = -Dk^2 dt. \quad (2.18)$$

Integrating both sides:

$$\int \frac{1}{f} df = \int -Dk^2 dt \implies \ln |f| = -Dk^2 t + C_0. \quad (2.19)$$

where C_0 is the constant of integration. Exponentiating both sides gives:

$$f(t) = e^{-Dk^2 t + C_0} = e^{C_0} e^{-Dk^2 t} = C e^{-Dk^2 t}. \quad (2.20)$$

We find the constant C using the initial condition $p(k, 0) = 1/(2\pi)$:

$$p(k, 0) = C e^0 = C \implies C = \frac{1}{2\pi}. \quad (2.21)$$

Thus, the solution in the Fourier space is:

$$p(k, t) = \frac{1}{2\pi} e^{-Dk^2 t}. \quad (2.22)$$

To find the solution in the original space, we apply the inverse Fourier transform to $p(k, t)$:

$$p(x, t) = \int_{-\infty}^{\infty} e^{ikx} p(k, t) dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} e^{-Dk^2 t} dk = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-Dk^2 t + ikx} dk. \quad (2.23)$$

To solve this Gaussian integral, we complete the square in the exponent:

$$\begin{aligned} -Dtk^2 + ikx &= -Dt \left(k^2 - \frac{ix}{Dt} k \right) \\ &= -Dt \left[\left(k - \frac{ix}{2Dt} \right)^2 - \left(\frac{ix}{2Dt} \right)^2 \right] \\ &= -Dt \left(k - \frac{ix}{2Dt} \right)^2 - Dt \left(\frac{x^2}{4D^2 t^2} \right) \\ &= -Dt \left(k - \frac{ix}{2Dt} \right)^2 - \frac{x^2}{4Dt}. \end{aligned} \quad (2.24)$$

Substituting this back into the integral:

$$p(x, t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \exp \left[-Dt \left(k - \frac{ix}{2Dt} \right)^2 - \frac{x^2}{4Dt} \right] dk = \frac{1}{2\pi} e^{-\frac{x^2}{4Dt}} \int_{-\infty}^{\infty} e^{-Dt \left(k - \frac{ix}{2Dt} \right)^2} dk. \quad (2.25)$$

The remaining integral is a standard Gaussian integral of the form $\int e^{-a(u-b)^2} du = \sqrt{\pi/a}$. Here, $a = Dt$. The complex shift $b = ix/(2Dt)$ does not change the result of the integral over the real line.

$$\int_{-\infty}^{\infty} e^{-Dt \left(k - \frac{ix}{2Dt} \right)^2} dk = \sqrt{\frac{\pi}{Dt}}. \quad (2.26)$$

Therefore, the solution is:

$$p(x, t) = \frac{1}{2\pi} e^{-\frac{x^2}{4Dt}} \sqrt{\frac{\pi}{Dt}} = \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}. \quad (2.27)$$

This is a Gaussian distribution with mean $\mu = 0$ and variance $\sigma_t^2 = 2Dt$.

A minimal and standard choice for the diffusion coefficient is $D = 1/2$. This specific case defines the standard Brownian motion, also known as the *Wiener process*. The probability density function becomes:

$$p(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}. \quad (2.28)$$

This process is a cornerstone of stochastic calculus. It exhibits scale invariance and is considered the first and most fundamental example of a 1D fractal process.

2.2.3 Physical interpretation and statistical consequences

The solution we found is a normal distribution:

$$p(x, t) = \frac{1}{\sqrt{2\pi\sigma_t^2}} e^{-\frac{x^2}{2\sigma_t^2}}. \quad (2.29)$$

where the variance σ_t^2 is time-dependent:

$$\sigma_t^2 = 2Dt. \quad (2.30)$$

We can calculate the key moments of this distribution.

- **Mean Position:** The average position of the particle is given by the expectation value $\langle x \rangle$. Since the Gaussian is centered at zero (it is an even function) and we are integrating over a symmetric domain, the mean is zero.

$$\langle x \rangle = \int_{-\infty}^{\infty} x p(x, t) dx = 0. \quad (2.31)$$

This confirms that the diffusion process has no drift; the particle is equally likely to be found on either side of the origin.

- **Variance:** The variance measures the spread of the distribution.

$$\begin{aligned} \text{Var}[x] &= \langle (x - \langle x \rangle)^2 \rangle = \langle x^2 \rangle - \langle x \rangle^2 = \langle x^2 \rangle \\ &= \int_{-\infty}^{\infty} x^2 p(x, t) dx = \sigma_t^2 = 2Dt. \end{aligned} \quad (2.32)$$

This is a very important result of diffusion theory: the variance of the particle's position grows linearly with time. This means the particle spreads out, and our uncertainty about its location increases as time passes. Standard diffusive processes are thus Gaussian.

The evolution of the probability density is a macroscopic description of the ensemble of particles, not the path of a single particle. It starts as an infinitely sharp Dirac delta at $t = 0$ and then "spreads out" as a Gaussian with increasing width, as illustrated below.

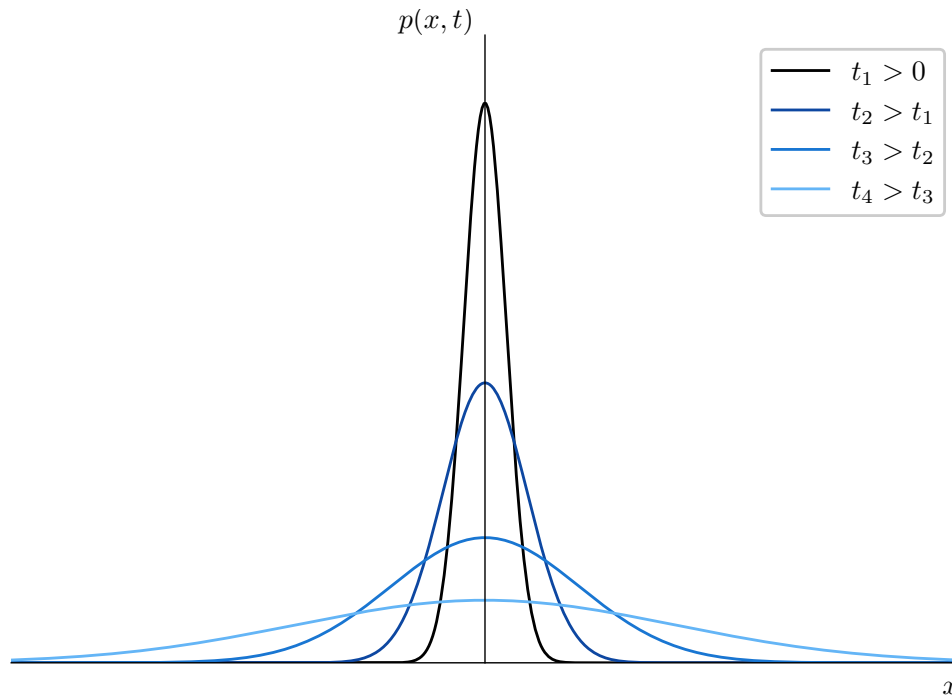


Figure 2.2: Evolution of the probability density $p(x, t)$ over time. Starting from a Dirac delta at $t = 0$, the distribution spreads out as a Gaussian with variance $\sigma_t^2 = 2Dt$ increasing linearly in time.

Another relevant quantity to characterize the displacement of the particle is the Mean Squared Displacement (MSD). It is defined as the average squared distance the particle has traveled from its starting point.

$$\text{MSD}(t) = \langle [x(t) - x(0)]^2 \rangle. \quad (2.33)$$

Let's consider two cases:

- If the particle starts at the origin, $x(0) = 0$:

$$\text{MSD} = \langle [x(t)]^2 \rangle = \langle x^2(t) \rangle = 2Dt. \quad (2.34)$$

The MSD is simply the variance.

- If the particle starts at a generic position $x(0) = x_0$:

$$\begin{aligned} \text{MSD} &= \langle [x(t) - x_0]^2 \rangle \\ &= \langle x^2(t) \rangle + x_0^2 - 2x_0 \langle x(t) \rangle \\ &= \langle x^2(t) \rangle + x_0^2 - 2x_0^2 \\ &= x_0^2 + 2Dt - x_0^2 \\ &= 2Dt. \end{aligned} \quad (2.35)$$

In both cases, the MSD grows linearly with time.

The typical distance traveled by the particle, the root-mean-square displacement λ_x , can be estimated as the root of the MSD.

$$\lambda_x = \sqrt{\text{MSD}} = \sqrt{2Dt}. \quad (2.36)$$

This shows that the characteristic displacement of a diffusing particle grows not with time t , but with its square root. This sub-linear growth is a direct consequence of the random, back-and-forth nature of the motion.

2.2.4 Einstein's second paper

Let us now investigate the physical significance of the diffusion coefficient D . We recall from the studies of R. Brown, that the motion of suspended (Brownian) particles becomes more intense as the temperature T rises and as the viscosity η of the liquid or the radius R of the particles decrease. Thus, we need to link these macroscopic parameters (the radius of the particle – $\sim 10^{-6}m$ – is sufficiently large to neglect quantum effects and consider it a macroscopic object) to the diffusion coefficient.

Let us consider a statistical ensemble of N identical Brownian particles of radius R , suspended in a three-dimensional tank liquid of temperature T and viscosity η in statistical equilibrium. We consider three main forces acting on each particle in the vertical direction (x):

- Gravitational pull:

$$F_G = -mg = -\frac{d}{dx}(V_G(x)), \quad V_G(x) = mgx. \quad (2.37)$$

- Archimede's buoyancy:

$$F_A(x) = \rho(x) g \text{Vol} \ll F_G \quad (\text{negligible}). \quad (2.38)$$

- Stokes drag:

$$F_S = -6\pi\eta Rv \quad (\text{valid for low Reynolds' number}), \quad (2.39)$$

where the factor $6\pi\eta R$ is also known as the *friction coefficient*; the mobility is its reciprocal.

Restricting our attention to the stationary regime we can define a drift velocity of

$$v_d = -\frac{mg}{6\pi\eta R}, \quad (2.40)$$

and a superficial current of matter (number of particles that cross the unit of surface in the unit of time)

$$J_{drift}(x) = v_d \rho(x). \quad (2.41)$$

Under the hypothesis of statistical equilibrium and classical physics, the Boltzmann formula¹ entails that

$$\rho(E) = N \frac{e^{-\beta E}}{\mathcal{Z}}, \quad \beta = (k_B T)^{-1}, \quad E = V(x) + \frac{mv^2}{2}. \quad (2.42)$$

Hence, if we have that $\rho(0) = Ne^{-\beta mv^2/2}$ up to the same normalization factor ($V(0) = 0$)

$$\frac{\rho(x)}{\rho(0)} = e^{-\beta V(x)}. \quad (2.43)$$

This is the case of the gravitational potential $V(x) = V_G(x) = mgx$, which recovers the so-called *barometric distribution*

$$\rho(x) = \rho(0)e^{-\beta mgx}. \quad (2.44)$$

Since the concentration increases with depth, diffusion produces an upward probability current, opposing gravitational drift. The analytic form of the current generated by this effect is nothing more than Fick's diffusion current

$$J_{diff}(x) = -D_{E.S.} \frac{d}{dx} \rho(x), \quad (2.45)$$

for some $D_{E.S.} \in \mathbb{R}^+$ (this is Fick's first law; the second is the diffusion equation itself). Considering the stationary regime, we demand $\rho(x)$ to be constant over time, *i.e.*, with zero net flow $J_{diff} + J_{drift} = 0$. Writing $V(x) = Fx$ with $F = mg > 0$ (drag is not a conservative force),

$$-D \frac{d\rho(x)}{dx} + v_d \rho(x) = 0, \quad (2.46)$$

that is,

$$\rho(0)e^{-\beta Fx} \frac{F}{6\pi\eta R} = -\rho(x)v_d = -D_{E.S.} \frac{d\rho(x)}{dx} = D_{E.S.} \beta F \rho(0)e^{-\beta Fx}, \quad (2.47)$$

and simplifying the common exponential factor,

$$D_{E.S.} = \frac{k_B T}{6\pi\eta R}, \quad (2.48)$$

where "E.S." stands for Einstein-Smoluchowski. As a consequence, the variance $\sigma_t^2 = 2D_{E.S.}t$ varies with the temperature T and the viscosity η , exactly as in Brown's qualitative observations. With this characterisation, we find all the properties of the chaotic motion described by R. Brown with the exception of self-similarity².

¹For the equilibrium background, see Sethna[38].

²See J. B. Perrin's work for an experimental verification of the Einstein-Smoluchowski formula[34]. Measuring D gives access to $k_B = R/N_A$, hence to an experimental measurement of Avogadro's number N_A , in great agreement with the chemical estimate; this earned Perrin the Nobel Prize in 1926.

Considering a statistical ensemble, we can identify $p(x, t) \leftrightarrow \rho(x, t)$. Recalling the following formula for Einstein's first work

$$\begin{aligned} \partial_t \rho(x, t) &= D \partial_x^2 \rho(x, t) \implies \\ 0 &= \partial_t \rho(x, t) - \partial_x \{D \partial_x \rho(x, t)\} = \partial_t \rho(x, t) + \partial_x J(x, t), \end{aligned} \quad (2.49)$$

recovering a continuity equation for $J(x, t) = -D \partial_x \rho(x, t)$. In the stationary regime we set $\rho(x, t) \rightarrow \rho_S(x)$, entailing that $\partial_t \rho_S = 0$ and consequently $\partial_x J_S = 0$. In this setting, we find $J(x) = -D \partial_x \rho_S(x)$. This observation fixes $D = D_{E.S.}$.

2.3 The Langevin equation

We now would like to extend Einstein's characterisation of the Brownian motion by describing the trajectory of the Brownian particles. This, for obvious reasons, cannot be done in a deterministic way, but an attempt to delineate an average trajectory was made by Langevin: his reasoning goes as follows. Langevin's equation (1908)[25] provides a *microscopic* description of the motion, and it is historically the first *stochastic differential equation* — an object which, at the beginning, had no precise mathematical meaning, and was formalized only later.

He considered a statistical ensemble of Brownian particles of mass m and radius R suspended in a 1-dimensional fluid of temperature T and viscosity η in statistical equilibrium. Langevin admits a non-symmetric "jump" distribution, whose unbalance between the probability of a right and a left jump generates an overall drift velocity v_d . The second principle of dynamics can be written as $ma = F = F^{(1)} + F^{(2)}$, with $F^{(1)} = F_S = -6\pi\eta Rv$ the deterministic Stokes force while $F^{(2)} = \xi(t)$ which models the stochastic particles scatterings as *White Noise*. We choose $\xi(t)$ such that for any time t the ensemble average reads

$$\langle \xi(t) \rangle_{\text{Ensemble}} = 0, \text{ for any } t \in \mathbb{R}. \quad (2.50)$$

If we consider $\xi(t) \equiv 0$, we find a deterministic differential equation $m \frac{dv(t)}{dt} = -6\pi\eta Rv(t)$, which has for a solution $v(t) = v(0)e^{-t/\tau}$, with $\tau = m/6\pi\eta R$. In the white-noise model the velocity is not differentiable; for nonzero mass the position, obtained by integrating that continuous velocity, is differentiable. The overdamped Wiener approximation for position is discussed in Chapter 3. We have

$$\begin{aligned} m \frac{d^2 x}{dt^2} &= -6\pi\eta R \frac{dx}{dt} + \xi(t) \\ mx \frac{d^2 x}{dt^2} &= -6\pi\eta Rx \frac{dx}{dt} + x\xi(t) \end{aligned} \quad (2.51)$$

using the fact that $\frac{d(x^2)}{dt} = 2x \frac{dx}{dt}$ and consequently $\frac{d(x^2)}{dt^2} = 2\left(\frac{dx}{dt}\right)^2 + 2x \frac{d^2 x}{dt^2}$, we have

$$\left(\frac{m}{2}\right) \left(2x \frac{d^2 x}{dt^2}\right) = \left(\frac{m}{2}\right) \left(\frac{d^2(x^2)}{dt^2} - 2v^2\right), \quad (2.52)$$

then, substituting in the previous equation to express everything in function of x^2

$$\left(\frac{m}{2}\right) \frac{d^2(x^2)}{dt^2} - mv^2 = -3\pi\eta R \frac{d(x^2)}{dt} + x\xi(t). \quad (2.53)$$

We observe that x is a stochastic variable since it depends on $\xi(t)$. In this spirit we evaluate the ensemble average of the differential equation, *i.e.*

$$\left(\frac{m}{2}\right) \frac{d^2}{dt^2} \langle x^2 \rangle - \langle mv^2 \rangle = -3\pi\eta R \frac{d}{dt} \langle x^2 \rangle + \langle x\xi(t) \rangle. \quad (2.54)$$

We recall that under the hypothesis statistical equilibrium, it holds the Equipartition Theorem[38], which entails that $\langle \frac{mv^2}{2} \rangle = \frac{k_B T}{2}$. In addition, since $\xi(t)$ is independent from the particle's position (*Markov process*), then we can factorise $\langle x\xi(t) \rangle = \langle x \rangle \langle \xi(t) \rangle = 0$. Collecting all of the above observations in one formula, gives

$$\frac{m}{2} \frac{d^2}{dt^2} \langle x^2 \rangle + 3\pi\eta R \frac{d}{dt} \langle x^2 \rangle = k_B T. \quad (2.55)$$

We observe that, despite in the initial differential equation a stochastic term was added, in the final equation it does not appear: this is because we supposed $\langle \xi(t) \rangle = 0$. However, the trace of its existence lies in the fact that to erase the stochastic term, we had to consider an ensemble-average equation, giving up the "determinability" of our result.

If we set $f(t) = \frac{d}{dt} \langle x^2 \rangle$, we have that

$$\begin{aligned} f'(t) + af(t) &= b, \quad a = 6\pi\eta R/m, \quad b = 2k_B T/m \implies \\ f(t) &= f_0(t) + f_p(t) = Ce^{-at} + b/a \implies \\ \frac{d}{dt} \langle x^2 \rangle &= Ce^{-\frac{6\pi\eta R t}{m}} + \frac{k_B T}{3\pi\eta R} = Ce^{-t/\tau} + \frac{k_B T}{3\pi\eta R}, \end{aligned} \quad (2.56)$$

with $\tau = m/6\pi\eta R \sim 10^{-8}s$. After the transient state (*i.e.* for $t \gg \tau$) $e^{-t/\tau} \approx 0$, as a consequence, if we integrate the above equation on $[0, t]$ we have that

$$\langle x^2(t) \rangle = \langle x^2(0) \rangle + C\tau(1 - e^{-t/\tau}) + 2tD_{E.S.} \sim 2tD_{E.S.} \quad (2.57)$$

This result should make the reader "jump on their chair" since it explicitly confirms the equivalence of the macroscopic (statistical – $p(x, t)$) derivation by Einstein-Smoluchowski with the microscopic (pseudo-deterministic – $\langle x(t) \rangle$) derivation by Langevin.

The solutions of the Langevin equation constitute a *family of time-dependent trajectories*. Figure 2.3 shows an ensemble of such trajectories: cutting the family at two successive times, the cross-sections of the cloud are Gaussians with larger and larger variance, exactly the $p(x, t) = \mathcal{N}(0, 2Dt)$ of the macroscopic description.

2.4 The random walk of Bachelier

The jump description of Brownian motion admits a discrete formulation of independent interest: the *random walk*. It was introduced for the first time by Bachelier in his PhD thesis *Théorie de la spéculation* (1900)[1] — five years *before* Einstein's papers, and with prices, not pollen grains, in mind.

The model rests on four hypotheses.

- i) The particle can occupy equispaced cells (of width Δx) on a one-dimensional lattice extending over $(-\infty, +\infty)$.
- ii) The initial position is zero at $t = 0$.

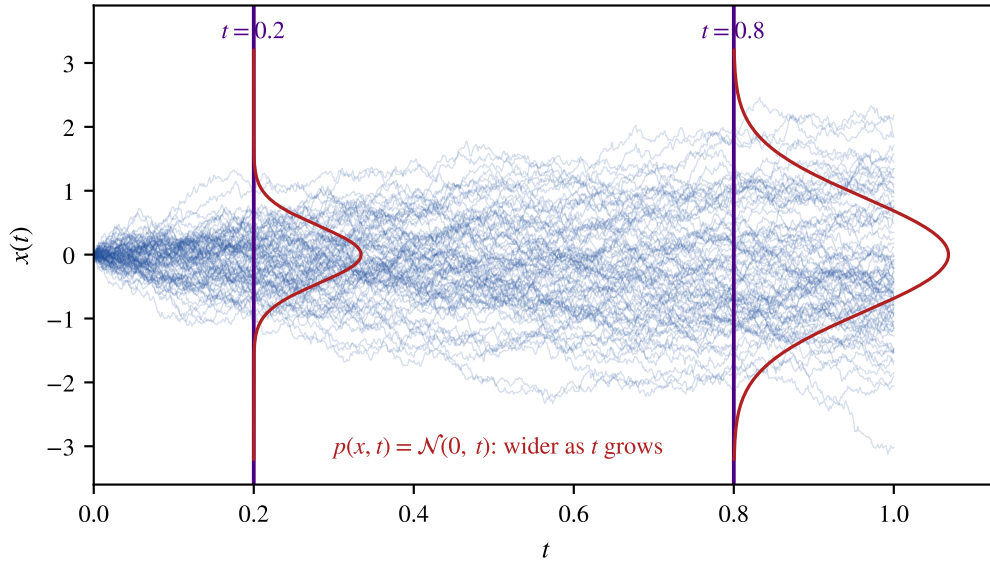
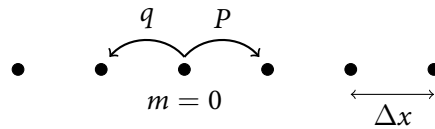


Figure 2.3: A family of Brownian trajectories, cut at two successive times $t_1 = 0.2$ and $t_2 = 0.8$ (vertical lines). The cross-section profiles (red) are Gaussians with variance growing linearly in time — the microscopic trajectories reconstruct the spreading density of Figure 2.2. Generated by `images/code_for_images/brownian_family.py`.

- iii) Calling P the probability of jumping to the right and q the one of jumping to the left, one has $P + q = 1$, and the jumps occur regularly, once every interval Δt . (For Einstein, $P = q$.)
- iv) The jumps are independent (Markov).

The position is labelled by $m \in \mathbb{Z}$ and $N \in \mathbb{N}$ is the number of jumps performed, so that $N\Delta t$ is the elapsed time.



The question is: what is the probability distribution $P_N(m)$ of being at position m after N jumps? If $P = \frac{1}{2} = q$ the model is the *symmetric* random walk; for $P \neq \frac{1}{2}$ and $q = 1 - P$ the individual jumps are Bernoulli variables, and their sum follows a *binomial distribution*. The counting is elementary: if the walker performs $m + \ell$ jumps to the right, it must perform ℓ jumps to the left, hence $N = m + 2\ell$ jumps in total; solving,

$$\ell = \frac{N - m}{2} \text{ jumps to the left,} \quad m + \frac{N - m}{2} = \frac{N + m}{2} \text{ jumps to the right,} \quad (2.58)$$

and the binomial distribution is then

$$P_N(m) = \binom{N}{\frac{N+m}{2}} P^{\frac{1}{2}(N+m)} q^{\frac{1}{2}(N-m)}. \quad (2.59)$$

Normalizing the price to zero at $t = 0$, this can be taken as a model for the evolution of prices — Bachelier's original purpose.

To compute the moments it is convenient to introduce the number of right jumps

$$r := \frac{N+m}{2}, \quad P_N(r) = \binom{N}{r} P^r q^{N-r} = \mathcal{B}(r; P, N), \quad (2.60)$$

for which the standard binomial results follow. The normalization is immediate from the binomial theorem,

$$\sum_{r=0}^{\infty} \binom{N}{r} P^r q^{N-r} = (P+q)^N = 1, \quad (2.61)$$

the mean is

$$\mathbb{E}[r] = \sum_{r=0}^{\infty} r P_N(r) = NP, \quad (2.62)$$

and, from $\mathbb{E}[r^2] = NP + N(N-1)P^2$, the variance is

$$\text{Var}[r] = NP + N(N-1)P^2 - N^2P^2 = NP(1-P). \quad (2.63)$$

Translating back to the position through $m = 2r - N$,

$$\langle m \rangle = 2\langle r \rangle - N = 2NP - N = 2N\left(P - \frac{1}{2}\right), \quad \text{Var}[m] = 4NP(1-P), \quad (2.64)$$

where we used $\langle m^2 \rangle = 4NP(1-P) + 4N^2(P - \frac{1}{2})^2$.

Both the mean and the variance depend linearly on N ; but $N = t/\Delta t$, so they are *linear in time* — recall Einstein's $\sigma_E^2 = 2Dt$, whose mean vanished because $P = q = \frac{1}{2}$. Note, moreover, that the distribution is a binomial, *not* a Gaussian.

The role of the asymmetry is best seen in three regimes, sketched in Figure 2.4. In the *maximally asymmetric* model ($P = 1, q = 0$, or $P = 0, q = 1$) one has $\text{Var}[m] = 0$ and $\mathbb{E}[m] = \pm N$: the walk is purely deterministic. With $P = \frac{3}{4}, q = \frac{1}{4}$ one finds $\langle m \rangle = N/2$ and $\text{Var}[m] = \frac{3}{4}N$: the variable fluctuates around a rising trend. Finally, with $P = q = \frac{1}{2}$, $\langle m \rangle = 0$ and $\text{Var}[m] = N$: pure fluctuation, with the $\sqrt{N}$ fan-out of diffusion.

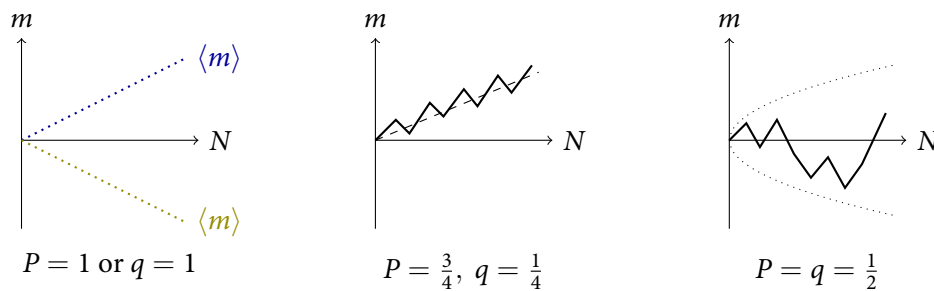


Figure 2.4: The three regimes of the random walk: purely deterministic ($P = 1$ or $q = 1$, zero variance), asymmetric (fluctuations around a linear trend), and symmetric (pure fluctuation inside the $\pm\sqrt{N}$ envelope).

But this — Samuelson observed — cannot be an acceptable model for the fluctuation of prices, because the fluctuations would decrease, which is not what happens. The objection deserves to be spelled out, because it is precisely the observation that will lead to the geometric model of Chapter 4. In an asymmetric random walk the mean position grows linearly, $\langle m \rangle \propto N$,

while the spread grows only as a square root, $\text{s.d.}[m] = \sqrt{4NPq} \propto \sqrt{N}$; the fluctuations *relative to the price level* therefore decay,

$$\frac{\text{s.d.}[m]}{\langle m \rangle} \propto \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}} \xrightarrow{N \rightarrow \infty} 0, \quad (2.65)$$

and an asset following this model would become, in proportion to its own value, ever quieter as it grows. A concrete example makes the failure vivid. Take a stock worth 100 € today, fluctuating by about 1 € per day (one percent), and suppose that over the years its price drifts up tenfold, to 1000 €. The arithmetic model predicts that its daily fluctuations are *still* of the order of 1 € — now a mere 0.1% of the price — whereas a real thousand-euro stock keeps fluctuating by percentage points, that is, by tens of euros per day. What is statistically stationary in real markets is the *relative* return, not the absolute increment: the fluctuations must be proportional to the price itself, which is exactly the multiplicative structure of the geometric Brownian motion of Chapter 4.

2.4.1 The continuum limit

The Gaussian is recovered from the binomial in the *continuum limit*. Define the physical variables

$$x := m\Delta x, \quad t := N\Delta t, \quad (2.66)$$

so that the mean position becomes

$$\langle x \rangle = \langle m \rangle \Delta x = (2P - 1)N\Delta x = (2P - 1)\frac{t}{\Delta t} \Delta x, \quad (2.67)$$

and, introducing the velocity

$$v := \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta t \rightarrow 0}} (2P - 1) \frac{\Delta x}{\Delta t} \quad \text{with the lattice parameters scaled to keep } v \text{ finite,} \quad (2.68)$$

one finds $\langle x \rangle = vt$: a *drift* term. For the variance,

$$\text{Var}[x] = \{ \langle m^2 \rangle - \langle m \rangle^2 \} (\Delta x)^2 = 4Pq (\Delta x)^2 \frac{t}{\Delta t}, \quad (2.69)$$

and defining the diffusion coefficient

$$D := \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta t \rightarrow 0}} 2Pq \frac{(\Delta x)^2}{\Delta t}, \quad \text{it follows that} \quad \text{Var}[x] = 2Dt. \quad (2.70)$$

The Gaussian solution is contained in Bachelier's model. Indeed, for $0 < P < 1$ fixed and $N \rightarrow \infty$, the local Gaussian approximation on the reachable sites $m \equiv N \pmod{2}$ is

$$P_N(m) \simeq 2P_G(m), \quad P_G(m) = \frac{1}{\sqrt{2\pi 4NPq}} \exp \left\{ -\frac{(m - N(2P - 1))^2}{2 4NPq} \right\}. \quad (2.71)$$

The factor two is the spacing between reachable sites; the other sites have zero probability. This is the local form of the *De Moivre-Laplace theorem*. Passing to $x = m\Delta x$ and $t = N\Delta t$, with $\Delta x^2/\Delta t \rightarrow 2D$ and $(2P - 1)\Delta x/\Delta t \rightarrow v$, gives

$$\frac{P_N(m)}{2\Delta x} \longrightarrow \frac{1}{\sqrt{4\pi Dt}} \exp \left\{ -\frac{(x - vt)^2}{4Dt} \right\} = P_G(x, t) \quad (2.72)$$

on the reachable lattice sites. The limiting density has $\langle x \rangle = vt$ and $\text{Var}[x] = 2Dt$: Einstein's solution, with a drift.

2.4.2 From the master equation to the Fokker–Planck equation

But where does this Gaussian come from? In the discrete case, the probability of being at x at time $t + \Delta t$ obeys the *master equation*

$$P(x, t + \Delta t) = P P(x - \Delta x, t) + q P(x + \Delta x, t), \quad (2.73)$$

since the walker must arrive either from the left (with probability P) or from the right (with probability q). Expanding the left side in time,

$$P(x, t + \Delta t) = P(x, t) + \Delta t \frac{\partial P}{\partial t}, \quad (2.74)$$

and each term on the right in space,

$$\begin{aligned} P P(x - \Delta x, t) + q P(x + \Delta x, t) &= P \left[P(x, t) - \Delta x \frac{\partial P}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 P}{\partial x^2} \right] \\ &\quad + q \left[P(x, t) + \Delta x \frac{\partial P}{\partial x} + \frac{\Delta x^2}{2} \frac{\partial^2 P}{\partial x^2} \right], \end{aligned} \quad (2.75)$$

which reassembles into

$$\underbrace{(P + q)}_{=1} P(x, t) + \underbrace{(-P + q)}_{1-2P} \Delta x \frac{\partial P}{\partial x} + \underbrace{(P + q)}_{=1} \frac{\Delta x^2}{2} \frac{\partial^2 P}{\partial x^2}. \quad (2.76)$$

Therefore, dividing by Δt (the terms $P(x, t)/\Delta t$ cancel between the two sides),

$$\frac{\partial P}{\partial t} = (1 - 2P) \frac{\Delta x}{\Delta t} \frac{\partial P}{\partial x} + \frac{1}{2} \frac{\Delta x^2}{\Delta t} \frac{\partial^2 P}{\partial x^2}, \quad (2.77)$$

and with the definitions of v and D given above (with $P = q = \frac{1}{2}$), in the continuum³

$$\frac{\partial}{\partial t} P(x, t) = \underbrace{D \frac{\partial^2}{\partial x^2} P(x, t)}_{\text{pure diffusion}} - \underbrace{v \frac{\partial}{\partial x} P(x, t)}_{\text{drift}}, \quad (2.78)$$

without the drift term when $\langle x \rangle = 0$. This is the *Fokker–Planck equation* — here obtained for the simplest of all processes, and derived in full generality in Chapter 3.

2.5 The central limit theorem

The Gaussian appeared twice in this chapter — as the solution of the diffusion equation and as the limit of the binomial — and this is no coincidence: behind both stands the *central limit theorem* (CLT), which we now state, prove, and (importantly for what comes later in this book) criticize.

Theorem (CLT). *Let $\{X_i\}_i$ be random variables and $S_N := \sum_{i=1}^N X_i$ their sum; then, for $N \rightarrow \infty$, $(S_N - N\mu)/(\sigma\sqrt{N})$ converges in distribution to $\mathcal{N}(0, 1)$, with $\mu = E[X_i]$ and $0 < \sigma^2 = \text{Var}(X_i) < \infty$, provided that (a) the X_i are i.i.d. (independent and identically distributed), and (b) the mean $\langle X_i \rangle$ and the variance $\text{Var}[X_i]$ exist finite for every i .*

The two hypotheses look innocent; the critique of Lévy and Mandelbrot, to which we return at the end of the section, is aimed precisely at them.

³A small bookkeeping subtlety: the expansion produces the diffusion coefficient $\frac{1}{2} \Delta x^2 / \Delta t$, while the variance-based definition given earlier is $D = 2Pq \Delta x^2 / \Delta t$; the two coincide exactly at $P = q = \frac{1}{2}$ ($4Pq = 1$), which is why the derivation specifies this case. In the joint limit $\Delta x, \Delta t \rightarrow 0$ with v and D finite, $P \rightarrow \frac{1}{2}$ automatically, so no inconsistency survives in the continuum.

2.5.1 Characteristic functions

The proof (due to Lindeberg and Lévy) rests on the *characteristic function*: given a density $P(x)$ on $\mathbb{R}$,

$$\varphi(k) := \int_{\mathbb{R}} dx e^{ikx} P(x) = \mathcal{F}[P] = \mathbb{E}[e^{ikx}], \quad P(x) = \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{-ikx} \varphi(k), \quad (2.79)$$

i.e. the Fourier transform of the density. Two properties make it the perfect tool.

a) Convolution theorem. Since $\mathcal{F}[f * g] = \mathcal{F}[f] \cdot \mathcal{F}[g]$, sums of independent variables become products of characteristic functions. Explicitly: for X_1 and X_2 i.i.d., the sum $S_2 := X_1 + X_2$ is again random, and its density is obtained by extracting the two variables, $P(x_1, x_2) = P(x_1)P(x_2) = P(S_2 - x_1)P(x_1)$, and summing over all the possible ways of realizing S_2 :

$$P_2(S_2) = \int_{\mathbb{R}} P(S_2 - x_1) P(x_1) dx_1, \quad (2.80)$$

the convolution of two identical densities. Transforming,

$$\mathcal{F}[P_2] = \mathcal{F}[P * P] = \varphi_2(k) = \mathcal{F}[P] \mathcal{F}[P] = \varphi^2(k), \quad (2.81)$$

and iterating,

$$\mathcal{F}[P_N] = \varphi_N(k) = \varphi^N(k). \quad (2.82)$$

b) Moment expansion. Expanding $e^{ikx} = \sum_{m=0}^{\infty} \frac{(ik)^m}{m!} x^m$ inside the transform,

$$\varphi(k) = \int_{\mathbb{R}} dx e^{ikx} P(x) = \sum_{m=0}^{\infty} \frac{(ik)^m}{m!} \int_{\mathbb{R}} dx x^m P(x) = \sum_{m=0}^{\infty} \frac{(ik)^m}{m!} \langle x^m \rangle, \quad (2.83)$$

provided interchange of the infinite series and integral is justified. Existence of all moments alone does not ensure this power-series representation; for the CLT only the second-order expansion with an $o(k^2)$ remainder is needed. Conversely, the moments are read off the derivatives at the origin:

$$\left. \frac{d^\ell}{dk^\ell} \varphi(k) \right|_{k=0} = i^\ell \int_{\mathbb{R}} dx x^\ell P(x) = i^\ell \langle x^\ell \rangle \implies \langle x^\ell \rangle = (-i)^\ell \left. \frac{d^\ell}{dk^\ell} \varphi(k) \right|_{k=0}, \quad \forall \ell, \quad (2.84)$$

so that from $\varphi(k)$ one reconstructs the whole distribution.

2.5.2 Proof of the theorem

We standardize first. Let $\mu := \langle X_i \rangle$ and $\sigma^2 := \text{Var}[X_i]$ (equal for all i because the variables are identically distributed, but the argument generalizes), and define

$$X_i \longrightarrow \xi_i := \frac{X_i - \mu}{\sigma}, \quad \langle \xi_i \rangle = 0, \quad \text{Var}[\xi_i] = 1, \quad (2.85)$$

rescaling then each variable by $\sqrt{N}$, $\xi_i \mapsto \xi_i/\sqrt{N}$. The standardized sum

$$S_N := \frac{1}{\sqrt{N}} \sum_{i=1}^N \xi_i, \quad \langle S_N \rangle = \frac{1}{\sqrt{N}} \sum_i \langle \xi_i \rangle = 0, \quad \text{Var}[S_N] = \langle S_N^2 \rangle = \frac{1}{N} \sum_{i=1}^N \langle \xi_i^2 \rangle = 1, \quad (2.86)$$

remains itself standardized. We must show convergence in distribution to $\mathcal{N}(0, 1)$; it follows from convergence of the characteristic functions.

Each rescaled variable contributes $\mathbb{E}[e^{ik\xi_i/\sqrt{N}}] = \varphi\left(\frac{k}{\sqrt{N}}\right)$ for every ξ_i , so by the convolution theorem

$$\varphi_N(k) = \left[\varphi\left(\frac{k}{\sqrt{N}}\right) \right]^N. \quad (2.87)$$

Expanding in moments,

$$\varphi\left(\frac{k}{\sqrt{N}}\right) \approx 1 + \frac{ik}{\sqrt{N}} \underbrace{\langle \xi_i \rangle}_{=0} - \frac{k^2}{2N} \underbrace{\text{Var}[\xi_i]}_{=1} - \frac{i}{3!} \frac{k^3}{N^{3/2}} \langle \xi_i^3 \rangle + \frac{1}{4!} \frac{k^4}{N^2} \langle \xi_i^4 \rangle + \dots, \quad (2.88)$$

where $\langle \xi_i^3 \rangle$ is the *skewness*, the parameter of asymmetry (zero if the distribution is symmetric), and $\langle \xi_i^4 \rangle$ is the *kurtosis*, the parameter of the behaviour of the tails (heavy tails). When those higher moments exist, the displayed terms describe higher-order corrections. Without assuming them, finite variance already gives $\varphi(k/\sqrt{N}) = 1 - k^2/(2N) + o(N^{-1})$, which is sufficient for the limit below:

$$\varphi_N(k) \approx \left(1 - \frac{k^2}{2N}\right)^N, \quad \lim_{N \rightarrow \infty} \varphi_N(k) = \lim_{N \rightarrow \infty} \left(1 - \frac{k^2}{2N}\right)^N = e^{-k^2/2}, \quad (2.89)$$

using $(1 + \frac{a}{x})^x \xrightarrow{x \rightarrow \infty} e^a$. The limiting characteristic function identifies a standard Gaussian law by the continuity theorem. Its density is obtained by completing the square as in the diffusion calculation,

$$\begin{aligned} P_G(S_N) &= \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{-k^2/2} e^{-ikS_N} \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{-\frac{1}{2}(k+iS_N)^2} e^{-S_N^2/2} = \frac{1}{\sqrt{2\pi}} e^{-S_N^2/2} : \end{aligned} \quad (2.90)$$

a Gaussian with mean 0 and variance 1. ■

2.5.3 Critical remarks

The theorem is powerful, but each of its hypotheses deserves interrogation — and the answers open the door to the econophysics of Chapter 7.

- *Does it hold if the variables are not independent?* Not in general; extensions require suitable restrictions on dependence.
- *If they are not identically distributed?* Yes — but no single variable must dominate the others.
- *Is the asymptotic limit necessary?* The theorem is asymptotic; finite- N bounds such as Berry–Esseen require additional assumptions (a finite third absolute moment in the i.i.d. case). The summands need not be stable.
- *Can it hold if the mean and/or the variance do not exist?* No. Do similar PDFs exist? Yes: the *Lévy-stable distributions* — which turn out to be better at describing real data. And does a form of the theorem exist for variables drawn from Lévy-stable laws? Yes: the *generalized CLT*, whose limit is not a Gaussian.

- If mean, variance and skewness exist, but the kurtosis does not, does the CLT hold? There exist distributions of this kind: e.g. the Student- t , $P_S(x, \nu)$, with ν degrees of freedom, for which $\nu \rightarrow \infty$ gives $P_S \rightarrow P_G(0, 1)$; or the Tsallis distribution, $P_t(x, q) \xrightarrow{q \rightarrow 1} \mathcal{N}(0, 1)$; here q, ν are shape parameters. Such distributions are *leptokurtic*: sharper at the centre and heavier in the tails than the Gaussian (Figure 2.5; the case $\nu = 3$ will star in Chapter 7). The sum of many P_S variables does reproduce a Gaussian *at the centre* — which makes sense, since mean and variance refer to the central part — but the convergence is very slow (sums of order 10^2).

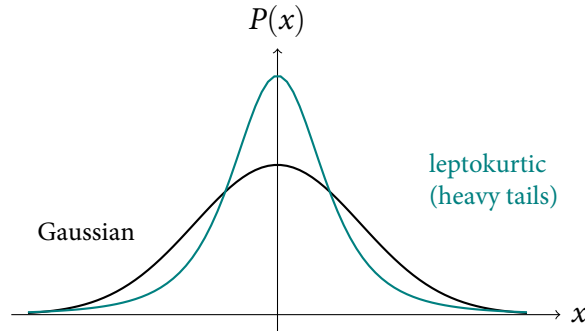


Figure 2.5: A leptokurtic distribution (teal) against a Gaussian of the same variance: sharper peak at the centre, heavier tails far out. For finite sums the central region can approach the Gaussian before the tails.

The most famous Lévy-stable law is the *Cauchy* distribution,

$$P_c(x) = \frac{\gamma}{\pi} \frac{1}{x^2 + \gamma^2}, \quad \gamma \text{ of scale}, \quad (2.91)$$

for which the pathology can be checked by hand: with $\gamma = 1$,

$$\begin{aligned} \langle x^2 \rangle &= \frac{1}{\pi} \int_{\mathbb{R}} x^2 \frac{1}{x^2 + 1} dx \\ &= \frac{2}{\pi} \left\{ \int_0^\infty dx \frac{x^2 + 1}{x^2 + 1} - \int_0^\infty dx \frac{1}{x^2 + 1} \right\} \\ &= \frac{2}{\pi} \left[x - \arctan x \right]_0^{+\infty} = +\infty, \end{aligned} \quad (2.92)$$

and the mean is undefined: the integral defining the mean is not absolutely convergent (its principal value is zero), so the mean is in fact *undefined* rather than infinite. More generally, a Lévy-stable density $P_\alpha(x)$ behaves asymptotically as

$$P_\alpha(x) \xrightarrow{|x| \rightarrow \infty} \frac{1}{|x|^{1+\alpha}}, \quad 0 < \alpha < 2, \quad (2.93)$$

interpolating between the Gaussian ($\alpha = 2$) and pure power laws: for $0 < \alpha \leq 1$ neither the mean nor the variance exist; for $1 < \alpha < 2$ the variance does not exist. And summing Cauchy variables? Still a Cauchy: the Lévy-stable laws *replicate themselves* under addition — they are the fixed points that the generalized CLT converges to, as the Gaussian is for the classical one. All of this is taken up quantitatively in Chapter 7, where these distributions meet financial data.

The handwritten notes summarize Mandelbrot's criticism as a warning against assuming Gaussian statistics universally; this is a lecture paraphrase, not a verified verbatim quotation.

Stochastic Processes

In the previous chapter Brownian motion appeared twice, and in two rather different guises: as a macroscopic diffusion process, governed by a partial differential equation for a density, and as the microscopic, erratic trajectory of a suspended particle. It is natural to ask what these two descriptions have in common, and whether they are instances of something more general. They are. In this chapter we step back and develop the general formalism of *stochastic processes*, of which Brownian motion is the most fundamental example. Our goal is to construct, in full generality, the equation that governs the evolution in time of the probability density of a random variable, and to identify the conditions under which this equation takes the manageable form of a *Fokker–Planck equation*. Everything we do later in this book — from option pricing to the distribution of wealth — rests on the machinery built here.

3.1 General formalism

A *stochastic process* is a system that evolves in time in a probabilistic way. Physically, it is the probabilistic counterpart of a dynamical system: the time evolution is still governed by a law, but the law prescribes probabilities rather than certainties. Formally, it is a family of random variables

$$\{X(t), t \in T\}, \quad (3.1)$$

indexed by a time parameter t that runs over an index set T . Two cases occur in practice. If time is *discrete*, $T = \mathbb{N}$, the process is a sequence $X(t = t_n) = X_n$; if time is *continuous*, $T = \mathbb{R}^+$, the process is a function $X(t) = x(t)$. We will mostly work in continuous time, although discretization will reappear whenever we need to actually compute something.

A particular sequence of values produced by the process is a *realization*, or *trajectory*. When the sequence comes from real observations, one speaks of a *historical time series*. The set of all possible realizations is called the *ensemble*, visualized in Figure 3.1: a stochastic process is an infinite family of functions evolving randomly in time, and cutting the family at a fixed time t_0 defines the random variable $X(t_0)$. The distinction matters: an experimentalist — or a trader — only ever observes *one* realization, whereas the theory speaks about the whole ensemble. We will meet the consequences of this tension more than once.

At the statistical level, the process is completely characterized by its *joint probability density*

$$P_n(x_n, t_n; x_{n-1}, t_{n-1}; \dots; x_1, t_1) \equiv P_n, \quad (3.2)$$

which gives the probability density of observing the value x_1 at time t_1 , the value x_2 at time t_2 , and so on. We adopt throughout the *forward* ordering of the times; the backward ordering gives an equivalent description. That the whole hierarchy $\{P_n\}$ really is all there is to know is the content of a classical result.

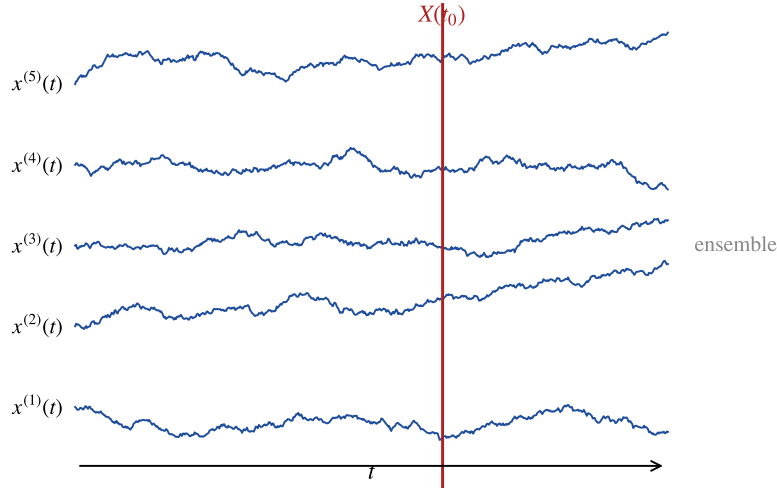


Figure 3.1: Realizations and ensemble. Each curve $x^{(k)}(t)$ is one realization of the same stochastic process; the collection of all of them is the ensemble. Fixing a time t_0 (vertical cut) turns the process into an ordinary random variable $X(t_0)$, whose statistics is taken *across* the realizations. Generated by `images/code_for_images/ensemble.py`.

Theorem (Kolmogorov extension).[15] *A consistent family of finite-dimensional probability distributions determines the law of a process on the product measurable space for every finite set of times $\{t_1, t_2, \dots, t_n\} \in T$.*

This does not by itself select a continuous version or other path regularity. When densities exist, their hierarchy cannot be arbitrary: it must satisfy four consistency conditions.

1. **Positivity:** each density is non-negative,

$$P_n \geq 0. \quad (3.3)$$

2. **Symmetry:** the density is invariant under any permutation of its (x_i, t_i) pairs,

$$P_n(\dots; x_i, t_i; \dots; x_j, t_j; \dots) = P_n(\dots; x_j, t_j; \dots; x_i, t_i; \dots). \quad (3.4)$$

3. **Completeness (marginalization):** integrating out the most recent variable lowers the order of the hierarchy,

$$\int_{-\infty}^{\infty} P_n(x_n, t_n; \dots; x_1, t_1) dx_n = P_{n-1}(x_{n-1}, t_{n-1}; \dots; x_1, t_1). \quad (3.5)$$

4. **Normalization:** the one-time density integrates to one,

$$\int_{-\infty}^{\infty} P_1(x, t) dx = 1. \quad (3.6)$$

These conditions look innocuous, but completeness in particular will do real work for us shortly: it is the seed of the Chapman–Kolmogorov equation.

Two special classes of processes deserve a name. A process is *stationary* when the joint density is invariant under a global translation of all the times,

$$P_n(x_n, t_n + \Delta t; \dots; x_1, t_1 + \Delta t) = P_n(x_n, t_n; \dots; x_1, t_1), \quad (3.7)$$

so that only time *differences* matter; in particular the one-time density becomes time independent, $P(x, t) = P(x)$. A process is instead *purely random* when there is no correlation at all between different times: for every m and every $t_1 \leq \dots \leq t_m$,

$$P(x_m, t_m \mid x_{m-1}, t_{m-1}; \dots; x_1, t_1) = P(x_m, t_m), \quad (3.8)$$

so that the joint density factorizes completely,

$$P_m(x_m, t_m; \dots; x_1, t_1) = \prod_{i=1}^m P(x_i, t_i). \quad (3.9)$$

A purely random process has no dynamics worth the name: each instant is drawn afresh, oblivious of everything that came before.

In writing the last definitions we have used the *conditional* density, defined as

$$P(x_m, t_m \mid x_{m-1}, t_{m-1}; \dots; x_1, t_1) := \frac{P_m(x_m, t_m; x_{m-1}, t_{m-1}; \dots; x_1, t_1)}{P_{m-1}(x_{m-1}, t_{m-1}; \dots; x_1, t_1)}, \quad (3.10)$$

from which the *Bayes theorem* follows immediately,

$$P_2(x_2, t_2; x_1, t_1) = P(x_2, t_2 \mid x_1, t_1) P_1(x_1, t_1). \quad (3.11)$$

The conditional density answers the question that any observer actually asks: *given* what I have seen so far, what happens next?

3.2 Markov processes and the Chapman–Kolmogorov equation

Between the two extremes of a purely random process (no memory at all) and a general process (arbitrarily long memory) lies the class that dominates applications. A process is a *Markov process* when the conditional density depends only on the most recent time,

$$P(x_m, t_m \mid x_{m-1}, t_{m-1}; \dots; x_1, t_1) = P(x_m, t_m \mid x_{m-1}, t_{m-1}), \quad \forall m, \forall t_1 \leq \dots \leq t_m. \quad (3.12)$$

In words: knowledge of the present determines the statistics of the future, with no memory of the past. A typical example is the tossing of coins in succession in time. Whether *prices* are Markovian is a far deeper question, and one to which we shall return in Chapter 5.

The Markov property has two consequences that between them carry the entire chapter.

1) Structure of the joint density. By definition of the conditional density,

$$\begin{aligned} P(x_m, t_m \mid x_{m-1}, t_{m-1}; \dots; x_1, t_1) &= \frac{P_m(x_m, t_m; x_{m-1}, t_{m-1}; \dots; x_1, t_1)}{P_{m-1}(x_{m-1}, t_{m-1}; \dots; x_1, t_1)} \\ &= P(x_m, t_m \mid x_{m-1}, t_{m-1}), \end{aligned} \quad (3.13)$$

from which

$$P_m(x_m, t_m; \dots; x_1, t_1) = P(x_m, t_m \mid x_{m-1}, t_{m-1}) P_{m-1}(x_{m-1}, t_{m-1}; \dots; x_1, t_1). \quad (3.14)$$

The same identity applied to P_{m-1} gives

$$P_{m-1}(x_{m-1}, t_{m-1}; x_{m-2}, t_{m-2}; \dots) = P(x_{m-1}, t_{m-1} \mid x_{m-2}, t_{m-2}) P_{m-2}(x_{m-2}, t_{m-2}; \dots), \quad (3.15)$$

and therefore

$$P_m = P(x_m, t_m \mid x_{m-1}, t_{m-1}) P(x_{m-1}, t_{m-1} \mid x_{m-2}, t_{m-2}) P_{m-2}. \quad (3.16)$$

Iterating all the way down to P_1 , we obtain

$$P_m(x_m, t_m; \dots; x_1, t_1) = \prod_{j=2}^m \underbrace{P(x_j, t_j \mid x_{j-1}, t_{j-1})}_{\text{transition probability (in } t_j - t_{j-1})} \underbrace{P(x_1, t_1)}_{\text{assigned condition}}. \quad (3.17)$$

This is a remarkable simplification. A Markov process is entirely built out of two ingredients: the *transition probabilities* $P(x_j, t_j \mid x_{j-1}, t_{j-1})$, which propagate the process one step at a time, and the initial one-time density. Instead of an infinite hierarchy of joint densities, we need to understand a single object — the transition probability — and everything else follows.

2) Linking points in time: the Chapman–Kolmogorov equation. The first non-trivial case of the product formula is $m = 3$, and it hides an integral equation. From the result above,

$$P_3(x_3, t_3; x_2, t_2; x_1, t_1) = P(x_3, t_3 \mid x_2, t_2) P(x_2, t_2 \mid x_1, t_1) P_1(x_1, t_1). \quad (3.18)$$

Let us integrate over the intermediate value x_2 and use the completeness condition of the previous section,

$$\begin{aligned} \int_{\mathbb{R}} dx_2 P_3(x_3, t_3; x_2, t_2; x_1, t_1) &\stackrel{\text{compl.}}{=} P_2(x_3, t_3; x_1, t_1) \\ &= P_1(x_1, t_1) \int_{\mathbb{R}} dx_2 P(x_3, t_3 \mid x_2, t_2) P(x_2, t_2 \mid x_1, t_1). \end{aligned} \quad (3.19)$$

On the other hand, Bayes gives $P_2(x_3, t_3; x_1, t_1) = P(x_3, t_3 \mid x_1, t_1) P_1(x_1, t_1)$, so that

$$P(x_3, t_3 \mid x_1, t_1) \cancel{P_1(x_1, t_1)} = \cancel{P_1(x_1, t_1)} \int_{\mathbb{R}} dx_2 P(x_3, t_3 \mid x_2, t_2) P(x_2, t_2 \mid x_1, t_1). \quad (3.20)$$

What remains is the *Chapman–Kolmogorov equation*,

$$P(x_3, t_3 \mid x_1, t_1) = \int_{\mathbb{R}} dx_2 P(x_3, t_3 \mid x_2, t_2) P(x_2, t_2 \mid x_1, t_1). \quad (3.21)$$

Its meaning is best read off Figure 3.2: to go from 1 to 3 the process must pass through *some* intermediate point 2, and since we do not know which one, we sum over all the possibilities. Iterating this decomposition over finer and finer intermediate intervals is precisely the idea that will lead us, at the end of the chapter, to the *path integral*.

The Chapman–Kolmogorov equation is exact, but it is an integral equation for an unknown function of four arguments — hardly something one solves directly. The way forward is to expand it for infinitesimal time steps and turn it into a differential equation. This is the Kramers–Moyal expansion, which we now derive.

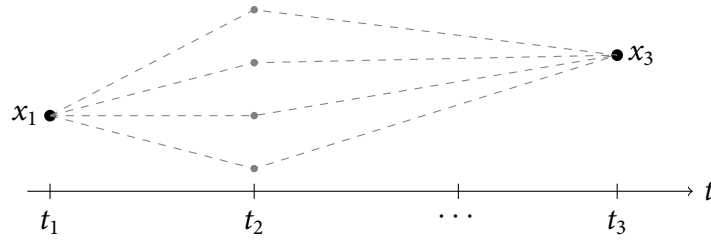


Figure 3.2: Chapman–Kolmogorov concatenation: every path from (x_1, t_1) to (x_3, t_3) is decomposed through all intermediate states at t_2 , and one sums over them.

3.3 The Kramers–Moyal expansion

We look for the equation according to which the density $P(x, t)$ evolves. It is convenient to change notation in the Chapman–Kolmogorov equation (3.21): we rename

$$\begin{aligned} x_1, t_1 &\longrightarrow x_0, t_0 && (x_0 \text{ fixed}), \\ x_2, t_2 &\longrightarrow x', t && (x' \text{ variable}), \\ x_3, t_3 &\longrightarrow x, t + \tau && (x \text{ fixed}), \quad \text{with } \tau \ll t - t_0, \end{aligned} \quad (3.22)$$

so that the roles are clear: the process started long ago at x_0 , we watch it at time t , and we ask what happens a short instant τ later. With this notation, subtracting $P(x, t \mid x_0, t_0)$ from both sides of Chapman–Kolmogorov gives

$$P(x, t + \tau \mid x_0, t_0) - P(x, t \mid x_0, t_0) = \int_{\mathbb{R}} dx' P(x, t + \tau \mid x', t) P(x', t \mid x_0, t_0) - P(x, t \mid x_0, t_0), \quad (3.23)$$

and dividing by τ and letting $\tau \rightarrow 0$ we recognize a time derivative on the left,

$$\frac{\partial}{\partial t} P(x, t \mid x_0, t_0) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left[\int_{\mathbb{R}} dx' P(x, t + \tau \mid x', t) P(x', t \mid x_0, t_0) - P(x, t \mid x_0, t_0) \right]. \quad (3.24)$$

Everything now hinges on the short-time transition density $P(x, t + \tau \mid x', t)$, which we expand. Following Einstein’s strategy from Chapter 2, define the increment $\Delta := x - x'$, so that $x = x' + \Delta$; the increment moves because x' moves. For each fixed increment Δ , introduce the function

$$f(x') := P(x' + \Delta, t + \tau \mid x', t) P(x', t \mid x_0, t_0), \quad (3.25)$$

the Taylor expansion around x (with $x' - x = -\Delta$) reads

$$f(x') = \sum_{m=0}^{\infty} \frac{1}{m!} \frac{\partial^m f}{\partial x'^m} \Big|_{x'=x} (x' - x)^m = \sum_{m=0}^{\infty} \frac{(-\Delta)^m}{m!} f^{(m)}(x). \quad (3.26)$$

Substituting into Equation (3.24) and integrating over Δ , we arrive at

$$\begin{aligned} &\frac{\partial}{\partial t} P(x, t \mid x_0, t_0) \\ &= \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left\{ \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} \frac{\partial^m}{\partial x^m} \int_{\mathbb{R}} d\Delta \Delta^m P(x + \Delta, t + \tau \mid x, t) P(x, t \mid x_0, t_0) \right. \\ &\quad \left. - P(x, t \mid x_0, t_0) \right\}. \end{aligned} \quad (3.27)$$

Let us examine the terms of the sum one by one. The $m = 0$ term is

$$\lim_{\tau \rightarrow 0} \frac{1}{\tau} \left\{ \int_{\mathbb{R}} d\Delta \underbrace{P(x + \Delta, t + \tau | x, t)}_{\text{normalized to 1}} P(x, t | x_0, t_0) - P(x, t | x_0, t_0) \right\} = 0, \quad (3.28)$$

because the transition density is normalized, and the two terms cancel exactly. The terms with $m \geq 1$ survive the limit, and they define the *Kramers–Moyal coefficients*

$$a_m(x, t) := \lim_{\tau \rightarrow 0} \frac{1}{\tau} \int_{\mathbb{R}} d\Delta \Delta^m P(x + \Delta, t + \tau | x, t), \quad (3.29)$$

which are nothing but the moments of the displacement per unit time. Dimensional analysis already whispers their physical meaning: when x is a position, $[a_1] = L T^{-1}$ and $[a_2] = L^2 T^{-1}$, so a_1 has the dimensions of a *velocity* and a_2 those of a *diffusion* coefficient. We will confirm this interpretation quantitatively in Section 3.6. Typically the coefficients do not depend on time, in which case the process is called *homogeneous*.

Putting everything together, we are left with the *Kramers–Moyal expansion*,

$$\frac{\partial}{\partial t} P(x, t | x_0, t_0) = \sum_{m=1}^{\infty} \frac{(-1)^m}{m!} \frac{\partial^m}{\partial x^m} [a_m(x, t) P(x, t | x_0, t_0)]. \quad (3.30)$$

The Chapman–Kolmogorov integral equation has become a differential equation[24] — but of *infinite* order. The next section explains why, in practice, two terms are all we are allowed to keep.

3.4 The Fokker–Planck equation

Equation (3.30) is a differential (in fact integro-differential) equation of infinite order, hardly tractable; moreover any acceptable solution must satisfy $P(x, t | x_0, t_0) \geq 0$, since it is a probability density. One would like to simply truncate the series, but truncation is a delicate business: a theorem tells us exactly which truncations are legitimate.

Theorem (Pawula).[33] *Only one of the following is admissible: the series truncated at first order, which gives the Liouville equation; the series truncated at second order, which gives the Fokker–Planck equation; or the full, untruncated series.*

Any truncation at third order or beyond produces an “evolution equation” whose solutions become negative somewhere — and a negative probability is no probability at all. The theorem is therefore a great piece of luck: either the process is deterministic in disguise (Liouville), or the *entire* statistics of its fluctuations is encoded in just two functions, a_1 and a_2 .

Truncating at $m = 1, 2$ we obtain

$$\frac{\partial}{\partial t} P(x, t | x_0, t_0) = -\frac{\partial}{\partial x} [a_1(x, t) P(x, t | x_0, t_0)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [a_2(x, t) P(x, t | x_0, t_0)], \quad (3.31)$$

which is the celebrated *Fokker–Planck equation*, also known as the *Kolmogorov forward equation* because it propagates the density forward in time, from (x_0, t_0) towards the future.[35, 15] This equation is the workhorse of the entire book.

Before moving on, one remark is in order. By Bayes,

$$P(x_2, t_2 | x_1, t_1) = \frac{P_2(x_2, t_2; x_1, t_1)}{P_1(x_1, t_1)}, \quad (3.32)$$

hence

$$\begin{aligned} \int_{\mathbb{R}} dx_1 P_2(x_2, t_2; x_1, t_1) &= P_1(x_2, t_2) \\ &\stackrel{\text{compl.}}{=} \int_{\mathbb{R}} dx_1 P(x_2, t_2 \mid x_1, t_1) P_1(x_1, t_1), \end{aligned} \quad (3.33)$$

from which one can derive the analogous expansion for the *marginal* density,

$$\frac{\partial}{\partial t} P(x, t) = \sum_{m=1}^{\infty} \frac{(-1)^m}{m!} \frac{\partial^m}{\partial x^m} \{a_m(x, t) P(x, t)\}. \quad (3.34)$$

This expansion requires a Markov transition law and sufficient moment and regularity conditions; Bayes extquotesingle formula alone does not establish it for an arbitrary process. The subtlety is what it is good for. For a Markov process the transition probability determines everything, and the equation genuinely closes the dynamics. For a non-Markov process the expansion still describes $P(x, t)$, but one would also need to know the entire past history to make predictions: the description is *incomplete*.

3.5 Probabilistic meaning of the Fokker–Planck equation

A partial differential equation is only as good as our understanding of what it conserves. Rewriting (3.31) for the marginal density and collecting the spatial derivative,

$$\frac{\partial}{\partial t} P(x, t) + \underbrace{\frac{\partial}{\partial x} \left\{ a_1(x, t) P(x, t) - \frac{1}{2} \frac{\partial}{\partial x} [a_2(x, t) P(x, t)] \right\}}_{\text{dimensions of a current} =: j(x, t)} = 0, \quad (3.35)$$

we discover that the Fokker–Planck equation is a *continuity equation*,

$$\frac{\partial}{\partial t} P(x, t) + \frac{\partial}{\partial x} j(x, t) = 0, \quad (3.36)$$

formally identical to the conservation laws of fluid dynamics or electromagnetism, with $j(x, t)$ playing the role of a *probability current*. To see what is being conserved, integrate over an interval $[a, b]$:

$$\frac{d}{dt} \int_a^b dx P(x, t) = - \int_a^b dx \frac{\partial}{\partial x} j(x, t) = j(a, t) - j(b, t), \quad (3.37)$$

so probability can only change inside the interval by flowing through its endpoints. It is physically reasonable to require $P(x, t) \xrightarrow{x \rightarrow \pm\infty} 0$ with fast decay, and likewise $j(x, t) \rightarrow 0$; therefore, letting $a \rightarrow -\infty$ and $b \rightarrow +\infty$,

$$\frac{d}{dt} \int_{\mathbb{R}} dx P(x, t) = 0, \quad (3.38)$$

that is $\int_{\mathbb{R}} dx P(x, t) = \text{const}$, normalized to 1. The Fokker–Planck equation *conserves probability*, just as the Schrödinger equation conserves the norm of the wave function. For $t \rightarrow \infty$ the density may settle down, $P(x, t) \rightarrow P(x)$, and then $\partial_x j(x, t) = 0$: this is the *stationary* Fokker–Planck equation, whose zero-flux form will be used explicitly for the income distributions of Chapter 8.

3.6 The drift and diffusion coefficients

We promised that a_1 acts as a velocity and a_2 as a diffusion coefficient; let us now prove it. The tool is the pair of ensemble moments

$$\langle x(t) \rangle = \int_{\mathbb{R}} dx x P(x, t), \quad \langle x^2(t) \rangle = \int_{\mathbb{R}} dx x^2 P(x, t), \quad (3.39)$$

and for the moment we set $a_1(x, t) =: v$ and $a_2(x, t) =: D$, both constant. Using the Fokker-Planck equation and integrating by parts (the boundary terms vanish thanks to the fast decay of P),

$$\begin{aligned} \frac{d}{dt} \langle x(t) \rangle &= \int_{\mathbb{R}} dx x \frac{\partial}{\partial t} P(x, t) \\ &\stackrel{\text{F-P}}{=} -v \int_{\mathbb{R}} dx x \frac{\partial}{\partial x} P(x, t) + \frac{D}{2} \int_{\mathbb{R}} dx x \frac{\partial^2}{\partial x^2} P(x, t) = v. \end{aligned} \quad (3.40)$$

This is the analogue of the Ehrenfest theorem of quantum mechanics: the diffusion coefficient D cannot contribute to the velocity of the mean, and

$$\frac{d}{dt} \langle x(t) \rangle = v \implies \langle x(t) \rangle = \langle x(0) \rangle + v t. \quad (3.41)$$

The density keeps its shape and is simply transported at constant speed, as sketched in Figure 3.3: this is why v is called a *drift* term.

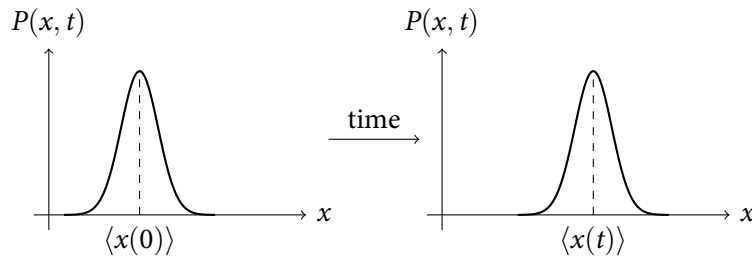


Figure 3.3: Pure drift: the density keeps the same shape and its mean is transported at constant velocity, $\langle x(t) \rangle = \langle x(0) \rangle + v t$.

The second moment tells the complementary story. For pure diffusion ($v = 0$), the same integration-by-parts routine gives

$$\begin{aligned} \frac{d}{dt} \langle x^2(t) \rangle &= \int_{\mathbb{R}} dx x^2 \frac{\partial}{\partial t} P(x, t) = \frac{D}{2} \int_{\mathbb{R}} dx x^2 \frac{\partial^2}{\partial x^2} P(x, t) \\ &= -\frac{D}{2} \int_{\mathbb{R}} dx 2x \frac{\partial}{\partial x} P(x, t) = D \int_{\mathbb{R}} dx P(x, t) = D, \end{aligned} \quad (3.42)$$

so it makes sense to call D a *diffusion* coefficient, with

$$\langle x^2(t) \rangle = \langle x^2(0) \rangle + D t. \quad (3.43)$$

The density now stays put but spreads, as in Figure 3.4.

If the motion is not purely diffusive there is also the drift contribution

$$\int dx x^2 (-v \partial_x P) = 2v \langle x \rangle,$$

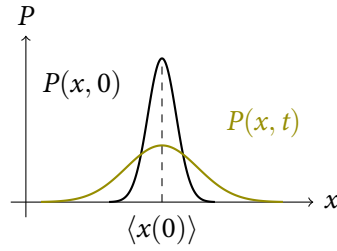


Figure 3.4: Pure diffusion ($v = 0$): the density keeps its centre $\langle x(0) \rangle$ fixed and spreads, its variance growing linearly as $\langle x^2(t) \rangle = \langle x^2(0) \rangle + Dt$.

so that

$$\frac{d}{dt} \langle x^2(t) \rangle = D + 2v \langle x(t) \rangle = D + 2v(\langle x(0) \rangle + vt), \quad (3.44)$$

and integrating in time,

$$\langle x^2(t) \rangle = \langle x^2(0) \rangle + Dt + 2v \langle x(0) \rangle t + v^2 t^2. \quad (3.45)$$

On the other hand $\langle x(t) \rangle^2 = [\langle x(0) \rangle + vt]^2 = \langle x(0) \rangle^2 + 2v \langle x(0) \rangle t + v^2 t^2$, and subtracting we find the clean statement

$$\text{Var}[x(t)] = \langle x^2(t) \rangle - \langle x(t) \rangle^2 = \text{Var}[x(0)] + Dt. \quad (3.46)$$

The drift drops out completely: the variance grows linearly in time, exactly as for diffusion, and only D controls its rate. Drift moves the crowd; diffusion spreads it. In the general case both happen at once, as summarized in Figure 3.5: the density is transported at velocity v and broadens as it travels.

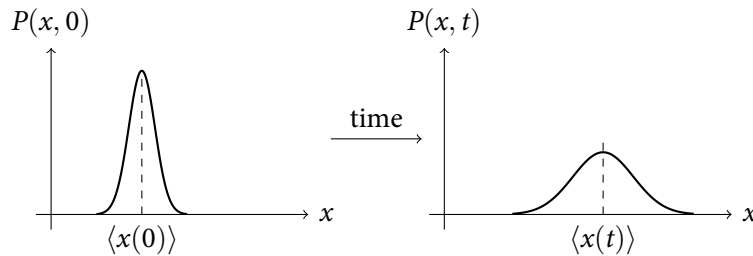


Figure 3.5: The general case, drift and diffusion together: the density is transported ($\langle x(t) \rangle = \langle x(0) \rangle + vt$) while it spreads ($\text{Var}[x(t)] = \text{Var}[x(0)] + Dt$).

For coefficients that genuinely depend on x and t , the same computation yields the general laws

$$\frac{d}{dt} \langle x(t) \rangle = \langle a_1(x, t) \rangle, \quad \frac{d}{dt} \langle x^2(t) \rangle = 2 \langle x a_1(x, t) \rangle + \langle a_2(x, t) \rangle. \quad (3.47)$$

How easy the Fokker–Planck equation is to solve depends entirely on the functional forms of $a_1(x, t)$ and $a_2(x, t)$; a small catalogue of the important cases is given in Section 3.9.

3.7 The Wiener process

The simplest non-trivial choice of coefficients is

$$a_1(x, t) = 0, \quad a_2(x, t) = 1, \quad (3.48)$$

that is, pure diffusion with unit diffusion coefficient: this defines the *Wiener process*. The Fokker–Planck equation to solve is

$$\frac{\partial}{\partial t}P(x, t | x_0, t_0) = \frac{1}{2} \frac{\partial^2}{\partial x^2}P(x, t | x_0, t_0), \quad P(x, t_0 | x_0, t_0) = \delta(x - x_0), \quad (3.49)$$

which is exactly the heat-equation problem already solved in Chapter 2, with solution

$$P(x, t | x_0, t_0) = \frac{1}{\sqrt{2\pi(t - t_0)}} e^{-\frac{(x-x_0)^2}{2(t-t_0)}}. \quad (3.50)$$

The process is often also called Brownian motion, although — as we shall see at the end of this section — somewhat improperly.

It is convenient to change the point of view from positions to *increments*. Introducing

$$\Delta t_i := t_i - t_{i-1}, \quad \Delta x_i := x_i - x_{i-1}, \quad (3.51)$$

the transition density reads

$$\begin{aligned} P(x_i, t_i | x_{i-1}, t_{i-1}) &= \frac{1}{\sqrt{2\pi(t_i - t_{i-1})}} e^{-\frac{(x_i - x_{i-1})^2}{2(t_i - t_{i-1})}} \\ &= \frac{1}{\sqrt{2\pi \Delta t_i}} e^{-\frac{\Delta x_i^2}{2\Delta t_i}} = P(\Delta x_i, \Delta t_i) = \mathcal{N}(0, \Delta t_i), \end{aligned} \quad (3.52)$$

so the *conditional* density of the positions has become the *marginal* density of the increments: each increment is a Gaussian of zero mean and variance equal to the elapsed time, regardless of where the process currently sits. This innocent-looking rewriting is the source of all the properties that follow.

a) Stationary increments. Sending $t_{i-1} \rightarrow t_{i-1} + \Delta t$ and $t_i \rightarrow t_i + \Delta t$ leaves the density unchanged: the increments are invariant under time translation. Note that the process itself is *not* stationary in the strong sense — its variance grows without bound — but its increments are.

b) Independent increments. By the Markov property,

$$\begin{aligned} P_n(x_n, t_n; \dots; x_0, t_0) &= \prod_{i=1}^n P(x_i, t_i | x_{i-1}, t_{i-1}) P(x_0, t_0) \\ &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi \Delta t_i}} e^{-\frac{\Delta x_i^2}{2\Delta t_i}} P(x_0, t_0) \\ &= \prod_{i=1}^n P(\Delta x_i, \Delta t_i) P(x_0, t_0), \end{aligned} \quad (3.53)$$

which is a product of n densities of identical functional form. Given x_0 , one reconstructs $x_1 = x_0 + \Delta x_1$, $x_2 = x_1 + \Delta x_2$, $\dots$, $x_i = x_{i-1} + \Delta x_i$; once x_0 is fixed, all the positions can be recovered from the increments, so that

$$P_n(x_n, t_n; \dots; x_0, t_0) = P_n(\Delta x_n, \Delta t_n; \dots; x_0, t_0), \quad (3.54)$$

but the factorized form holds *only because* the increments are independent. If, moreover, the spacing is uniform, $\Delta t := (t_m - t_0)/m$, then $P(\Delta x_i, \Delta t) = \mathcal{N}(0, \Delta t)$ for every i , and the increments become independent *and* identically distributed — the i.i.d. setting in which all of elementary probability applies.

c) **Scale invariance.** Start again from

$$P(x, t \mid x_0, t_0) = \frac{1}{\sqrt{2\pi(t-t_0)}} e^{-\frac{(x-x_0)^2}{2(t-t_0)}}, \quad (3.55)$$

let $b \in \mathbb{R}^+$ be a scaling factor, and stretch time and space in the coordinated way

$$t - t_0 \rightarrow b(t - t_0) = \hat{t} - \hat{t}_0, \quad x - x_0 \rightarrow \sqrt{b}(x - x_0) = \hat{x} - \hat{x}_0. \quad (3.56)$$

Then

$$\begin{aligned} \hat{P}(\hat{x}, \hat{t} \mid \hat{x}_0, \hat{t}_0) &= \frac{1}{\sqrt{2\pi b(t-t_0)}} e^{-\frac{b(x-x_0)^2}{2b(t-t_0)}} \\ &= b^{-1/2} \frac{1}{\sqrt{2\pi(t-t_0)}} e^{-\frac{(x-x_0)^2}{2(t-t_0)}} = b^{-1/2} P(x, t \mid x_0, t_0), \end{aligned} \quad (3.57)$$

and checking the normalization,

$$\int d\hat{x} \hat{P}(\hat{x}, \hat{t} \mid \hat{x}_0, \hat{t}_0) = \int dx b^{1/2} b^{-1/2} P(x, t \mid x_0, t_0) = 1, \quad (3.58)$$

so the rescaled density is still a normalized density with the same moments. Zooming into a Wiener trajectory with the right anisotropic magnification ($\sqrt{b}$ in space for every factor b in time) reproduces a statistically identical trajectory: the Wiener process is *self-similar*, a one-dimensional fractal process. This is why plots of stock prices look qualitatively the same at the scale of a day or of a year.

d) Continuity of the trajectories. We now ask whether an individual trajectory is continuous. Let $t = t_{i-1}$, $x(t) = x_{i-1}$, $\Delta t = t_i - t_{i-1}$, $\Delta x = x_i - x_{i-1}$, so that $t_i = t + \Delta t$ and $x_i = x + \Delta x$. We first test continuity in probability at a fixed time by asking whether finite increments retain nonzero probability as the time step vanishes. Writing the increment $W := \Delta x$, the probability of such a jump is

$$\text{Prob}\{|W| > k\} = \int_{|W|>k} dW \frac{1}{\sqrt{2\pi \Delta t}} e^{-\frac{|W|^2}{2\Delta t}}, \quad (3.59)$$

over the symmetric integration domain $(-\infty, -k) \cup (k, +\infty)$, hence

$$\text{Prob}\{|W| > k\} = \frac{2}{\sqrt{2\pi \Delta t}} \int_k^{+\infty} dW e^{-\frac{W^2}{2\Delta t}}. \quad (3.60)$$

Recalling the error function and its complement,

$$\text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x dt e^{-t^2}, \quad \text{erfc}(x) = 1 - \text{erf}(x) = \frac{2}{\sqrt{\pi}} \int_x^\infty dt e^{-t^2}, \quad (3.61)$$

and substituting $W \rightarrow W' = W/\sqrt{2\Delta t}$, $dW = \sqrt{2\Delta t} dW'$, we obtain

$$\text{Prob}\{|W| > k\} = \frac{2}{\sqrt{\pi}} \int_{k/\sqrt{2\Delta t}}^{+\infty} dW' e^{-W'^2} = \text{erfc}\left(\frac{k}{\sqrt{2\Delta t}}\right) \xrightarrow{\Delta t \rightarrow 0} 0. \quad (3.62)$$

This proves continuity in probability, not yet almost-sure path continuity. For Gaussian increments, $E[|W_t - W_s|^4] = 3|t-s|^2$; the Kolmogorov continuity criterion then ensures a continuous modification, which is the version used here.

e) **Differentiability of the trajectories.** Continuity secured, the natural next question is whether the trajectories have a velocity. Consider

$$\lim_{\Delta t \rightarrow 0} \frac{x(t + \Delta t) - x(t)}{\Delta t} = \lim_{\Delta t \rightarrow 0} \frac{|W|}{\Delta t}; \quad (3.63)$$

we examine the size of the difference quotient at a fixed deterministic time. The same computation as in (d), with k replaced by $k \Delta t$, gives

$$\begin{aligned} \text{Prob} \left\{ \frac{|W|}{\Delta t} > k \right\} &= \frac{2}{\sqrt{2\pi} \Delta t} \int_{k\Delta t}^{+\infty} dW e^{-\frac{W^2}{2\Delta t}} \\ &= \text{erfc} \left(\frac{k\Delta t}{\sqrt{2\Delta t}} \right) = \text{erfc} \left(k\sqrt{\frac{\Delta t}{2}} \right) \xrightarrow{\Delta t \rightarrow 0} 1, \quad \forall k \in \mathbb{R}^+. \end{aligned} \quad (3.64)$$

The probability that the absolute difference quotient exceeds any fixed threshold tends to one, excluding a finite derivative at a fixed deterministic time. The stronger theorem that Brownian paths are almost surely nowhere differentiable requires a separate pathwise argument; it is not proved by this one-time calculation.

This last property deserves a pause. If the Wiener process were the physical trajectory of Einstein's Brownian particle, the particle would have no well-defined instantaneous velocity in this idealized overdamped model. The Wiener process is therefore an admirable mathematical idealization, but it *cannot* be the literal physical representation of Brownian motion; and its non-differentiability is precisely what will force us, in the next chapter, to invent a new calculus.

3.8 A rigorous definition of the Wiener process

So far the Wiener process has been defined through its Fokker–Planck equation. Modern probability theory prefers an axiomatic characterization, which we record here because it is the form used in the mathematical finance literature. From a purely stochastic point of view, the *Wiener process* $W(t)$ (also denoted $B(t)$) is a continuous-time stochastic process $\{W(t)\}_{t \geq 0}$ such that:

- (i) $W(0) = 0$ almost surely;
- (ii) the trajectories are continuous almost surely;
- (iii) the process has *independent increments*: for $s' \leq t' \leq s \leq t$, the increments $W(t) - W(s)$ and $W(t') - W(s')$ are independent;
- (iv) $W(t) - W(s) \sim \mathcal{N}(0, t - s)$ for $t > s$.

One checks easily that these axioms reproduce everything derived in the previous section.

The increments are independent; but are the *positions* independent as well? Intuition says no — where the process can be now depends on where it was — and the autocorrelation function confirms it. Writing $W(t) = W(s) + [W(t) - W(s)]$ and using $W(s) = W(s) - W(0)$,

$$\begin{aligned} \langle W(t) W(s) \rangle &= \langle [W(t) - W(s)] W(s) \rangle + \langle W^2(s) \rangle \\ &= \underbrace{\langle [W(t) - W(s)] [W(s) - W(0)] \rangle}_{\text{independent increments}} + \langle W^2(s) \rangle. \end{aligned} \quad (3.65)$$

The two increments $W(t) - W(s)$ and $W(s) - W(0)$ live on disjoint time intervals, so they are independent and both have zero mean; the first term vanishes and only the second moment survives. Now, with $\Delta W(s) = W(s) - W(0)$ and $\langle \Delta W(s) \rangle = 0$,

$$\begin{aligned} \text{Var}[\Delta W(s)] &= \langle (\Delta W(s) - \langle \Delta W(s) \rangle)^2 \rangle \\ &= \langle (W(s) - W(0))^2 \rangle = \langle W^2(s) \rangle, \quad \text{Var}[W(s)] = s, \end{aligned} \quad (3.66)$$

hence $\langle W(t)W(s) \rangle = s$ — but only for $s < t$; exchanging the roles of s and t gives the symmetric result, and in general

$$\begin{aligned} \langle W(t)W(s) \rangle &= \min(t, s), \\ \text{Cov}[W(t), W(s)] &= \langle W(t)W(s) \rangle - \langle W(s) \rangle \langle W(t) \rangle = \min(t, s). \end{aligned} \quad (3.67)$$

The corresponding correlation coefficient is

$$\rho = \frac{\text{Cov}[W(t), W(s)]}{\sigma_{W(t)} \sigma_{W(s)}} = \frac{\min(s, t)}{\sqrt{s} \sqrt{t}} \xrightarrow{t/s \rightarrow \infty} 0 \quad (t \geq s > 0). \quad (3.68)$$

The positions are therefore *not* independent: they are the more correlated the closer the two times are, and the memory fades — but only slowly — as the times drift apart. The curious formula $\langle W(t)W(s) \rangle = \min(t, s)$ will return in the next chapter, where its distributional derivatives generate the white noise.

3.9 General solution of the Fokker–Planck equation

How the Fokker–Planck equation (3.31) is solved depends on the functional forms of the coefficients $a_1(x, t)$ and $a_2(x, t)$, and it is worth having a small catalogue of the cases that will matter to us.

1. $a_1 = 0, a_2 \in \{1, D\}$: pure diffusion, i.e. the heat equation, which we have already solved.
2. $a_1 = v, a_2 \in \{1, D\}$: constant drift plus diffusion, solved by Fourier transform.
3. $a_1(x, t) = -\gamma x, a_2 = D$ with $\gamma > 0$: a linear restoring drift; this is the *Ornstein–Uhlenbeck process*, [41] harder to solve but feasible, and ubiquitous in what follows.
4. $a_1(x) = \kappa(\theta - x), a_2(x) = \sigma^2 x$: the *CIR process*, [6] with a linear mean-reverting drift and a diffusion that switches off as $x \rightarrow 0$.
5. $a_1(x, t) = ax, a_2(x, t) = \sigma^2 x^2$: the *geometric Brownian motion* (GBM), whose solution is not Gaussian; it will be the standard model for stock prices from Chapter 5 onwards.

Case-by-case tricks are all well and good, but one would like a *general* strategy, valid for arbitrary coefficients. There is one, and it proceeds in two steps: first we find the transition probability over an infinitesimally short time, where everything can be computed explicitly; then we chain infinitely many short steps together with Chapman–Kolmogorov, which produces a *path integral* in the style of Wiener and Kac. We take the two steps in turn.

3.9.1 Short-time transition probability

It is convenient to write the Fokker–Planck equation in operator form,

$$\frac{\partial}{\partial t} P(x, t | x', t_0) = L_{\text{FP}}(x, t) P(x, t | x', t_0), \quad L_{\text{FP}}(x, t) = -\frac{\partial}{\partial x} a_1(x, t) + \frac{1}{2} \frac{\partial^2}{\partial x^2} a_2(x, t), \quad (3.69)$$

where L_{FP} is called the Fokker–Planck operator. Integrating formally in time,

$$P(x, t | x', t_0) = P(x, t_0 | x', t_0) + \int_{t_0}^t dt' L_{\text{FP}}(x, t') P(x, t' | x', t_0), \quad (3.70)$$

which is an integral equation whose unknown appears on both sides — an invitation to iterate. Taking the initial condition $P(x, t_0 | x', t_0) := \delta(x - x')$ and substituting the equation into itself repeatedly,

$$\begin{aligned} P(x, t | x', t_0) &= \delta(x - x') + \int_{t_0}^t dt' L_{\text{FP}}(x, t') \delta(x - x') \\ &\quad + \int_{t_0}^t dt' \int_{t_0}^{t'} dt'' L_{\text{FP}}(x, t') L_{\text{FP}}(x, t'') \delta(x - x') + \dots, \end{aligned} \quad (3.71)$$

which the reader may recognize as a *Dyson expansion*, the same structure that appears in time-dependent perturbation theory in quantum mechanics. If L_{FP} does not depend on time, $a_i(x, t) = a_i(x)$, the series resums neatly into an exponential,

$$P(x, t | x', t_0) = \exp[L_{\text{FP}}(x) (t - t_0)] \delta(x - x'), \quad (3.72)$$

while at first order

$$P(x, t | x', t_0) \approx \delta(x - x') + \int_{t_0}^t dt' L_{\text{FP}}(x, t') \delta(x - x'). \quad (3.73)$$

Setting $\tau := t - t_0$, each term of the series carries a factor $t - t_0$: if the interval $[t_0, t]$ is very small, the area under $L_{\text{FP}}(x, t')$ is approximately a rectangle,

$$\int_{t_0}^t dt' L_{\text{FP}}(x, t') \approx \tau L_{\text{FP}}(x, t) + \text{higher-order infinitesimals}, \quad (3.74)$$

and the truncation at first order becomes accurate. Then for a generic short transition $t \rightarrow t + \tau$,

$$P(x, t + \tau | x', t) \approx \delta(x - x') + L_{\text{FP}}(x, t) \tau \delta(x - x') = \{1 + L_{\text{FP}}(x, t) \tau\} \delta(x - x'), \quad (3.75)$$

or, writing the operator out explicitly,

$$P(x, t + \tau | x', t) \approx \delta(x - x') - \frac{\partial}{\partial x} \{a_1(x, t) \delta(x - x')\} \tau + \frac{1}{2} \frac{\partial^2}{\partial x^2} \{a_2(x, t) \delta(x - x')\} \tau. \quad (3.76)$$

This is a formula built out of derivatives of delta functions — can one pass from this distributional form to an honest analytic one? Yes, and legitimately so, because the expression only ever appears under an integral. For any test function $f(x)$ the delta acts as $f(x) \delta(x - x') = f(x') \delta(x - x')$; applying this with $f(x) = a_i(x, t)$ frees the coefficients from the derivatives,

$$P(x, t + \tau | x', t) \approx \left\{ 1 - \frac{\partial}{\partial x} a_1(x', t) \tau + \frac{1}{2} \frac{\partial^2}{\partial x^2} a_2(x', t) \tau \right\} \delta(x - x'). \quad (3.77)$$

Now we use the Fourier representation $\delta(x - x') = \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{ik(x-x')}$, under which each ∂_x becomes a factor ik , and then re-exponentiate ($e^z = 1 + z + \dots$, accurate to the same order in τ):

$$\begin{aligned} P(x, t + \tau \mid x', t) &\approx \frac{1}{2\pi} \int_{\mathbb{R}} dk \left\{ 1 - ik a_1(x', t) \tau - \frac{1}{2} k^2 a_2(x', t) \tau \right\} e^{ik(x-x')} \\ &\approx \frac{1}{2\pi} \int_{\mathbb{R}} dk e^{ik(x-x' - a_1(x', t)\tau) - \frac{1}{2} k^2 a_2(x', t)\tau} \\ &= \frac{1}{\sqrt{2\pi a_2(x', t) \tau}} \exp \left\{ -\frac{[x - x' - a_1(x', t) \tau]^2}{2 a_2(x', t) \tau} \right\}, \end{aligned} \quad (3.78)$$

where the last step is the standard Gaussian integral in k . The result is beautifully transparent: over a short time τ , any diffusion process looks Gaussian, with mean $x' + a_1(x', t) \tau$ (the particle drifts) and variance $a_2(x', t) \tau$ (the particle diffuses). Without drift ($a_1 = 0$) one recovers exactly the Einstein–Wiener solution. All the complications of a general Fokker–Planck equation come from chaining these elementary Gaussian steps together — which is precisely what we do next.

3.9.2 Path integral and the Wiener measure

Can the short-time result be applied to a generic, not necessarily small, interval $[t_0, t]$? Not directly; but Chapman–Kolmogorov tells us how to compose short steps into long ones. Introducing an intermediate time t_1 ,

$$P(x, t \mid x_0, t_0) = \int_{\mathbb{R}} dx_1 P(x, t \mid x_1, t_1) P(x_1, t_1 \mid x_0, t_0), \quad (3.79)$$

and iterating with a second intermediate time t_2 ,

$$P(x, t \mid x_0, t_0) = \int_{\mathbb{R}} dx_1 \int_{\mathbb{R}} dx_2 P(x, t \mid x_2, t_2) P(x_2, t_2 \mid x_1, t_1) P(x_1, t_1 \mid x_0, t_0), \quad (3.80)$$

with three sub-intervals and three conditional densities. Continuing with uniform spacing $\Delta t = (t - t_0)/N$, so that $t_0 + N\Delta t = t = t_N$ and $t_0 < t_1 < \dots < t_N = t$,

$$P(x, t \mid x_0, t_0) = \int_{\mathbb{R}} dx_1 \cdots \int_{\mathbb{R}} dx_{N-1} \prod_{k=1}^N P(x_k, t_k \mid x_{k-1}, t_{k-1}), \quad (3.81)$$

which we write compactly as a *functional integral* over all trajectories,

$$P(x, t \mid x_0, t_0) = \int \mathcal{D}_\alpha x \prod_{k=1}^N P(x_k, t_k \mid x_{k-1}, t_{k-1}). \quad (3.82)$$

The formal limit

$$N \rightarrow \infty, \quad \Delta t \rightarrow 0, \quad N\Delta t = t - t_0 \text{ finite}, \quad (3.83)$$

defines the *Wiener measure* on the space of trajectories. Each factor in the product is now a *short-time* propagator, for which we may insert the Gaussian found above (valid for $\Delta t \ll t - t_0$),

$$P(x_k, t_k \mid x_{k-1}, t_{k-1}) \approx \frac{1}{\sqrt{2\pi a_2(x_{k-1}, t_{k-1}) \Delta t}} \exp \left\{ -\frac{[x_k - x_{k-1} - a_1(x_{k-1}, t_{k-1}) \Delta t]^2}{2 a_2(x_{k-1}, t_{k-1}) \Delta t} \right\}, \quad (3.84)$$

and using $\prod_k e^{S_k} = e^{\sum_k S_k}$ the product of exponentials becomes the exponential of a sum,

$$P(x, t \mid x_0, t_0) = \lim_{N \rightarrow \infty} \int \mathcal{D}_\alpha x \prod_{k=1}^N \frac{1}{\sqrt{2\pi a_2(x_{k-1}, t_{k-1}) \Delta t}} \exp \left\{ \sum_{k=1}^N S_k \right\}, \quad (3.85)$$

with

$$S := \sum_{k=1}^N S_k = - \sum_{k=1}^N \frac{1}{2 a_2(x_{k-1}, t_{k-1})} \left[\frac{x_k - x_{k-1}}{\Delta t} - a_1(x_{k-1}, t_{k-1}) \right]^2 \Delta t. \quad (3.86)$$

Thus $P(x, t \mid x_0, t_0) \sim e^S$, and S plays the role of an *action*: each term has the structure

$$-\frac{[x_k - x_{k-1} - a_1 \Delta t]^2}{2 a_2 \Delta t} = -\frac{1}{2 a_2} \left[\frac{\Delta x}{\Delta t} - a_1 \right]^2 \Delta t, \quad (3.87)$$

a squared “velocity mismatch” weighted by the inverse diffusion. In the continuum limit $\sum_{k=1}^N (\cdots) \Delta t \rightarrow \int_{t_0}^t dt'$ and $\Delta x / \Delta t \rightarrow \dot{x}(t)$, hence

$$S[x(t)] = \int_{t_0}^t dt' \left\{ -\frac{1}{2 a_2(x, t)} (\dot{x} - a_1(x, t))^2 \right\}, \quad (3.88)$$

and finally

$$P(x, t \mid x_0, t_0) = \int \mathcal{D}[x(t)] \exp \left\{ - \int_{t_0}^t \mathcal{L}_{\text{OM}}(x, \dot{x}, t) dt' \right\}, \quad (3.89)$$

where

$$\mathcal{L}_{\text{OM}} = \frac{(\dot{x} - a_1(x, t))^2}{2 a_2(x, t)} \quad (3.90)$$

is the *Onsager–Machlup Lagrangian*. [30] The interpretation is worth savouring: every conceivable trajectory from (x_0, t_0) to (x, t) contributes, but trajectories whose velocity strays far from the local drift a_1 are exponentially suppressed, the more severely the smaller the diffusion a_2 . With $a_1 = 0$ and $a_2 = 1/m$ one gets $\mathcal{L} = \frac{1}{2} m \dot{x}^2$, formally the Lagrangian of a free particle in classical mechanics.

Properties and equivalence. The continuum expression is formal: the prepoint time-sliced expression specifies the Itô convention, and for state-dependent a_2 its normalization must remain inside the discretized integral. It is not an ordinary integral over differentiable paths or a convention-independent Onsager–Machlup action. Under suitable smoothness, nondegeneracy and convergence assumptions, it is called the *Wiener integral solution* of the Fokker–Planck equation, and three of its features deserve emphasis. First, it is automatically normalized and positive definite: the exponential is positive, and the construction preserves normalization at every step — so the constraint $P \geq 0$, which made truncating Kramers–Moyal so delicate, here comes for free. Second, it is a genuine functional integral, formally defined with the Wiener measure $\mathcal{D}x(t)$. Third, and most importantly, it closes a conceptual loop: starting from Chapman–Kolmogorov there are *two equivalent routes* to the dynamics — the Kramers–Moyal expansion, which leads to the Fokker–Planck *differential* equation, and the short-time transition probability, which leads to the path-integral, i.e. *integral*, representation. One must also be able to travel back and recover the Fokker–Planck equation from the path integral; this can indeed be shown, using the fact that the contributing trajectories are continuous and non-differentiable.

We close with a remark for the reader coming from quantum mechanics. There is a profound difference between the Wiener–Kac construction, whose weight e^S is *real* and hence directly a

probability density, and the Feynman construction, whose weight e^{iS} is a complex *amplitude* (which becomes classical only in the limit $\hbar \rightarrow 0$). Here the weight is e^S , and by integrating over all trajectories one can reconstruct any distribution of interest. With this machinery — Fokker–Planck on one side, trajectories on the other — we are ready to ask the question that opens the next chapter: what is the differential equation obeyed by a *single* trajectory?

Stochastic Calculus

In the previous chapter the Fokker–Planck equation provided a macroscopic, deterministic description of a stochastic process: deterministic not in the trajectories, but in the probability density, which evolves according to an ordinary partial differential equation. We now look at the same physics from the complementary point of view of the *trajectory* itself, encoded in a stochastic differential equation of the Langevin type. The two descriptions turn out to be perfectly equivalent — and we will build the dictionary between them — but giving a precise meaning to the equation of the trajectory is surprisingly subtle. The culprit is a fact we proved in Chapter 3: the trajectories of the Wiener process are nowhere differentiable. Making sense of a “differential equation” for a non-differentiable function forces us to construct a new calculus — *stochastic calculus* — in which the ordinary rules of differentiation no longer hold, and to replace them with Itô’s rules. By the end of the chapter we will be solving stochastic differential equations as comfortably as ordinary ones, including the two that dominate quantitative finance: geometric Brownian motion and the Ornstein–Uhlenbeck process.

4.1 From the Fokker–Planck equation to the Langevin equation

Two questions guide this section. First, in what sense is the Fokker–Planck equation equivalent to a Langevin-type stochastic differential equation (SDE) for the trajectory? Second, what exactly is the relation between the *white noise* $\xi(t)$ that appears in the Langevin equation and the Wiener process $W(t)$ of the previous chapter?

Recall the Langevin equation of Chapter 2,

$$\frac{dv}{dt} = -\gamma v + \xi(t), \quad \gamma := \frac{6\pi\eta R}{m}, \quad (4.1)$$

which described the velocity of a Brownian particle subject to viscous damping and to a random force. We now promote it to the general form

$$\frac{dx(t)}{dt} = a + b \xi(t), \quad a \in \mathbb{R}, \quad b \in \mathbb{R}^+, \quad (4.2)$$

where the constants a and b will turn out to be directly connected to the coefficients a_1, a_2 of the Fokker–Planck equation.

First we must say what we mean by the noise. Here ξ denotes normalized Gaussian white noise, a generalized random process rather than a function with pointwise values. Its covariance formulas are interpreted distributionally; its amplitude need not equal that of the physical random force in Chapter 2. It has three formal properties:

- (i) it has zero mean, $\langle \xi(t) \rangle = 0$ for all t ;
- (ii) it is delta-correlated, $\langle \xi(t)\xi(t') \rangle = \delta(t - t')$ for all t, t' : the noise at two distinct instants, however close, is completely uncorrelated;
- (iii) $\xi(t)$ is Gaussian.

Properties (i) and (ii) are enough for everything we compute in this section; the third is needed because uncorrelated variables are independent only in the Gaussian case, and independence is what we will eventually use. The name “white” comes from the flat power spectrum implied by (ii): all frequencies contribute equally, like white light.

Let us see what this SDE predicts. Integrating,

$$dx(t) = a dt + b \xi(t) dt \implies x(t) = x(0) + a \int_0^t dt' + b \int_0^t \xi(t') dt'. \quad (4.3)$$

Taking the ensemble average and using property (i),

$$\mathbb{E}[x(t)] = \langle x(0) \rangle + a \int_0^t dt' + b \int_0^t \langle \xi(t') \rangle dt' = \langle x(0) \rangle + a t, \quad (4.4)$$

which coincides with the Fokker–Planck result $\langle x(t) \rangle = \langle x(0) \rangle + \nu t$ provided we identify $a = \nu = a_1$: the coefficient a is the *drift*. For the second moment, squaring the solution and averaging,

$$\mathbb{E}[x^2(t)] = \langle x^2(0) \rangle + 2a t \langle x(0) \rangle + a^2 t^2 + b^2 \int_0^t dt' \int_0^t dt'' \underbrace{\langle \xi(t')\xi(t'') \rangle}_{\delta(t'-t'')}, \quad (4.5)$$

and since the double integral collapses on the diagonal, $\int_0^t dt' \int_0^t dt'' \delta(t' - t'') = t$, we find

$$\text{Var}[x(t)] = \mathbb{E}[x^2(t)] - \mathbb{E}^2[x(t)] = \text{Var}[x(0)] + b^2 t. \quad (4.6)$$

Comparing with the Fokker–Planck result $\text{Var}[x(t)] = \text{Var}[x(0)] + D t$, we identify $b^2 = D = a_2$: the squared noise amplitude is the *diffusion* coefficient.

This establishes the dictionary between the Fokker–Planck equation and the (Itô) SDE:

$$a_1(x, t) \longleftrightarrow a(x, t), \quad a_2(x, t) \longleftrightarrow b^2(x, t). \quad (4.7)$$

Explicitly: given the SDE $\frac{dx(t)}{dt} = a(x, t) + b(x, t) \xi(t)$, with $\langle \xi(t) \rangle = 0$ and $\langle \xi(t)\xi(t') \rangle = \delta(t - t')$, the associated Fokker–Planck equation is

$$\frac{\partial}{\partial t} P(x, t) = -\frac{\partial}{\partial x} [a(x, t) P(x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [b^2(x, t) P(x, t)], \quad (4.8)$$

and, conversely, given the Fokker–Planck equation

$$\frac{\partial}{\partial t} P(x, t) = -\frac{\partial}{\partial x} [a_1(x, t) P(x, t)] + \frac{1}{2} \frac{\partial^2}{\partial x^2} [a_2(x, t) P(x, t)], \quad (4.9)$$

the equivalent SDE is

$$\frac{dx(t)}{dt} = a_1(x, t) + \sqrt{a_2(x, t)} \xi(t), \quad \langle \xi(t) \rangle = 0, \quad \langle \xi(t)\xi(t') \rangle = \delta(t - t'). \quad (4.10)$$

The macroscopic and the microscopic descriptions carry exactly the same information, and one can move freely between them.

There is, however, a difficulty — and it is not a pedantic one: what we have just written is, strictly speaking, meaningless. Take the Wiener process itself, with $a_1(x, t) = 0$ and $a_2(x, t) = 1$; the equivalent SDE would read

$$\frac{dW(t)}{dt} = 0 + \xi(t), \quad (4.11)$$

but we proved in Chapter 3 that the trajectories of $W(t)$ have *no derivative* almost surely, so the left-hand side does not exist. The noise itself is no healthier:

$$\text{Var}[\xi(t)] = \lim_{t' \rightarrow t} \langle \xi(t) \xi(t') \rangle = \delta(0) = \infty, \quad (4.12)$$

an object with infinite variance at every instant, which exists nowhere in nature. Both pathologies have the same cure, to which we now turn: stop dividing by dt , and work directly with the increments.

4.2 White noise and the Wiener differential

To give a rigorous meaning to the formal manipulations above, we look for the precise relation between $\xi(t)$ and $W(t)$. The starting point is the autocorrelation of the Wiener process computed in Chapter 3,

$$\langle W(t) W(t') \rangle = \min(t, t') = t \Theta(t' - t) + t' \Theta(t - t'), \quad (4.13)$$

where in the second equality we have rewritten the minimum through the Heaviside step function Θ — a form that can be differentiated. Differentiating with respect to t in the distributional sense (the derivative of Θ being a delta, whose two contributions cancel),

$$\frac{d}{dt} [t \Theta(t' - t) + t' \Theta(t - t')] = \Theta(t' - t), \quad (4.14)$$

and differentiating once more, now with respect to t' ,

$$\frac{d}{dt} \frac{d}{dt'} \langle W(t) W(t') \rangle = \delta(t - t') \stackrel{!}{=} \langle \xi(t) \xi(t') \rangle. \quad (4.15)$$

The autocorrelation of the “derivative” of the Wiener process is exactly the autocorrelation of white noise. This justifies, in the distributional sense, the two key identities on which everything else rests. First,

$$\frac{dW(t)}{dt} = \xi(t), \quad \text{so that} \quad \int_0^t \xi(t') dt' = \int_0^t dW(t') = W(t) - W(0); \quad (4.16)$$

white noise is the (distributional) derivative of the Wiener process, and if $\xi(t)$ is Gaussian, so is $W(t)$. Second, at the level of increments,

$$\langle dW(t) dW(t') \rangle = \delta(t - t') dt dt' \implies \langle dW^2(t) \rangle = dt, \quad (4.17)$$

while $\langle dW(t) dW(t') \rangle = 0$ for $t \neq t'$, which — the process being Gaussian — means full independence of increments at different times. The scaling $\langle dW^2 \rangle = dt$ is the single most consequential formula of this chapter; we will meet it again as $[dW]^2 = dt$.

The practical moral is this: the ill-defined product $b \xi(t) dt$ must everywhere be replaced by the perfectly well-defined *Wiener differential* $b dW(t)$,

$$dx(t) = a dt + b \xi(t) dt \longrightarrow dx(t) = a dt + b dW(t). \quad (4.18)$$

No derivative of W ever appears; only its increments do, and those are honest Gaussian random variables.

4.3 The stochastic integral

Having agreed to write SDEs in differential form, we must say what it means to *integrate* them.

Definition. Let $x(t)$ be a stochastic process; the associated SDE is

$$dx(t) = a(x, t) dt + b(x, t) dW(t) \quad (4.19)$$

(an abbreviated notation: $dW(t)$ standing alone is meaningless), and for every t it holds that

$$x(t) = x(t_0) + \int_{t_0}^t a(x, t') dt' + \underbrace{\int_{t_0}^t b(x, t') dW(t')}_{\text{stochastic integral}}. \quad (4.20)$$

The first integral is an ordinary one; all the novelty sits in the second, the *stochastic integral*, in which the integration variable is itself a random process. One can show that this integral exists and is unique — but, as we are about to discover, only after one extra choice has been made.

Let $G(t)$ be the integrand and ask: what is the meaning of $\int_{t_0}^t G(t') dW(t')$? We proceed as Riemann taught us, by discretization. Partition $[t_0, t]$ into small intervals $[t_{i-1}, t_i]$, choose in each an evaluation point $\tau_i \in [t_{i-1}, t_i]$, and form the Riemann–Stieltjes-like sum

$$S_m = \sum_{i=1}^m G(\tau_i) \{W(t_i) - W(t_{i-1})\}. \quad (4.21)$$

To see whether the choice of τ_i matters — for an ordinary integral it would not — take as a probe the simplest non-trivial integrand, $G(\tau_i) = W(\tau_i)$. The average of the sum is then computable from the autocorrelation $\langle W(t)W(s) \rangle = \min(t, s)$:

$$\langle S_m \rangle = \sum_{i=1}^m \langle W(\tau_i) \{W(t_i) - W(t_{i-1})\} \rangle = \sum_{i=1}^m \{ \min(\tau_i, t_i) - \min(\tau_i, t_{i-1}) \} = \sum_{i=1}^m \{\tau_i - t_{i-1}\}. \quad (4.22)$$

Parametrizing the evaluation point by its relative position in the interval,

$$\tau_i = \alpha t_i + (1 - \alpha) t_{i-1}, \quad \alpha \in [0, 1] \quad (\alpha = 1 \Rightarrow \tau_i = t_i, \quad \alpha = 0 \Rightarrow \tau_i = t_{i-1}), \quad (4.23)$$

we obtain

$$\langle S_m \rangle = \sum_{i=1}^m \{ \alpha t_i + (1 - \alpha) t_{i-1} - t_{i-1} \} = \alpha \sum_{i=1}^m (t_i - t_{i-1}) = \alpha (t - t_0), \quad (4.24)$$

which ranges over the whole interval $0 \leq \langle S_m \rangle \leq t - t_0$ as α varies.

This is the crucial — and disconcerting — discovery: the value of a stochastic integral *depends on where inside each interval the integrand is evaluated*. The roughness of the Wiener path is to blame: within a single interval $[t_{i-1}, t_i]$ the integrand moves by $\sim \sqrt{\Delta t}$, which is too much to be washed out in the limit. Different choices of α therefore define genuinely different integrals, and two conventions have survived:

- $\alpha = 0, \tau_i = t_{i-1}$: the *pre-point* rule, defining the **Itô** integral;

- $\alpha = \frac{1}{2}$, $\tau_i = \frac{1}{2}(t_i + t_{i-1})$: the *mid-point* rule, defining the **Stratonovich** integral.

The two conventions are inter-translatable, hence equivalent in content. In finance one invariably uses Itô, and for a good reason: the pre-point rule evaluates the integrand *before* the increment arrives, i.e. it requires knowledge of the present only — no peeking into the future, which is exactly the situation of a trader. With Itô understood, we write

$$S = \int_{t_0}^t G(t') dW(t'), \quad (4.25)$$

where the convergence of S_m to S is meant as a *mean-square limit*,

$$S = \text{ms-lim}_{m \rightarrow \infty} S_m, \quad \langle S \rangle = \text{ms-lim}_{m \rightarrow \infty} \langle S_m \rangle, \quad \text{Var}[S] = \text{ms-lim}_{m \rightarrow \infty} \text{Var}[S_m], \quad (4.26)$$

i.e. convergence both in mean and in variance, $\lim_{m \rightarrow \infty} \langle (S_m - S)^2 \rangle = 0$.

It is worth spelling out the trade-off between the two conventions, because it explains why both survive in the literature. For the *Stratonovich* integral the standard rules of calculus (chain rule, Leibniz rule, and so on) continue to hold — which is why physicists often prefer it — but in the SDE $dx = a dt + b dW$ the coefficient a is *not* the drift of the process, which is counter-intuitive. For the *Itô* integral it is the other way around: a gives the mean and b^2 the variance, with a transparent probabilistic meaning, but the standard rules of calculus *fail*, and one needs a corrected change-of-variable formula (Itô's formula, Section 4.5). A simple correspondence rule connects the two conventions by shifting the drift, so nothing is lost either way.

4.4 Itô integrals

Before stating Itô's formula we collect[22] the small toolbox of results about Itô integrals that makes every later computation a few lines long.

Quadratic variation. For the Wiener increment $\Delta W = W(t) - W(s)$, we already know that $\langle \Delta W \rangle = 0$ and $\text{Var}[\Delta W] = \Delta t$ (where $\Delta t = t - s$, $t > s$). We now need the statistics of the *squared* increment $\Delta W^2 = [W(t) - W(s)]^2$, which will be summed over a partition to obtain quadratic variation. Using $\langle W(t)W(s) \rangle = \min(t, s) = s$,

$$\mathbb{E}[\Delta W^2] = \mathbb{E}[W^2(t)] + \mathbb{E}[W^2(s)] - 2\langle W(t)W(s) \rangle = t + s - 2s = t - s = \Delta t, \quad (4.27)$$

and, using the Gaussian moment identity $\mathbb{E}[\Delta W^4] = 3 \mathbb{E}[\Delta W^2]^2$,

$$\text{Var}[\Delta W^2] = \mathbb{E}[\Delta W^4] - \mathbb{E}[\Delta W^2]^2 = 3\Delta t^2 - \Delta t^2 = 2\Delta t^2. \quad (4.28)$$

Keep these two numbers in mind: the squared increment has mean of order Δt but variance of order Δt^2 (standard deviation of order Δt). That mismatch of orders is about to do a lot of work.

(a) Mean of an Itô integral. The two basic examples are

$$\begin{aligned} \int_{t_0}^t dW(t') &= W(t) - W(t_0)|_{t_0=0} = W(t), \\ \int_{t_0}^t W(t') dW(t') &= \frac{1}{2} \{ W^2(t) - W^2(t_0) \} - \frac{1}{2}(t - t_0). \end{aligned} \quad (4.29)$$

The second result should give the reader pause: ordinary calculus would produce only the first term, $\frac{1}{2}(W^2(t) - W^2(t_0))$, whereas the Itô integral carries the *anomalous* extra term $-\frac{1}{2}(t - t_0)$, with no classical counterpart. Both integrals have zero mean:

$$\mathbb{E} \left[\int_{t_0}^t dW \right] = 0, \quad \mathbb{E} \left[\int_{t_0}^t W dW \right] = \frac{1}{2} \mathbb{E}[W^2(t)] - \frac{1}{2}t = \frac{1}{2}t - \frac{1}{2}t = 0. \quad (4.30)$$

This is no accident. Take G progressively measurable with respect to the Brownian filtration and satisfying $E[\int_{t_0}^t G(s)^2 ds] < \infty$. This is the non-anticipation and square-integrability condition used for the following mean and isometry statements.¹ Then

$$\mathbb{E} \left[\int_{t_0}^t G(t') dW(t') \right] = 0. \quad (4.31)$$

The proof is one line and shows exactly where the pre-point rule enters: discretizing, $G_{i-1} = G(\tau_i = t_{i-1})$ and $\Delta W_i = W(t_i) - W(t_{i-1})$ are independent (the integrand is frozen *before* the increment arrives), so

$$\mathbb{E} \left[\int_{t_0}^t G dW \right] = \text{ms-lim}_{m \rightarrow \infty} \sum_{i=1}^m \langle G_{i-1} \Delta W_i \rangle = \text{ms-lim}_{m \rightarrow \infty} \sum_{i=1}^m \langle G_{i-1} \rangle \langle \Delta W_i \rangle = 0. \quad (4.32)$$

(b) Correlation of two Itô integrals (Itô isometry). For G, H satisfying those same conditions,

$$\left\langle \int_{t_0}^t G(t') dW(t') \int_{t_0}^t H(t') dW(t') \right\rangle = \int_{t_0}^t dt' \langle G(t') H(t') \rangle; \quad (4.33)$$

the cross terms $\langle G_{i-1} H_{j-1} \Delta W_i \Delta W_j \rangle$ with $i \neq j$ vanish by independence of the increments, while the diagonal terms reduce, through $\langle \Delta W_i^2 \rangle = \Delta t$, to an ordinary Riemann integral. The isometry converts the second moment of a stochastic integral into a deterministic integral — it is the tool we will use every time we need a variance.

(c) Integration against the quadratic variation. For every non-anticipating $G(t)$,

$$\int_{t_0}^t G(t') [dW(t')]^2 = \int_{t_0}^t G(t') dt', \quad \text{i.e.} \quad [dW(t)]^2 = dt. \quad (4.34)$$

The notation on the right expresses $d[W]_t = dt$ for the quadratic variation; it is not a pathwise equality $(W_{t+dt} - W_t)^2 = dt$ for an individual finite increment. To see why, take $G = 1$; the claim amounts to $t = \text{ms-lim}_{m \rightarrow \infty} Q_m$ with $Q_m = \sum_{i=1}^m (\Delta W_i)^2$. Using the two moments computed above with uniform spacing $\Delta t = t/m$,

$$\begin{aligned} \mathbb{E}[Q_m] &= \sum_{i=1}^m \mathbb{E}[\Delta W_i^2] = \sum_{i=1}^m \frac{t}{m} = t, \\ \text{Var}[Q_m] &= \sum_{i=1}^m \text{Var}[\Delta W_i^2] = \sum_{i=1}^m 2 \frac{t^2}{m^2} = \frac{2t^2}{m} \xrightarrow{m \rightarrow \infty} 0, \end{aligned} \quad (4.35)$$

so the mean of Q_m is already exactly t , and its fluctuations die out as $m \rightarrow \infty$: $\langle (Q_m - t)^2 \rangle = \text{Var}[Q_m] \rightarrow 0$, which is precisely $\text{ms-lim } Q_m = t$. Each individual $(\Delta W_i)^2$ is random, but their sum self-averages. As a corollary of the same order counting, $dW(t) dt = 0$ and $[dW(t)]^{2+N} = 0$ for every $N > 0$: in stochastic calculus, dW behaves as an object of order $\sqrt{dt}$, and only the combinations dt and dW^2 survive at first order.

¹Björk[2], Definition 4.4, Proposition 4.8 and Corollary 4.9. The full martingale definition is given in Section 5.11.

4.5 Itô's formula

The fact that $[dW]^2 = dt$ is of *first* order in dt has a dramatic consequence: the chain rule of ordinary calculus must be modified. The corrected rule is *Itô's formula* (Itô's lemma), the change-of-variable formula of stochastic calculus, and it is the single most used result in mathematical finance.

Let $x(t)$ obey the Itô SDE

$$dx(t) = a(x, t) dt + b(x, t) dW(t), \quad (4.36)$$

and consider a smooth function $f[x(t)]$ of the process — for instance $f = \log x(t)$, the function that turns prices into returns. Since $dW \sim \sqrt{dt}$, a first-order Taylor expansion is not enough: the second-order term contains $(dx)^2 \supset b^2 dW^2 = b^2 dt$, which is of the same order as the first-order drift. We must therefore expand to second order,

$$df[x(t)] = f[x(t) + dx(t)] - f[x(t)] = \frac{\partial f}{\partial x} dx + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} (dx)^2, \quad (4.37)$$

where, by the multiplication table of the previous section,

$$(dx)^2 = a^2 dt^2 + 2ab dW dt + b^2 dW^2 = b^2 dt, \quad (4.38)$$

because dt^2 and $dW dt$ vanish while $dW^2 = dt$. Collecting terms, and allowing f to depend explicitly on time as well, $f = f(x, t)$, we obtain

$$df(x, t) = \left[\frac{\partial f}{\partial t} + a(x, t) \frac{\partial f}{\partial x} + \frac{1}{2} b^2(x, t) \frac{\partial^2 f}{\partial x^2} \right] dt + b(x, t) \frac{\partial f}{\partial x} dW(t). \quad (4.39)$$

This is Itô's formula. Compared with the ordinary chain rule, it contains one intruder: the term $\frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2}$, the *anomalous Itô term*, through which the noise leaks into the deterministic drift of any nonlinear function of the process. Stopping at second order is not an approximation but an exact statement in the mean-square sense, because $dt dW(t) \stackrel{\text{ms-lim}}{=} 0$ and $dt^2 \rightarrow 0$: no higher term survives. Whenever in the following chapters we “apply Itô's lemma”, it is equation (4.39) that is being invoked.

4.6 Solving the main stochastic differential equations

Armed with Itô's formula, we can now solve, one by one, the stochastic differential equations that will accompany us for the rest of the book. The strategy is always the same: find a change of variables that removes the awkward part of the equation, integrate, and translate back.

4.6.1 The Wiener process

The simplest SDE of all is

$$dx(t) = dW(t), \quad (a = 0, b = 1). \quad (4.40)$$

Integrating, $\int_0^t dx(t') = \int_0^t dW(t') = W(t) - W(0)$, so that (with $x(0)$ almost surely fixed)

$$x(t) = x(0) + W(t) : \quad (4.41)$$

the process is simply a Wiener process shifted by the initial condition. Its mean is inherited unchanged,

$$\mathbb{E}[x(t)] = x(0), \quad \mathbb{E}\left[\int_0^t dW\right] = 0, \quad (4.42)$$

while for the second moment,

$$\mathbb{E}[x^2(t)] = \langle x(0)x(0) \rangle + 2 \underbrace{\langle x(0) \int_0^t dW(s) \rangle}_0 + \left\langle \int_0^t dW(t') \int_0^t dW(s) \right\rangle = x^2(0) + t, \quad (4.43)$$

where the last term used the Itô isometry with $G = H = 1$. Therefore

$$\text{Var}[x(t)] = \text{Var}[x(0)] + t, \quad (4.44)$$

in agreement with the Fokker–Planck computation of Chapter 3. For $x(0) = 0$ the density solves $\partial_t P = \frac{1}{2} \partial_x^2 P$, giving once more the familiar Gaussian

$$P(x, t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}. \quad (4.45)$$

The SDE form has a practical virtue that the Fokker–Planck form lacks: it can be *simulated* directly. Introducing a time slicing $dt = T/N$, the increments of $W(t)$ are generated as

$$dW(t) = \varepsilon \sqrt{dt}, \quad \varepsilon \sim \mathcal{N}(0, 1), \quad (4.46)$$

one independent Gaussian draw per step; iterating produces a Monte Carlo trajectory, and an ensemble of such trajectories reconstructs the density. Figure 4.1 shows the result: three hundred simulated paths fanning out within the $\pm\sqrt{t}$ envelope, with the histogram of the endpoints reproducing the Gaussian $\mathcal{N}(0, T)$. This is the bridge between the microscopic (trajectory) and macroscopic (density) descriptions made algorithmic.

4.6.2 Arithmetic Brownian motion (Bachelier)

Adding a constant drift produces the *generalized Wiener process*, or arithmetic (additive) Brownian motion, with Itô SDE[1]

$$dx(t) = a dt + b dW(t), \quad b \in \mathbb{R}^+, a \in \mathbb{R}, \quad (4.47)$$

called additive because the increments add up iteratively. Integrating and taking moments exactly as before,

$$\mathbb{E}[x(t)] = a t \implies a \text{ is the drift}, \quad \mathbb{E}[x^2(t)] = a^2 t^2 + b^2 t \implies \text{Var}[x(t)] = b^2 t, \quad (4.48)$$

with standard deviation $\text{s.d.}[x(t)] = b\sqrt{t}$ — the quantity that finance calls the *volatility* — and the full distribution is Gaussian at all times,

$$\text{PDF}(x, t) = \mathcal{N}(a t, b^2 t). \quad (4.49)$$

A sample path is sketched in Figure 4.2: the trajectory rattles around the straight line $\langle x(t) \rangle = a t$, with excursions that grow like $\sqrt{t}$.

This was, in fact, the very first mathematical model of stock prices, proposed by Bachelier in his 1900 thesis[1] — five years before Einstein’s paper on Brownian motion. Historically

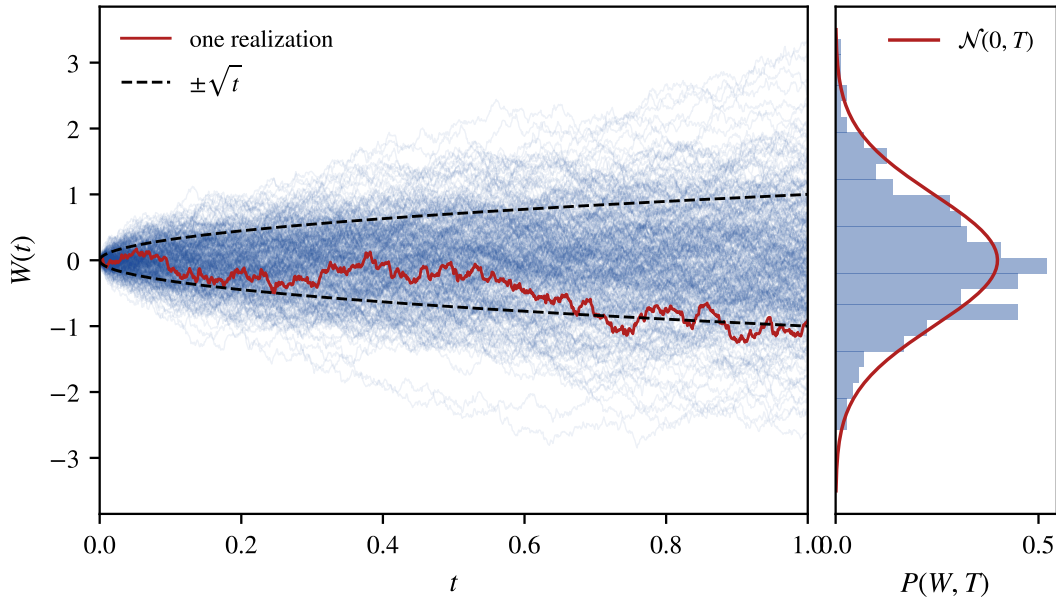


Figure 4.1: Monte Carlo simulation of 300 Wiener trajectories ($dW = \varepsilon\sqrt{dt}$, $\varepsilon \sim \mathcal{N}(0, 1)$), spreading as $\sqrt{t}$ (dashed envelope). Right: histogram of the endpoints $W(T)$ against the exact density $\mathcal{N}(0, T)$. Generated by `images/code_for_images/wiener_ensemble.py`.

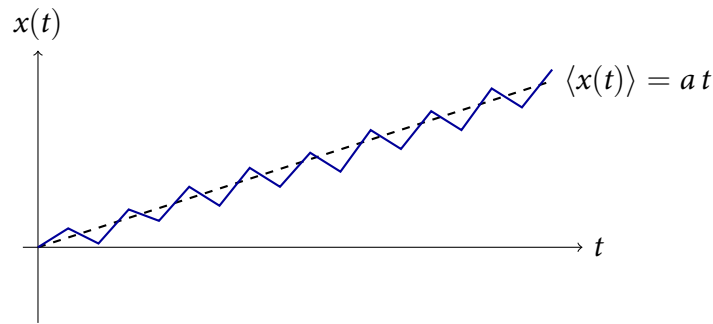


Figure 4.2: Arithmetic Brownian motion: a sample path fluctuating around the linear mean $\langle x(t) \rangle = at$.

glorious, but as a price model it has a fatal flaw: since the Gaussian has unbounded support and the fluctuations grow in probability, there is a nonzero probability that the price becomes *negative*, which is not admissible for a stock. Its autocovariance, for the record, is

$$\text{Cov}[x(t), x(s)] = \mathbb{E}[x(t)x(s)] - \mathbb{E}[x(t)]\mathbb{E}[x(s)] = b^2 \langle W(t)W(s) \rangle = b^2 \min(t, s). \quad (4.50)$$

Note that so far we have not needed Itô's formula at all: with constant coefficients, everything is linear. The next example is where the new calculus earns its keep.

4.6.3 Multiplicative linear noise

Consider now an Itô process in which the noise enters *multiplicatively*, proportional to the process itself,

$$dx(t) = b x(t) dW(t) \implies \frac{dx(t)}{x(t)} = b dW(t) \neq d \ln x(t), \quad (4.51)$$

and the inequality on the right is the whole point: in ordinary calculus dx/x would equal $d \ln x$, but here it does not, and Itô's formula tells us by how much it fails. Take $f(x) = \ln x$ (with

$x(0) > 0$); applying (4.39) with $a = 0$, $\partial_x f = 1/x$ and $\partial_x^2 f = -1/x^2$,

$$d \ln x(t) = \left[0 \cdot \frac{1}{x} + \frac{1}{2} b^2 x^2 \left(-\frac{1}{x^2} \right) \right] dt + b x \frac{1}{x} dW(t) = -\frac{b^2}{2} dt + b dW(t). \quad (4.52)$$

The anomalous term has generated a *negative drift* $-b^2/2$ out of pure noise. In logarithmic variables the process is now simple — an additive Brownian motion — with

$$\mathbb{E}[\ln x(t)] = \ln x_0 - \frac{b^2}{2} t, \quad \text{Var}[\ln x(t)] = b^2 t. \quad (4.53)$$

Integrating and exponentiating back,

$$\ln x(t) = \ln x(0) - \frac{b^2}{2} t + b W(t) \implies x(t) = x_0 \exp \left\{ -\frac{b^2}{2} t + b W(t) \right\}, \quad (4.54)$$

which is manifestly positive at all times. One more check is instructive. Setting $z = \ln x(t) \sim \mathcal{N}(\ln x_0 - \frac{b^2}{2} t, b^2 t)$ and using the Gaussian identity $\langle e^z \rangle = e^{\langle z \rangle + \frac{1}{2} \text{Var}[z]}$,

$$\begin{aligned} \mathbb{E}[x(t)] &= \langle e^z \rangle = e^{\langle z \rangle + \frac{1}{2} \text{Var}[z]} \\ &= e^{\ln x_0 - \frac{b^2}{2} t} e^{\frac{b^2}{2} t} = x_0. \end{aligned} \quad (4.55)$$

The two effects cancel exactly: the anomalous drift pushes the *typical* trajectory down, while the fat upper tail of the exponential pulls the *mean* up, and the mean is preserved despite the multiplicative noise. This tug-of-war between median and mean is a recurring theme in finance.

Geometric Brownian motion. Adding a linear drift to the multiplicative noise gives the *geometric Brownian motion* (GBM), with Itô SDE

$$dx(t) = a x(t) dt + b x(t) dW(t), \quad b > 0, a \in \mathbb{R}, \quad (4.56)$$

which is best read after dividing by x : the *percentage* change $\frac{dx(t)}{x(t)} = a dt + b dW(t)$ is an arithmetic Brownian motion. Percentage changes, not absolute ones, are what investors actually care about — a 1 € move means something very different for a 10 € stock and a 1000 € one — and this is the economic rationale for the multiplicative structure. By Itô's formula, exactly as in the previous subsection but keeping the drift,

$$d \ln x(t) = \left(a - \frac{b^2}{2} \right) dt + b dW(t), \quad (4.57)$$

so the increments $\ln(x(t+dt)/x(t))$ — the *log-returns* — are Gaussian, and the arithmetic Brownian motion of the logarithm becomes the geometric motion of the price. As before,

$$\mathbb{E}[\ln x(t)] = \ln x(0) + \left(a - \frac{b^2}{2} \right) t, \quad \text{Var}[\ln x(t)] = b^2 t, \quad (4.58)$$

$$\text{PDF}(\ln x(t), t) = \mathcal{N} \left(\ln x_0 + \left(a - \frac{b^2}{2} \right) t, b^2 t \right), \quad (4.59)$$

and translating back to the price itself,

$$\mathbb{E}[x(t)] = x_0 e^{at}, \quad \text{Var}[x(t)] = x_0^2 e^{2at} (e^{b^2 t} - 1). \quad (4.60)$$

The process never becomes negative and its mean grows exponentially at rate a , which makes GBM a *plausible model for spot prices* — indeed, the standard one, as we will see at length in the next chapter. It was introduced in finance by Osborne in 1959[31], and was later given a proper economic rationale by Samuelson (1965)[36]. Since the model is Markovian, the prices before $x(0)$ are irrelevant to the future — an assumption still debated, to which we return in Chapter 5.

4.6.4 The Ornstein–Uhlenbeck process

Our last and richest example is the *Ornstein–Uhlenbeck* (OU) process. It occupies a special place in the theory: by *Doob’s theorem*[8, 41] under the usual nondegeneracy and regularity assumptions it characterizes the stationary scalar Gaussian Markov case (in the strong sense, $P_t(x, t) = P_t(x)$). Uniqueness makes it the default model for any quantity that fluctuates around a stable level: short-term interest rates, volatility, or the velocity of a Brownian particle. Its Itô SDE is

$$dx(t) = -\kappa x dt + \sqrt{D} dW(t), \quad \kappa > 0, [\kappa] = T^{-1}, \tau = \frac{1}{\kappa}, \quad (4.61)$$

a random walk tethered by a linear restoring force: whenever x wanders away from zero, the drift $-\kappa x$ pulls it back, on the characteristic time scale $\tau = 1/\kappa$.

To solve it we use the substitution $f(x, t) = x e^{\kappa t}$ (an integrating factor), designed precisely to absorb the restoring drift. By Itô’s formula,

$$df(x, t) = \left[\frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right] dt + b \frac{\partial f}{\partial x} dW(t) = \sqrt{D} e^{\kappa t} dW(t), \quad (4.62)$$

because the two drift contributions cancel identically ($\kappa x e^{\kappa t} - \kappa x e^{\kappa t} = 0$) and f is linear in x , so no anomalous term appears. What remains is a pure (time-modulated) noise, which we integrate at sight:

$$\int_0^t df(x(t'), t') = \sqrt{D} \int_0^t e^{\kappa t'} dW(t'), \quad (4.63)$$

and since $x(t) = f e^{-\kappa t}$ with $x(0)$ almost surely fixed,

$$x(t) = x_0 e^{-\kappa t} + \sqrt{D} \int_0^t e^{-\kappa(t-t')} dW(t'). \quad (4.64)$$

The structure is transparent: the initial condition decays exponentially, while the noise accumulates — but each past kick $dW(t')$ is remembered only with the fading weight $e^{-\kappa(t-t')}$. Taking the mean (the Itô integral averages to zero),

$$\mathbb{E}[x(t)] = x_0 e^{-\kappa t}, \quad (4.65)$$

and, computing the variance with the Itô isometry,

$$\begin{aligned} \text{Var}[x(t)] &= \left\langle (x(t) - \langle x(t) \rangle)^2 \right\rangle = D e^{-2\kappa t} \int_0^t (e^{\kappa t'})^2 dt' \\ &= D e^{-2\kappa t} \frac{e^{2\kappa t} - 1}{2\kappa} = \frac{D}{2\kappa} (1 - e^{-2\kappa t}). \end{aligned} \quad (4.66)$$

The two limiting regimes are revealing. For long times, $t \gg 1/\kappa$, the memory of the initial condition is gone and the variance saturates,

$$\mathbb{E}[x(t)] \rightarrow 0, \quad \text{Var}[x(t)] \rightarrow \frac{D}{2\kappa}, \quad (4.67)$$

so the distribution converges to a stationary law; a process initialized with fixed x_0 is not stationary at finite times: the injection of noise (D) and the dissipation of the restoring force (κ) balance. For short times, $t \ll 1/\kappa$, expanding the exponentials gives

$$\mathbb{E}[x(t)] \rightarrow x_0 \{1 - \kappa t + \dots\}, \quad \text{Var}[x(t)] \rightarrow \frac{D}{2\kappa} \{1 - 1 + 2\kappa t + \dots\} = D t, \quad (4.68)$$

i.e. before the restoring force has had time to act, the process simply *diffuses* like a free Brownian motion. The full distribution at all times is Gaussian,

$$\mathcal{N}\left(x_0 e^{-\kappa t}, \frac{D}{2\kappa} [1 - e^{-2\kappa t}]\right) \xrightarrow{t \rightarrow \infty} \mathcal{N}\left(0, \frac{D}{2\kappa}\right) = P^{\text{O.U.}}(x). \quad (4.69)$$

In the original physical setting, where x is the velocity of a Brownian particle, the stationary law $P^{\text{O.U.}}(v) = P^{\text{M.B.}}(v)$ is nothing but the Maxwell–Boltzmann distribution: the OU process describes, in real time, how an out-of-equilibrium velocity distribution relaxes to thermal equilibrium, and matching the two stationary variances yields the Einstein fluctuation–dissipation relation, $D/(2\kappa) = k_B T/m$ for the velocity process. Figure 4.3 sketches the characteristic noisy relaxation of a single path, and Figure 4.4 shows the full Monte Carlo picture: an ensemble of paths hugging the analytic mean, with the fluctuation band saturating at $\pm\sqrt{D/2\kappa}$.

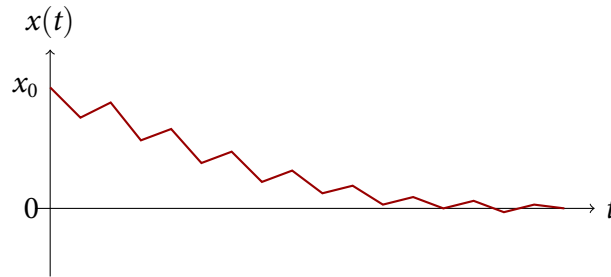


Figure 4.3: Ornstein–Uhlenbeck process: a noisy, exponentially damped relaxation of $x(t)$ from x_0 towards the stationary level, with asymptotic variance $D/2\kappa$.

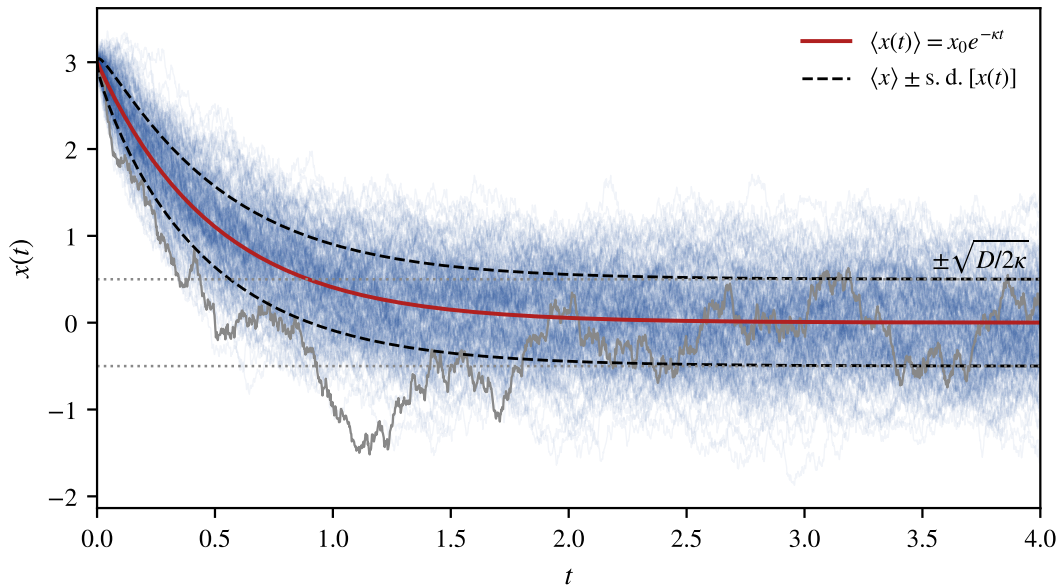


Figure 4.4: Monte Carlo simulation of 200 Ornstein–Uhlenbeck paths ($\kappa = 2$, $D = 1$, $x_0 = 3$), relaxing from the common initial condition to the stationary band. The red curve is the analytic mean $x_0 e^{-\kappa t}$; the dashed curves are the ± 1 standard-deviation envelope, saturating at $\pm\sqrt{D/2\kappa}$ (dotted). Generated by `images/code_for_images/ou_ensemble.py`.

With the Wiener process, geometric Brownian motion, and the Ornstein–Uhlenbeck process in hand, our toolbox is complete. The next chapter puts it to work on the problem that made stochastic calculus famous: the fair pricing of financial derivatives.

Financial Derivatives and Pricing

The machinery of stochastic processes and stochastic calculus developed in the previous chapters finds its most celebrated application in the theory of *financial derivatives*. The problem it solves is easy to state and was, for most of a century, believed to have no objective answer: what is the *fair price* of a contract whose value depends on the unknown future price of something else? This chapter builds the answer from the ground up. We first describe the structure of financial markets and the main asset classes; we then introduce options and the no-arbitrage principle; and we finally derive the Black–Scholes equation and solve it, obtaining the formula that in 1973 turned derivative pricing from an art into a science.

5.1 Financial markets

A *financial market* is a physical or virtual venue in which agents exchange *assets* at a *spot price* (instantaneous price) $S(t)$, determined by the law of supply and demand. The participants are heterogeneous — banks, hedge funds, brokers, individual traders — and so are the venues. *Stock markets* in particular come in three flavours: the *official exchanges* (NYSE, LSE, ...), of which there are about 200 worldwide, historically paper-based; the fully *electronic* exchanges (Nasdaq, Euronext, ...), where trading is algorithmic; and *hybrid* venues such as today’s NYSE, which combine both. Alongside the official markets there exist non-official ones, distinguished by the absence of intermediaries.

Since this book is written for physicists, it is worth pausing on how a trade actually happens, because the microscopic mechanism is what gives the spot price its meaning. At any instant, an exchange maintains for each asset an *order book*: a list of offers to buy at stated prices (the *bids*) and offers to sell at stated prices (the *asks*). The highest bid and the lowest ask face each other across a small gap, the *bid–ask spread*; a trade occurs when someone accepts the best price on the opposite side, and the price at which this happens is the quoted price of the asset. The market is called *liquid* when the book is thick — when there are so many standing offers that one can buy or sell a large quantity at any moment without moving the price appreciably. Liquidity is the market’s analogue of a good thermal bath: it is what allows a single agent to exchange with the collective without perturbing it, and most of the idealizations of this chapter (continuous trading, negligible spread) are statements about high liquidity.

Two more pieces of jargon will recur. An investor is *long* an asset after buying it, profiting if the price rises; and is *short* after selling an asset it does not own — borrowing it, selling it, and planning to buy it back later — profiting if the price falls. Short selling looks exotic but is essential: it is what allows negative positions, and therefore hedges, to exist; we will meet a short position at the heart of the Black–Scholes argument.

The central object of the theory is the spot price, and two of its properties deserve to be

stated explicitly. First, the spot price is *unique*: at any given time there is a single price in the whole market — there is no “price here” and “price there”. (Why this is so is a question we defer for a few pages; the answer is arbitrage.) Second, it is a price of *equilibrium*: it arises from the matching of supply and demand, i.e. from the meeting of the best ask and the best bid, with a bid–ask spread $\Delta(\text{ask}, \text{bid}) \rightarrow 0$. In reality $\Delta > 0$, because of transaction costs and related frictions, but the idealization is a good one for liquid markets.

The equilibrium and pricing assumptions must be distinguished. In an Arrow–Debreu economy, equilibrium existence requires assumptions on preferences, endowments and production. For the derivative-pricing arguments below, no arbitrage and replication are the relevant conditions: completeness means that the specified contingent payoffs can be replicated using traded assets. Neither condition implies that information asymmetries disappear immediately. Expected returns may contain compensation for risk as well as the risk-free rate.

5.2 Risk and the Efficient Market Hypothesis

Types of risk. Since risk is what one is paid to bear, we should say what kinds there are. Financial risk is traditionally classified into three categories. *Operational risk* covers human errors, computer crashes, and similar accidents of implementation — a category whose weight has grown with automation: the course slides quote an undated 60–75% algorithmic-trading share, without specifying a market or volume measure; it should not be read as a verified current worldwide statistic. *Credit risk* is the risk that the counterparty does not honour the obligations of the contract — the risk of insolvency. *Market risk*, finally, is the one that will concern us: the random fluctuation of prices themselves, caused partly by the internal dynamics of supply and demand, partly by exogenous factors (politics, disasters, ...), and partly by psychological factors — imitation, panic — whose study belongs to *behavioural economics*. The Black–Scholes theory we develop below deals with market risk only; credit risk is assumed away, an assumption whose limits became painfully visible in 2008.

The Efficient Market Hypothesis. The fundamental question, on which the applicability of everything we have built so far depends, is this:

Is the dynamics of an asset a Markov process?

Figure 5.1 shows why the question is not rhetorical: the trajectory of a major index looks like the noisy paths of the previous chapters, but it visibly responds to identifiable exogenous shocks — trade disputes, a pandemic, central-bank decisions — and one may legitimately ask whether such a history can be modelled by a memoryless Itô SDE driven by market risk alone.

The EMH, formulated by Fama[14], concerns how prices reflect a specified information set. The Markov property concerns the conditional law of future states given the past. These are different assertions: market efficiency does not imply a scalar Markov price process, and the Markov property does not imply market efficiency. Likewise, the fundamental theorems of asset pricing concern no arbitrage and martingale measures, not a proof of the EMH (Section 5.11). We assume a Markov GBM as a tractable stock-price model, rather than as a consequence of those theorems. A non-Markov observed price can also be part of a larger Markov state, as in stochastic-volatility models. Whether a chosen model describes real returns is an empirical question, taken up in Chapter 7.

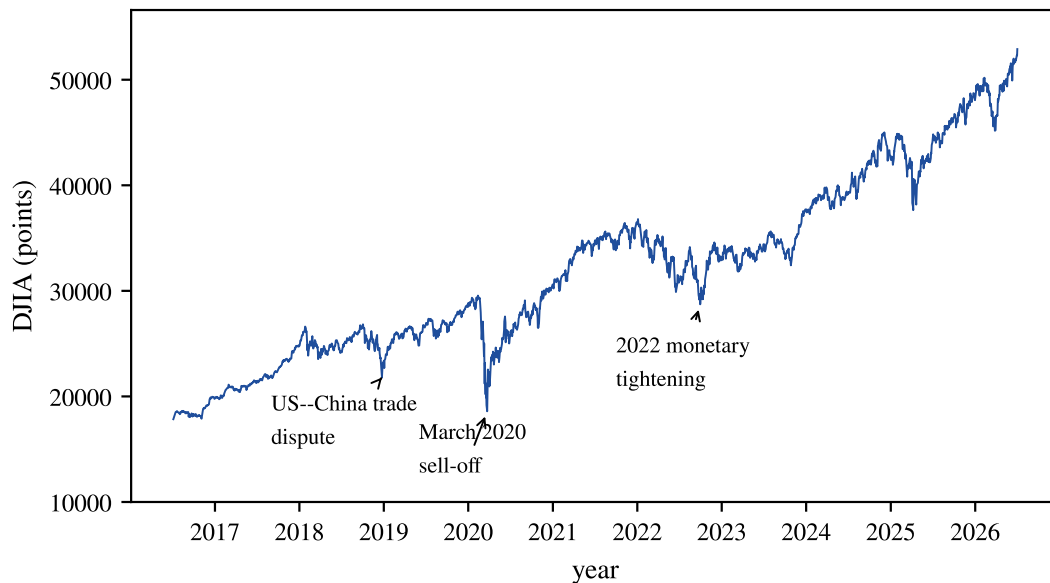


Figure 5.1: The Dow Jones Industrial Average, daily closes 2016–2026 (FRED series DJIA), with some major exogenous events annotated. The fundamental question of this section: can this dynamics be modelled as a Markov process? Generated by `images/code_for_images/djia_events.py`.

5.3 Financial assets

Before specializing to derivatives, it is worth surveying what is actually traded. The main asset classes are the following.

- **Currencies:** money itself, traded against other money.
- **Commodities:** energy, precious metals, agricultural products, and so on.
- **Stocks (equities):** securities issued by a company to raise capital by selling a fraction of its ownership, without any obligation of repayment. For the buyer they offer visibility, a voice in company decisions, and possible dividends. A *stock index* is the average (arithmetic or weighted) value of a group of equity titles — the S&P 500, which averages five hundred large American companies weighted by capitalization, will reappear several times in this book; other examples are the Dow Jones (USA), the FTSE 100 (UK), the CAC 40 (France), the DAX (Germany), the FTSE MIB (Italy), and the Nikkei (Japan).
- **Bonds (obligations):** debt securities issued by states or companies to raise large sums; they have a *maturity* T at which the capital is repaid with interest, and are comparatively low-risk. They are the subject of Chapter 6.
- **Derivatives:** contracts whose value derives from other assets — the subject of the present chapter.
- **Cryptocurrencies:** assets living outside the official markets.

Portfolios mixing these classes (gold, currencies, stocks, bonds, ETFs...) are the standard tool of diversification and, in general, of speculation. A *portfolio* is simply a weighted collection of assets held together, and *diversification* — not putting all the eggs in one basket — has a rationale a physicist will recognize at once: if the fluctuations of N assets were independent, the relative

fluctuation of their sum would shrink as $1/\sqrt{N}$, exactly like the relative energy fluctuations of N weakly coupled subsystems. Mixing imperfectly correlated assets therefore reduces risk at no cost in average return; the catch, to which we return in Chapter 7, is that in a crisis the correlations rise and the $\sqrt{N}$ protection evaporates precisely when it is needed. To appreciate the scale of all this: the handwritten notes quote about \$461 trillion of global financial assets and roughly \$100 trillion of annual world GDP, without a reference year; these are undated lecture estimates (a stock and an annual flow). Finance is not a small correction to the real economy; by market value it is several times larger.

5.4 Derivatives and options

A *derivative* is a contract between a buyer and a seller whose value *derives* from the value of other financial instruments (currencies, stocks, etc.). The idea is old: derivatives appeared in the Netherlands in the 1600s for the tulip trade, and the first recorded example is usually attributed to Aristotle (*Politics* I.xi), where Thales, expecting a good olive harvest, secured in advance the right to use the local olive presses. Then as now, derivatives serve two purposes: *hedging*, i.e. insurance against unfavourable price movements, and *speculation*, i.e. profiting from anticipated ones.

One important type of derivative is the (*stock*) **option**. An option is a contract between two parties — the **writer**, who sells the contract, and the **holder**, who buys it — granting the holder the right, but not the obligation, to buy (**call**) or sell (**put**) from/to the writer the *underlying asset* (the stock) at a fixed price X , the *strike*, at a future time T , the *maturity*.

The contract is fundamentally *asymmetric*, and the asymmetry is what makes pricing non-trivial. The holder may exercise the option or walk away, at will; the writer, if the holder exercises, *must* sell (or buy) at the agreed price, whatever the market is doing. The holder buys optionality; the writer sells certainty and takes on risk.

Options come in various types. *European* options are exercisable only at maturity T ; within this family live the *plain vanilla* options, but also asiatic options, barrier options, and other so-called exotics. *American* options are exercisable at any time $t \leq T$, a freedom that complicates the analysis considerably; we will be able to price them only numerically, in Section 5.16.

5.5 Option payoffs and fair pricing

We can now state the problem of the chapter. The *option pricing problem* is to find the unique fair (equilibrium) price $O(S, t)$ of the contract at the moment of signing — fair in the sense that neither party makes a riskless profit at the expense of the other, at any $t \leq T$:

$$O(S, t) = C_{BS}(S, t) \quad \forall t \leq T. \quad (5.1)$$

Solving this problem is the achievement of Black and Scholes; before getting there, we need to understand what an option is worth in the one situation where no theory is required, namely *at maturity*.

Call option. Consider first the hedging use of a call, and let us put numbers on it, because the insurance logic is easiest to absorb on a concrete case. Suppose the holder runs a large energy company that consumes gas, currently at 90 € per unit, and fears that the price will slip upwards over the next year; buying a call option at strike $X = 100$ €, for a premium of, say, 5 €, insures against exactly this. If at maturity gas trades at 130 €, the company exercises and buys at 100:

the insurance has capped its cost at 105 € per unit (strike plus premium) instead of 130. If instead gas trades at 80 €, the company simply buys on the market and lets the option expire: the premium of 5 € is the cost of the peace of mind, exactly like an unused fire-insurance policy. In general, at maturity T two scenarios are possible.

- a) If $S(T) < X$, buying at the market price is cheaper than exercising; the holder does not exercise, and simply loses the premium paid for the option — an insurance policy that turned out not to be needed.
- b) If $S(T) > X$, the holder exercises, buys at X what the market values at $S(T)$, and gains $S(T) - X$.

For purely speculative purposes the same logic applies, with the asset sold immediately after exercise: if $S(T) > X$ the speculator pockets a guaranteed $S_T - X$ (minus the premium, of course).

Both scenarios are summarized by the **payoff** of the call,

$$C(S_T, T) = \begin{cases} S_T - X & \text{if } S_T > X, \\ 0 & \text{if } S_T \leq X, \end{cases} = \max\{S_T - X, 0\} = \{S_T - X\}_+, \quad (5.2)$$

the “hockey-stick” function of Figure 5.2: flat at zero below the strike, rising with unit slope above it.

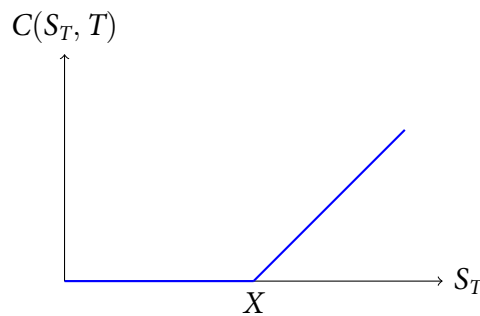


Figure 5.2: Payoff of a European call at maturity: worthless below the strike X , linear in S_T above it.

The payoff tells us what the option is worth at T ; the problem, of course, is that we would like to know its value *now*.

5.5.1 Put option

A put is the mirror image: a bet on a price *decrease*. The holder expects the asset to lose value and secures the right to sell it at $X > S_T$, gaining $X - S_T$ if the fall indeed materializes; if instead $S_T \geq X$ the right is worthless, since selling at the market price is better. The **payoff** of the put is therefore

$$P(S_T, T) = \begin{cases} X - S_T & \text{if } S_T < X, \\ 0 & \text{if } S_T \geq X, \end{cases} = \max\{X - S_T, 0\} = \{X - S_T\}_+, \quad (5.3)$$

the reflected hockey stick of Figure 5.3.

A caveat: these clean payoff formulas hold only for *plain vanilla* options, which depend solely on T and on the terminal price S_T . For *exotic* options the payoff involves the whole price

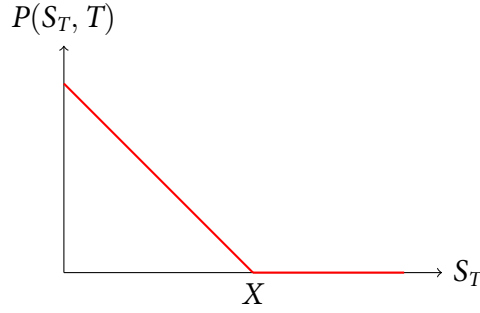


Figure 5.3: Payoff of a European put at maturity: linear below the strike, worthless above it.

history; in the *asiatic* case, for instance, it depends on the average of the prices during the life of the contract,

$$\max\left\{\frac{1}{n} \sum_{i=1}^n S_{t_i} - X, 0\right\}, \quad (5.4)$$

and for such contracts no closed formula exists — one resorts to numerical (Monte Carlo) simulations.

The no-arbitrage argument. The terminal price S_T is a stochastic process — but which one, and priced how? We will model it with the geometric Brownian motion of Chapter 4; but before any model, a purely logical argument already pins down the value of the option *at maturity*. Follow the money:

1. the holder H exercises the option and acquires the stock at X at time T ;
2. the writer W is obligated to sell at X ;
3. W observes that $S_T > X$ and buys the stock in the market, selling it immediately to H ; the round trip costs W exactly $S_T - X$.

For neither party to be robbed, the price of the option at maturity must satisfy $X + C(S_T) = S_T$, i.e. $C(S_T) = S_T - X$; any other value would create an instantaneous riskless profit — an *arbitrage* — for one of the two parties, and would be arbitrated away immediately. To avoid being at a disadvantage in the future, then, the parties can already agree that at maturity $C(S_T) = S_T - X$ is the fair value.

The real problem is to propagate this terminal value *backwards*: the fair value in the present must be deduced from the future. One models the future between $t = 0$ and $t = T$ with a stochastic process, and this is where the GBM enters. The GBM was introduced as a price model by Osborne in 1959[31] and applied to economics in 1965, notably by Samuelson[36]. The older Bachelier model (arithmetic Brownian motion, Chapter 4) is ruled out for the reason we found there: it assigns positive probability to negative prices, which is unphysical.

5.6 The risk-free rate and the bank account

To understand why the GBM is the right model, let us first understand the alternative to investing at all: the bank. Consider a *bank account*: the agent starts outside the market, with money in

the bank, at $t_0 = 0$ with capital $S(t_0) = S_0$. The bank pays a risk-free interest rate r , assumed constant, so that with *simple interest*

$$S_t := S(t) = S_0 + S_0 r t \quad \forall t. \quad (5.5)$$

The rate r represents the cost (or reward) of money over time: a euro today is worth more than a euro tomorrow, and r measures by how much.

Simple interest, however, undercounts: nothing prevents the agent from cashing in at an intermediate time and reinvesting. Splitting $[0, t]$ into m steps of length t/m and reinvesting the accrued interest at each step,

$$S_t = S_0 \left(1 + \frac{r t}{m}\right)^m, \quad (5.6)$$

and in the limit of continuous reinvestment, $m \rightarrow \infty$ (for fixed r and t),

$$S_t \longrightarrow S_0 e^{r t}, \quad (5.7)$$

which is *compound interest in the continuum* — sometimes called mathematical interest. Exponential growth at rate r is thus the baseline against which every investment must be measured.

The adjective *risk-free* deserves a comment. It means a guaranteed return with zero credit risk — and it is, of course, an abstraction. Even banks deemed “too big to fail” can fail, as the crisis of 2008 (and the state-financed rescues that followed) made clear; and the rate r itself is not constant in time but fluctuates, following its own SDE $r(t) \rightarrow dr(t)$ with unknown future values — the subject of Chapter 6. Can one at least find a good *proxy* for the ideal risk-free rate? In practice, yes: rates on high-grade government bills with short maturity, between one and three months, whose default risk is negligible. This identification gives the initial condition $dB_t = r B_t dt$ for the riskless asset, and hence, in the absence of risk,

$$dS_t = r S_t dt \implies S_t = S_0 e^{r t}. \quad (5.8)$$

5.7 Geometric Brownian Motion for stock prices

Now let the agent leave the bank and enter the market; the risk is no longer zero, and the dynamics must say so. Two ingredients are needed.

The first is the reward. One introduces

$$\mu = \text{percentage return rate (historical rate of return)}, \quad (5.9)$$

a number extracted from real historical time series — and *not* universal: each asset has its own. The deterministic part of the price dynamics becomes

$$dS_t = S_t \mu dt, \quad (5.10)$$

formally identical to the bank account but with μ in place of r . An expected excess return may compensate investors for risk, but $\mu > r$ is not a mathematical requirement of GBM or no arbitrage. The difference $\mu - r$ is called the *risk premium*; its sign is not fixed by the model.

The second ingredient is the risk itself. The model represents unpredictable price changes through a stochastic term $\sigma S_t dW_t$, where σ is the *historical volatility*, with the meaning of an instantaneous standard deviation. Assembling the two pieces,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad \mu \in \mathbb{R}, \sigma \in \mathbb{R}^+, \quad (5.11)$$

which is precisely the SDE of *Geometric Brownian Motion* (GBM), solved in Chapter 4. The stock price, in this model, is a bank account whose interest rate has been made noisy.

Log-returns. The financially meaningful quantity is not the absolute change dS_t but the *percentage* return dS_t/S_t , which is adimensional — it treats a 10 € stock and a 1000 € one on the same footing. By Itô's lemma applied to $\ln S_t$ (Chapter 4),

$$d \ln S_t = \left(\mu S_t \frac{1}{S_t} + \frac{1}{2} \sigma^2 S_t^2 \frac{-1}{S_t^2} \right) dt + \sigma S_t \frac{1}{S_t} dW_t = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t, \quad (5.12)$$

that is, $d \ln S_t = \frac{dS_t}{S_t} - \frac{1}{2} \sigma^2 dt$: the log-return differs from the percentage return by the anomalous Itô term. These increments $d \ln S_t$ are the *log-returns* (financial returns), and they are the quantities whose statistics one actually measures on market data. Integrating,

$$\ln \frac{S_t}{S_0} = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \implies S_t = S_0 \exp\left\{ \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right\} > 0, \quad (5.13)$$

manifestly positive at all times, as a price should be. It is immediate to verify that

$$\mathbb{E}[\ln \frac{S_t}{S_0}] = \left(\mu - \frac{\sigma^2}{2} \right) t, \quad \text{Var}[\ln \frac{S_t}{S_0}] = \sigma^2 t, \quad (5.14)$$

so the PDF of the log-returns is Gaussian — a prediction we will test against reality, with instructive results, in Chapter 7.

5.8 The log-normal distribution

If the log-returns are Gaussian, what is the distribution of the price itself? One could attack the Fokker–Planck equation for $P(S, t \mid S_0, t_0)$ directly, but it is far more complicated than the one for $P(\ln S_t \mid \ln S_0, t_0)$, because $\ln S_t$ is the variable that diffuses Gaussianly; one can change variables in the probability density.

Let $P(\ln S_t \mid \ln S_0, t_0) := P(\ln S_t)$ be the Gaussian density of the logarithm; its normalization reads

$$\int_{-\infty}^{+\infty} d \ln S_t P(\ln S_t) = 1, \quad \text{with} \quad d \ln S_t = \frac{dS_t}{S_t} \quad (5.15)$$

(a normal change of variables — no Itô here, because we are integrating over values, not following a trajectory). Since $\ln S_t \rightarrow +\infty$ when $S_t \rightarrow +\infty$ and $\ln S_t \rightarrow -\infty$ when $S_t \rightarrow 0$, the integration limits transform accordingly, and

$$\int_0^{+\infty} \frac{dS_t}{S_t} P(\ln S_t) = \int_0^{+\infty} dS_t P_{\ln}(S_t), \quad (5.16)$$

where the density picks up a Jacobian factor $1/S_t$: this defines the *log-normal probability density*,

$$P_{\ln}(S_t, t \mid S_0, t_0) = \frac{1}{S_t \sqrt{2\pi\sigma^2(t-t_0)}} \exp\left\{ -\frac{\left[\ln \frac{S_t}{S_0} - \left(\mu - \frac{\sigma^2}{2} \right) (t-t_0) \right]^2}{2\sigma^2(t-t_0)} \right\}. \quad (5.17)$$

Note that the moments of S_t can no longer be read off by inspection, as they could for a Gaussian; they must be computed. Using the Gaussian identity $\langle e^z \rangle = e^{\langle z \rangle + \frac{1}{2} \text{Var}[z]}$ with $z = \ln S_t$, one finds

$$\mathbb{E}[S_t] = \int_0^\infty dS_t S_t P_{\ln}(S_t) = e^{\langle \ln S_t \rangle} \cdot e^{\frac{1}{2} \text{Var}[\ln S_t]} = S_0 e^{\mu(t-t_0)}, \quad (5.18)$$

$$\text{Var}[S_t] = e^{2\langle \ln S_t \rangle} \cdot e^{\text{Var}[\ln S_t]} (e^{\text{Var}[\ln S_t]} - 1) = S_0^2 e^{2\mu(t-t_0)} (e^{\sigma^2(t-t_0)} - 1). \quad (5.19)$$

The mean grows exponentially at the full rate μ — the $\sigma^2/2$ correction of the typical trajectory is compensated, as in Chapter 4, by the fat upper tail.

That fat tail is visible in the shape of the distribution, sketched in Figure 5.4: the log-normal is skewed to the right, with the *mode* to the left of the *mean*. The financial reading is worth remembering: most trajectories end up below the average (the mode is low), while a few end up far above it (the tail is long) — high gains come with large risks.

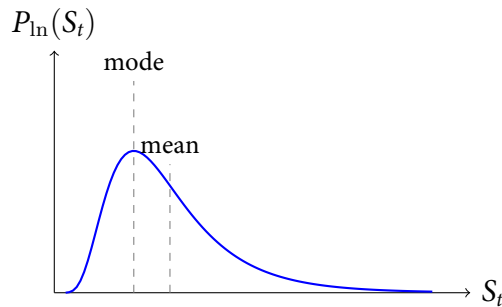


Figure 5.4: The log-normal density of the price S_t : right-skewed, with the mode below the mean. Small gains are typical; large gains are rare but heavy.

5.9 Limits of the GBM model

Before using GBM for pricing, we examine its empirical limitations. Three questions summarize the situation.

1. *Do prices really follow a Markov process?* This is a modelling question distinct from whether the EMH holds. Markets digest information quickly, but not instantaneously, and traces of memory remain.
2. *Is the distribution of log-returns Gaussian at every time scale?* That is, is

$$\text{PDF}\left[\ln \frac{S(t + \Delta t)}{S(t)}\right] = \mathcal{N} \quad \forall t? \quad (5.20)$$

At longer horizons Gaussian approximations can improve, but there is no universal one-month threshold; but at high frequencies — daily, and today down to seconds — it fails visibly: the empirical distribution exhibits *fat tails*, with extreme events far more probable than any Gaussian allows. Chapter 7 is devoted to this discrepancy.

3. *Is the volatility constant?* The model predicts $\text{s.d.}[\ln r(\Delta t)] = \sigma\sqrt{\Delta t}$ with a fixed σ . The data say no: σ itself depends on time (Engle[12]), and quiet periods alternate with turbulent ones — the phenomenon of *volatility clustering*, also taken up in Chapter 7.

One can generalize the model to address these failures, as Chapter 7 illustrates. Some alternatives also have analytic or transform solutions; the simplicity of the GBM explains much of its continuing use.

5.10 The Black–Scholes equation

We now have all the ingredients: a model for the underlying (the GBM), a terminal condition (the payoff), and a guiding principle (no arbitrage). The Black–Scholes model (Black & Scholes, 1973; Merton, 1973)[3, 29] assembles them into a partial differential equation for the option price — work recognized with the 1997 Nobel Prize in Economics, awarded to Scholes and Merton. Given the terminal payoff $C(S_T, t = T) = \{S_T - X\}_+$, the PDE propagates the value backwards and yields $C(S, t)$ for all $t \leq T$.

5.10.1 Model hypotheses

The derivation rests on seven hypotheses, which we number for later reference.

- i) The only risk is *market risk* (no credit risk); price fluctuations are driven by supply and demand.
- ii) The stock is assumed to follow the Markov GBM:

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (5.21)$$

- iii) The volatility σ is constant (relaxing this leads to Heston-type models), and the risk-free rate r is constant as well (not a stochastic process).
- iv) No dividends are paid during the life of the contract. (Dividends can be accommodated by shifting the drift, $\mu \rightarrow \tilde{\mu} = \mu + q$ with $q < 0$, because paying dividends reduces the capital of the company.)
- v) The underlying is *infinitely divisible*: one can hold any real fraction $\Delta \cdot S_t$, with

$$\Delta = \frac{\# \text{ shares held}}{\# \text{ shares in the contract}}, \quad \Delta \in \mathbb{R}. \quad (5.22)$$

- vi) *Continuous trading* is possible: the portfolio can be adjusted at every instant.
- vii) **No arbitrage**: it is impossible to realize risk-free profits whose return exceeds r . Equivalently — and this is the operative form we will use — every risk-free investment earns exactly the rate r (*no free lunch*).

Since hypothesis (vii) does all the heavy lifting in what follows, an everyday example may help fix ideas. Suppose oranges cost 9 €/kg in Bologna and 10 €/kg in Miami. A trader buys 1000 kg in Bologna for 9000 €, sells them in Miami for 10 000 €, and gains 1000 € without effort or risk. But many other traders think the same way; their collective buying raises the price in Bologna and their selling lowers it in Miami, and the two prices converge to a single value — essentially instantly in a liquid market. (If they did not, the two markets would be permanently decoupled, which is not what we observe.) Hence the principle: no risk-free profit above r can persist, and its corollary, that all risk-free investments return exactly r .

5.10.2 The hedging portfolio

The genius of the Black–Scholes argument is to look at the problem from the point of view of the *writer*, and to ask how the writer can protect — *hedge* — the position. The writer has sold a call option (European, on a stock) and is therefore exposed: if before maturity the price climbs to $S \gg X$, exercise becomes very probable, and the writer, who does not hold the shares, would have to buy them dearly. The protection is to hold, at all times, a suitable fraction $\Delta \in \mathbb{R}$ of shares — a *dynamic* quantity, continuously adjusted — alongside a cash position $\Pi(S, t)$.

The total *wealth* of the writer’s position, seen from the holder’s side of the ledger, must replicate the option:

$$W(S, t) = O(S, t) = \Delta(S, t) S + \Pi(S, t), \quad (5.23)$$

where $O(S, t)$ is the option price we are trying to determine. Note what this equation says: the option can be *replicated* using only instruments already present in the market (the stock and cash) — this is completeness at work — and by no-arbitrage the replicating portfolio must have the same value as the option itself. Solving for the cash position,

$$\Pi(S, t) = O(S, t) - \Delta(S, t) S, \quad (5.24)$$

which is the money received from selling the option minus the cost of the hedging shares; it is typically negative (a debt), i.e. a *short position*.

5.10.3 Derivation of the Black–Scholes PDE

We now follow the portfolio over an infinitesimal time step and demand that it be riskless. Its variation is $d\Pi(S, t) = dO(S, t) - d[\Delta(S, t) S]$, where S is stochastic, $dS(t) = \mu S(t) dt + \sigma S(t) dW(t)$, and the two terms are computed as follows.

For the first term, Itô’s formula (4.39) applied to $O(S, t)$ gives

$$dO(S, t) = \left(\frac{\partial O}{\partial t} + \mu S \frac{\partial O}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 O}{\partial S^2} \right) dt + \sigma S \frac{\partial O}{\partial S} dW_t. \quad (5.25)$$

For the replicating portfolio the self-financing condition is

$$dO = \Delta dS + r\Pi dt.$$

Rebalancing is funded within the portfolio.¹ Write $dG = dO - \Delta dS$ for the gain on the currently held hedged positions. Self-financing implies $dG = r\Pi dt$; no terms in the product rule are discarded.

Therefore

$$dG = \left(\frac{\partial O}{\partial t} + \mu S \frac{\partial O}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 O}{\partial S^2} \right) dt + \sigma S \frac{\partial O}{\partial S} dW_t - \Delta(S, t) (\mu S dt + \sigma S dW_t). \quad (5.26)$$

Now comes the decisive move, the procedure known as *delta hedging*. The portfolio is risky only through the terms proportional to dW_t ; but their coefficient,

$$\sigma S \frac{\partial O}{\partial S} - \Delta(S, t) \sigma S, \quad (5.27)$$

¹Björk[2], Definition 6.10 and Chapter 8. For the cash value $\Pi = O - \Delta S$, the product rule gives $d\Pi = r\Pi dt - S d\Delta - d[\Delta, S]$. Thus $d\Pi$ cannot be identified with $dO - \Delta dS$ when Δ varies.

is entirely under the writer's control, through the choice of Δ . Choosing

$$\Delta(S, t) = \frac{\partial O(S, t)}{\partial S} \quad (5.28)$$

makes the stochastic terms cancel *exactly*: the market risk has been engineered away, and with it — remarkably — also every occurrence of the drift μ . What survives is deterministic:

$$dG = \left(\frac{\partial O}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 O}{\partial S^2} \right) dt. \quad (5.29)$$

But a riskless portfolio, by hypotheses (vi) and (vii), must earn exactly the risk-free rate, with r constant by (iii):

$$dG = r \Pi(S, t) dt. \quad (5.30)$$

Equating the two expressions and substituting $\Pi(S, t) = O(S, t) - \frac{\partial O}{\partial S} S$, we obtain

$$\boxed{\frac{\partial O}{\partial t} + r S \frac{\partial O}{\partial S} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 O}{\partial S^2} = r O(S, t)} \quad (5.31)$$

the celebrated **Black–Scholes equation**.

Properties of the Black–Scholes equation. A few structural remarks before we solve it. The Black–Scholes equation is a *parabolic, linear homogeneous* PDE, and a suitable change of variables reduces it to the heat equation — the same equation Einstein derived for Brownian motion, now propagating option values instead of pollen grains. Its solution $O(S, t)$, with the appropriate terminal condition, is the price of either a put or a call: it is the *no-arbitrage* price.

The most striking feature of (5.31) is a symbol that is *missing*: the equation contains r , σ , and (through the boundary conditions) X and T , but not μ . Two stocks with the same volatility but wildly different expected returns carry identical option prices. The interpretation is deep: the pricing is done *not* under the historical GBM (with drift μ) but under the *risk-neutral* one,

$$dS = r S dt + \sigma S dW_t, \quad (5.32)$$

in which μ has been replaced by r . Delta hedging removes the market risk, and with it the reward for bearing it; what is left is a world where every asset drifts at the risk-free rate — the risk-neutral world — and it is there that options are priced.

5.11 Risk-neutral valuation and Girsanov's theorem*

The replacement of the historical drift μ by r can be expressed as a change of probability measure. This section supplies the probabilistic framework for the hedging calculation.²

²This is a supplement to the handwritten derivation, not a transcription. The statements and notation follow Björk, fourth edition[2]: Definition 4.7 and Proposition 4.8 (p. 50), Theorem 12.3 (pp. 177–178), Definition 12.4 and Lemma 12.5 (p. 179), and Chapter 11 on arbitrage pricing. Risk-neutral valuation itself is in the course program; Girsanov's theorem is an added explanation.

Information and martingales. Fix a finite horizon T and a filtered probability space

$$(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$$

satisfying the usual conditions. Here $\mathcal{F}_t$ represents all information available at time t , and W is a Wiener process under P . A process X is a martingale if it is adapted, $E^P[|X_t|] < \infty$, and

$$E^P[X_t | \mathcal{F}_s] = X_s, \quad 0 \leq s \leq t \leq T.$$

Conditioning on $\mathcal{F}_s$ includes the entire observed history, rather than just the current value of X . For a progressively measurable integrand g with $E^P[\int_0^T g_s^2 ds] < \infty$, the Itô integral $\int_0^t g_s dW_s$ is a square-integrable martingale. Adaptedness alone does not give this conclusion.

Changing the measure. Let φ be progressively measurable, with $\int_0^T \varphi_s^2 ds < \infty$ almost surely, and define the likelihood process

$$L_t = \exp \left\{ \int_0^t \varphi_s dW_s - \frac{1}{2} \int_0^t \varphi_s^2 ds \right\}, \quad dL_t = \varphi_t L_t dW_t, \quad L_0 = 1. \quad (5.33)$$

Assume $E^P[L_T] = 1$. A sufficient condition is Novikov's condition,

$$E^P \left[\exp \left\{ \frac{1}{2} \int_0^T \varphi_s^2 ds \right\} \right] < \infty.$$

Define Q on $\mathcal{F}_T$ by $dQ/dP = L_T$. Girsanov's theorem[18] then states that

$$W_t^Q = W_t - \int_0^t \varphi_s ds \quad \text{or, equivalently,} \quad dW_t = \varphi_t dt + dW_t^Q \quad (5.34)$$

is a Wiener process under Q . The measures are equivalent: they have the same null events, although they assign different probabilities to other events. The positive exponential is only a local martingale without the expectation condition; it cannot in general be used as a probability density merely because the stochastic integral exists.

Application to the Black–Scholes market. Use Björk's notation B for the bank account in this section, with $dB_t = rB_t dt$ and $B_0 = 1$. For the stock,

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad \mu, r \in \mathbb{R}, \quad \sigma > 0.$$

Set $\varphi = (r - \mu)/\sigma$; Novikov's condition holds for these constant coefficients on $[0, T]$. Substitution in the stock equation gives

$$dS_t = rS_t dt + \sigma S_t dW_t^Q, \quad d(S_t/B_t) = \sigma(S_t/B_t) dW_t^Q. \quad (5.35)$$

The discounted stock is a true Q -martingale in this model. The market price of risk is $\lambda = (\mu - r)/\sigma = -\varphi$. The volatility coefficient and the paths are unchanged; the distribution of those paths changes.

For an integrable replicable claim X paid at T , the valuation formula is

$$\Pi_t[X] = B_t E^Q \left[\frac{X}{B_T} \middle| \mathcal{F}_t \right]. \quad (5.36)$$

Here X and $\Pi_t[X]$ are Björk's claim and price notation, not the strike X and cash holding $\Pi(S, t)$ used in the preceding derivation. For a European payoff $\Phi(S_T)$, the bridge to that notation is

$$O(S_t, t) = \Pi_t[\Phi(S_T)] = e^{-r(T-t)} E^Q[\Phi(S_T) | \mathcal{F}_t].$$

Since S is Markov in this model, this conditional expectation can also be written as a function of S_t and t . Replication gives this formula when the discounted replicating wealth is a martingale; appropriate integrability is part of that assertion.

No arbitrage and completeness. In the general locally bounded semimartingale setting, the precise condition is *no free lunch with vanishing risk*, for admissible strategies whose wealth is bounded below. It is equivalent to the existence of an equivalent *local* martingale measure for discounted prices, not necessarily a true martingale measure; continuous-trading martingale methods were developed by Harrison and Pliska[20].³ For the Brownian Black–Scholes market with the Brownian filtration and nonzero σ , the equivalent martingale measure is unique and the usual square-integrable claims are replicable. The Heston model introduced later has a second, untraded source of risk when $|\rho| < 1$; trading only the stock and bank account does not make that market complete. A volatility risk premium must then be specified to select a pricing measure.

5.12 The Feynman–Kac formula

The Black–Scholes equation is a PDE; the payoff is a terminal condition; what we want is the solution. The key that unlocks it is the Feynman–Kac theorem, a beautiful bridge between partial differential equations and stochastic expectations: it says that solving the PDE is the same as *averaging the payoff over the trajectories of an SDE*. We first present the result in general, then apply it to Black–Scholes.

Statement and conditional expectation. Use Björk’s notation $F(t, x)$ for the deterministic solution and X_s for the stochastic state.⁴ Suppose that X solves

$$dX_s = \mu(s, X_s) ds + \sigma(s, X_s) dW_s, \quad X_t = x,$$

and that $F \in C^{1,2}$ solves the terminal-value problem

$$F_t(t, x) + \mu(t, x)F_x(t, x) + \frac{1}{2}\sigma^2(t, x)F_{xx}(t, x) = 0, \quad F(T, x) = \Phi(x). \quad (5.37)$$

Assume that the SDE is well posed and that $E_{t,x}[\int_t^T (\sigma(s, X_s)F_x(s, X_s))^2 ds] < \infty$. Itô’s formula, evaluated along the stochastic trajectory, gives

$$\Phi(X_T) - F(t, x) = \int_t^T \sigma(s, X_s)F_x(s, X_s) dW_s.$$

The conditional expectation of the integral is zero under the stated square-integrability assumption. Consequently,

$$F(t, x) = E_{t,x}[\Phi(X_T)] = \int \Phi(y)P(y, T | x, t) dy, \quad (5.38)$$

when a transition density exists. This is the Feynman–Kac representation: the solution at (t, x) is the expected terminal value for the process starting there.

³Björk[2], Theorem 11.10 (p. 159); the completeness and uniqueness statement is Theorem 11.17 (p. 163), within the market and claim classes specified there. See also the numeraire valuation formulation in Sections 11.8–11.9.

⁴The argument follows Björk[2], Propositions 5.5–5.6 (pp. 74–75). Relative to the handwritten notation, $F(t, x) = f(x, t)$, $\mu(t, x) = a(x, t)$ and $\sigma(t, x) = b(x, t)$. The coefficients in this paragraph are generic functions, not the constant stock parameters.

Discounting. For a constant r , the PDE $F_t + \mu F_x + \frac{1}{2}\sigma^2 F_{xx} - rF = 0$ remains linear and homogeneous; $-rF$ is a zeroth-order term, not an external source. Applying the same argument to $e^{-rs}F(s, X_s)$, with the corresponding square-integrability condition, yields the discount factor. In the option notation and under the risk-neutral dynamics this reads

$$C(S, t) = e^{-r(T-t)} E^Q[\Phi(S_T) \mid S_t = S]. \quad (5.39)$$

This connects the terminal-value PDE to the valuation formula in Section 5.11.

Application to Black–Scholes. For the Black–Scholes equation the relevant SDE is the risk-neutral GBM, $dS = rS dt + \sigma S dW_t$, and the terminal condition is the call payoff $\Phi(S_T) = \{S_T - X\}_+$. The transition density of the risk-neutral GBM is the log-normal⁵

$$P_{\ln}^N(S_T, T \mid S, t) = \frac{1}{S_T \sqrt{2\pi\sigma^2(T-t)}} \exp\left\{-\frac{\left[\ln \frac{S_T}{S} - (r - \frac{\sigma^2}{2})(T-t)\right]^2}{2\sigma^2(T-t)}\right\}, \quad (5.40)$$

an expression that is extremely awkward to integrate analytically — but not impossible, as the next section shows. The call price is therefore given by the Feynman–Kac representation[23]

$$C(S, t) = e^{-r(T-t)} \int_0^\infty dS_T \{S_T - X\}_+ P_{\ln}^N(S_T, T \mid S, t). \quad (5.41)$$

Everything that remains is one Gaussian integral.

5.13 The Black–Scholes formulas

5.13.1 Solution for the European call

We now evaluate (5.41). The payoff $\{S_T - X\}_+$ vanishes for $S_T \leq X$, so only the region $S_T > X$ contributes; and the log-normal density begs for the substitution that makes it Gaussian. Introduce therefore the standardized variable

$$Q = \frac{\ln S_T - m}{\tilde{\sigma}}, \quad \mathbb{E}[Q] = 0, \quad \text{Var}[Q] = 1, \quad (5.42)$$

where

$$m = \ln S + \left(r - \frac{\sigma^2}{2}\right)(T-t), \quad \tilde{\sigma}^2 = \sigma^2(T-t) \quad (5.43)$$

are the mean and variance of $\ln S_T$ under the risk-neutral measure. Then $\ln S_T = Q\tilde{\sigma} + m$, so $S_T = e^{Q\tilde{\sigma}+m}$, and the density of Q is the standard Gaussian

$$h(Q) dQ = \frac{1}{\sqrt{2\pi}} e^{-Q^2/2} dQ. \quad (5.44)$$

The constraint $S_T > X$ translates into $Q > Q_{\min}$ with $Q_{\min} = \frac{\ln X - m}{\tilde{\sigma}}$, and the payoff splits the integral into two pieces,

$$\int_{Q_{\min}}^{+\infty} dQ (e^{Q\tilde{\sigma}+m} - X) h(Q) = \textcircled{1} - \textcircled{2}, \quad (5.45)$$

⁵Obtained from (5.17) by the replacement $\mu \rightarrow r$.

which we evaluate in turn.

For integral ①, complete the square in the exponent,

$$e^{Q\tilde{\sigma}} e^{-Q^2/2} = e^{-(Q^2 - 2Q\tilde{\sigma} + \tilde{\sigma}^2)/2} e^{\tilde{\sigma}^2/2}, \quad (5.46)$$

and substitute $u = Q - \tilde{\sigma}$, which gives

$$\int_{Q_{\min}}^{+\infty} dQ e^{Q\tilde{\sigma}+m} h(Q) = e^{m+\tilde{\sigma}^2/2} \int_{Q_{\min}-\tilde{\sigma}}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du = e^{m+\tilde{\sigma}^2/2} N(\tilde{\sigma} - Q_{\min}), \quad (5.47)$$

where

$$N(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x dt e^{-t^2/2} \quad (5.48)$$

is the *cumulative standard normal distribution*, the S-shaped function of Figure 5.5, with values in $[0, 1]$. It is not known in closed form, but it is tabulated everywhere and is, in practice, as good as elementary. Note also, from $1 = N(+\infty) = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$, the reflection identity $N(-x) = 1 - N(x)$, which we will use freely.

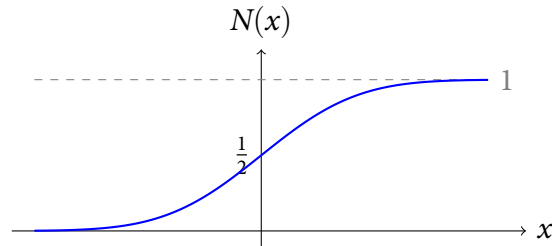


Figure 5.5: The cumulative standard normal distribution $N(x)$, which interpolates from 0 to 1; in the Black–Scholes formulas it plays the role of a probability of exercise.

Integral ② is immediate:

$$\int_{Q_{\min}}^{+\infty} dQ X h(Q) = X N(-Q_{\min}). \quad (5.49)$$

It remains to tidy up the arguments. Define

$$d_2 = -Q_{\min} = \frac{\ln \frac{S}{X} + (r - \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}, \quad (5.50)$$

$$d_1 = d_2 + \sigma\sqrt{T-t} = \frac{\ln \frac{S}{X} + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}. \quad (5.51)$$

Since $e^{m+\tilde{\sigma}^2/2} = S e^{r(T-t)}$, reassembling the pieces with the overall discount factor gives

$$C(S, t) = e^{-r(T-t)} \{ S e^{r(T-t)} N(\tilde{\sigma} - Q_{\min}) - X N(-Q_{\min}) \} = S N(d_1) - X e^{-r(T-t)} N(d_2), \quad (5.52)$$

that is,

$$\boxed{C(S, t) = S N(d_1) - X e^{-r(T-t)} N(d_2)} \quad (5.53)$$

with $d_i = \frac{\ln \frac{S}{X} + (r \pm \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}$, $i = 1, 2$ (plus sign for d_1 , minus for d_2). This is the **Black–Scholes formula for the European call** — the formula that governs the pricing of options on every exchange

in the world. Its structure is telling: the price is the current stock value weighted by $N(d_1)$, minus the discounted strike weighted by $N(d_2)$, and the weights, as we will see in Section 5.14, are probabilities under different numeraires.⁶

5.13.2 Limiting cases

A formula this important should be stress-tested in the regimes where we already know the answer. As $t \rightarrow T$ with $S_T > X$, both $d_i \rightarrow +\infty$, so $N(d_i) \rightarrow 1$ and $C \rightarrow S_T - X$: the payoff is recovered. ✓ As $S \rightarrow 0$, both $d_i \rightarrow -\infty$, so $N(d_i) \rightarrow 0$ and $C(S, t) \rightarrow 0$: a call on a worthless stock is worthless. As $S \rightarrow +\infty$ (deep in the money), $d_i \rightarrow +\infty$, $N(d_i) \rightarrow 1$, and $C(S, t) \approx S - X e^{-r(T-t)} \approx S$: exercise is a near-certainty, and the option is worth nearly the stock itself (one buys at X what one can sell for much more). Figure 5.6 interpolates between these regimes; market jargon distinguishes the option *out of the money* ($S < X$), *at the money* ($S \approx X$), and *in the money* ($S > X$).

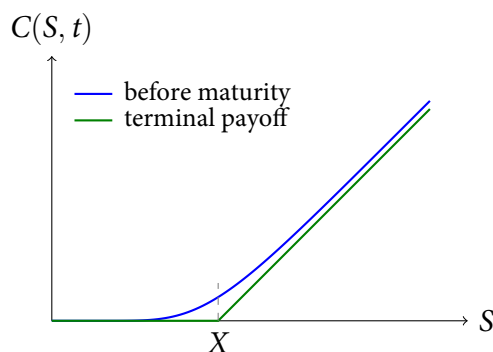


Figure 5.6: The Black–Scholes call value $C(S, t)$ before maturity (blue), smoothing the terminal hockey-stick payoff (green); as $t \rightarrow T$ the curve collapses onto the payoff.

The European put and put-call parity. The same computation with the put payoff $\{X - S_T\}_+$ gives

$$P(S, t) = X e^{-r(T-t)} N(-d_2) - S N(-d_1). \quad (5.54)$$

(The great practical problem of both formulas, to which we return in Section 5.15, is the volatility of the underlying, which is not directly observable.)

Puts and calls are not independent instruments, however. Subtracting the two formulas and using $N(-x) = 1 - N(x)$,

$$P(S, t) - C(S, t) = -S + X e^{-r(T-t)}, \quad (5.55)$$

or equivalently,

$$P(S, t) = C(S, t) - S + X e^{-r(T-t)}. \quad (5.56)$$

This is the *put-call parity*: a deterministic, model-independent relation between the two prices. It is a direct consequence of the no-arbitrage principle — the portfolio “long a call, short a put, short the stock” has a deterministic payoff at maturity, and must therefore earn exactly r — and it does not require GBM, but does require matched European strike and maturity, consistent discounting, and the no-dividend/frictionless assumptions used here. Any quoted pair of put and call prices violating (5.56) is free money, and traders make sure it never lasts.

⁶For a non-dividend-paying stock, $N(d_2) = Q(S_T > X \mid \mathcal{F}_t)$ under the bank-account measure, whereas $N(d_1)$ is the corresponding probability under the stock-numeraire measure. It is also the call delta; it is not the exercise probability under Q . See Björk[2], the chapter on change of numeraire.

5.14 The Greeks

The Black–Scholes price is a function of many variables, and its partial derivatives — universally known as *the Greeks* — measure the sensitivity of the option to each of them. They are the risk-management dashboard of every derivatives desk.

5.14.1 Delta

The first and most important Greek is the hedge ratio itself,

$$\Delta_C(S, t) = \frac{\partial O(S, t)}{\partial S} = \frac{\partial C(S, t)}{\partial S}, \quad (5.57)$$

the quantity of stock the writer must hold at every instant. Differentiating the call formula (5.53),

$$\Delta_C(S, t) = N(d_1) + \left\{ S \frac{\partial N(d_1)}{\partial d_1} \frac{\partial d_1}{\partial S} - X e^{-r(T-t)} \frac{\partial N(d_2)}{\partial d_2} \frac{\partial d_2}{\partial S} \right\}. \quad (5.58)$$

The term in braces looks alive, but it is not: using $N'(d_i) = \frac{1}{\sqrt{2\pi}} e^{-d_i^2/2}$ together with the relation $d_1 = d_2 + \sigma\sqrt{T-t}$, a careful calculation shows that the two contributions cancel identically, leaving the remarkably clean result

$$\Delta_C(S, t) = N(d_1). \quad (5.59)$$

For the put, differentiating the parity relation (5.56) term by term,

$$\Delta_P(S, t) = -(1 - N(d_1)) = N(d_1) - 1. \quad (5.60)$$

The function $N(d_1)$ is approximately a step function centred at X , but broadened by the volatility: deep out of the money the writer holds almost no stock, deep in the money almost the full share, and near the strike the hedge transitions smoothly. Delta also gives the trader's rule of thumb for small price moves,

$$C(S + dS) \approx C(S) + \Delta dS. \quad (5.61)$$

Other Greeks. In the model r and σ are constants; in reality they fluctuate, and their own sensitivities get names too:

$$\rho := \frac{\partial O}{\partial r}, \quad (5.62)$$

$$\Theta := \frac{\partial O}{\partial t} = -\frac{\partial O}{\partial \tau} \quad (\tau = T - t), \quad (5.63)$$

$$\nu := \frac{\partial O}{\partial \sigma} \quad (\text{vega}). \quad (5.64)$$

Rho measures exposure to interest-rate moves, Theta the change with calendar time (not always negative), and Vega — the most watched of the three — the exposure to changes in volatility.

Interpretation of the writer's portfolio. The Greeks make the writer's cash position interpretable. For a **call**, substituting $\Delta_C = N(d_1)$,

$$\begin{aligned}\Pi(S, t) &= C(S, t) - \Delta(S, t) S = S N(d_1) - X e^{-r(T-t)} N(d_2) - N(d_1) S \\ &= -X e^{-r(T-t)} N(d_2).\end{aligned}\tag{5.65}$$

Every factor now speaks: X is the amount the holder will pay the writer at T if the call is exercised; $e^{-r(T-t)}$ discounts that payment from maturity to the present; and $N(d_2)$ is the probability that the exercise actually happens. The writer's debt is exactly the discounted, probability-weighted strike. For a **put**, the same substitution gives

$$\begin{aligned}\Pi(S, t) &= P(S, t) - \Delta_P(S, t) S = X e^{-r(T-t)} N(-d_2) - S N(-d_1) + N(-d_1) S \\ &= X e^{-r(T-t)} N(-d_2),\end{aligned}\tag{5.66}$$

with the mirror-image reading: the writer pays X to the holder and receives the stock upon exercise, and $N(-d_2)$ is the probability of exercise of the put.

5.15 Volatility

The Black–Scholes price $C_{BS} = C_{BS}(S, t, X, T, r, \sigma)$ depends on six quantities. Five of them are either written in the contract (X, T), read off the market (S, t), or proxied by government bonds (r). The sixth, σ , is different: it is the most critical input and the only *hidden* one — no screen displays the volatility. Estimating it is a subject in itself, and there are two philosophies.

5.15.1 Historical volatility

The first philosophy looks backwards: the *historical* (or *realized*) volatility is extracted from a statistical analysis of real time series. The GBM model says that log-returns over a step Δt satisfy

$$d \ln S(t) = \ln \frac{S(t + \Delta t)}{S(t)} = \mu^* \Delta t + \sigma dW_t, \quad \mu^* = \mu - \frac{\sigma^2}{2}, \tag{5.67}$$

where the most common choice is daily returns ($\Delta t = 1$ day), although today one can go below the minute (*high-frequency* data). The relevant moments are

$$\mathbb{E} \left[\ln \frac{S(t + \Delta t)}{S(t)} \right] = \mu^* \Delta t, \tag{5.68}$$

$$\text{Var} \left[\ln \frac{S(t + \Delta t)}{S(t)} \right] = \sigma^2 \Delta t \implies \sigma = \sqrt{\frac{1}{\Delta t} \text{Var}[\dots]}, \tag{5.69}$$

where the first is the historical instantaneous return and the second yields the volatility, with the tell-tale dimension $[\sigma] = t^{-1/2}$ (typically days $^{-1/2}$).

Here, however, we run into the conceptual problem announced in Chapter 3. The expectations above are *ensemble* averages, over infinitely many copies of the market; what we possess is a single time series $S_k, k = 0, \dots, N$ — *one* realization (the same predicament as in meteorology). Time averages equal ensemble averages only for stationary processes, and the market is *not* stationary: both μ^* and σ drift over the years. Strictly speaking, then, the procedure is unjustified;

but no better tool exists, so one proceeds anyway, with eyes open, using the sample estimators

$$\mu^* \longrightarrow m = \frac{1}{\Delta t} \frac{1}{N} \sum_{k=0}^{N-1} \ln \frac{S_{k+1}}{S_k}, \quad (5.70)$$

$$\sigma^2 \longrightarrow S^2 = \frac{1}{\Delta t} \frac{1}{N-1} \sum_{k=0}^{N-1} \left(\ln \frac{S_{k+1}}{S_k} - m \Delta t \right)^2, \quad S = \sqrt{S^2} = \sigma_{\text{historical}}. \quad (5.71)$$

Even then, judgement is required: there is no scientific criterion for the choice of the window N . Traders often take one year of data — go too far back and the market was a different, possibly turbulent, animal. For greater accuracy one can use high-frequency returns, obtaining σ_{hf} . In all cases σ_{hist} depends critically on Δt and N : two analysts with different windows will disagree.

Implied volatility. The second philosophy looks forwards, and inverts the logic. Historical volatility summarizes the past; but options are being traded *now*, at prices the market has already agreed on, and those prices embody the market's collective expectation of *future* volatility. The *implied volatility* is defined by *calibration*: it is the value σ_{implied} that makes the model reproduce the market,

$$C_{\text{market}} = C_{\text{BS}}(S, t, X, T, r, \sigma_{\text{implied}}). \quad (5.72)$$

The Black–Scholes formula cannot be inverted for σ analytically, so one solves

$$C_m - C_{\text{BS}}(S, t, X, T, r, \sigma_{\text{implied}}) = 0 \quad (5.73)$$

numerically, by iteration (e.g. Newton's method) — fast and routine. Implied volatility is thus a market-quoted forecast, reflecting expectation rather than history.

5.15.2 The volatility smile

Implied volatility also hands us a clean, self-contained test of the model. If Black–Scholes were exactly right, options with different strikes X on the same underlying would all imply the *same* σ — the volatility is a property of the stock, not of the contract. Plotting σ_{impl} against X , the data disagree: one finds a roughly parabolic shape, lowest near the current price and rising on both wings, universally known as the **volatility smile** (Figure 5.7).

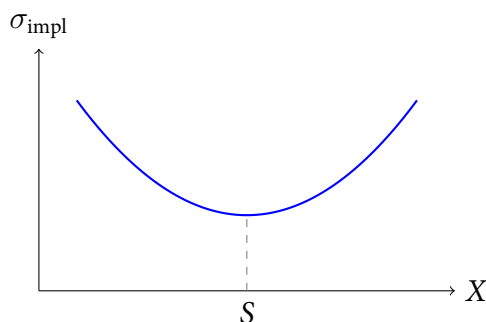


Figure 5.7: The volatility smile: implied volatility as a function of the strike X , minimal near the spot price S and rising on the wings. A constant- σ model would predict a horizontal line.

The smile is the market telling us that the constant- σ , Gaussian world of Black–Scholes underprices extreme events: buyers of far out-of-the-money options (bets on large moves) are willing to pay more than the Gaussian tail would justify, and the excess shows up as higher implied

volatility. The same message appears in the time series: with a truly constant σ , the log-returns $\ln(S_{k+1}/S_k)$ would look like a homogeneous random walk, whereas real returns display *intermittency* — bursts of high volatility separated by calm stretches, the volatility clustering already mentioned. The conclusion is twofold: σ is the quantity that must be watched most carefully, and the Black–Scholes model, while analytically wonderful, has identified its own critical weakness — volatility. Chapter 7 takes up the models that promote σ to a stochastic process of its own.

5.16 The CRR binomial model

The Black–Scholes formula covers European plain vanilla options; for anything more exotic — and for American options in particular — closed formulas rarely exist, and one needs a numerical method. The most elegant one is the *binomial tree* of Cox, Ross, and Rubinstein (1979)[7], the CRR model: a discretization of the GBM so natural that it doubles as an independent derivation of Black–Scholes.

5.16.1 The binomial tree

On the interval $[0, T]$, introduce a time step $\Delta t = T/N$ with $N \gg 1$, chosen by a convergence check, so that time becomes discrete. The brutal simplification — and the source of all the model’s tractability — is this: at each step, the stock price is allowed only *two* moves,

$$S \longrightarrow \begin{cases} S_u = Su & \text{(up move), with probability } p, \\ S_d = Sd & \text{(down move), with probability } 1 - p, \end{cases} \quad d < 1 < u. \quad (5.74)$$

Three unknowns, then: p , u , and d . Three conditions will pin them down, each with a clear meaning.

The first condition matches the *mean*: after one step, the expected stock value must grow at the risk-neutral rate,

$$\mathbb{E}[S(\Delta t)] = p S_u + (1 - p) S_d = p Su + (1 - p) Sd = S e^{r\Delta t}. \quad (5.75)$$

The second condition matches the *variance*: the log-returns over one step must have the GBM variance,

$$\text{Var}\left[\ln \frac{S(t + \Delta t)}{S(t)}\right] = \sigma^2 \Delta t. \quad (5.76)$$

Since $\frac{\Delta S}{S} \approx \ln(1 + \frac{\Delta S}{S})$ for $\Delta S \ll S$, we may work with the two-point distribution

$$\tilde{P}\left(\frac{\tilde{S}}{S}\right) = p \delta\left(\frac{\tilde{S}}{S} - u\right) + (1 - p) \delta\left(\frac{\tilde{S}}{S} - d\right), \quad (5.77)$$

whose variance is computed in two lines:

$$\text{Var}\left[\frac{\Delta S}{S}\right] = p u^2 + (1 - p) d^2 - [p u + (1 - p) d]^2 = p(1 - p)(u - d)^2 = \sigma^2 \Delta t + O(\Delta t^2). \quad (5.78)$$

Equations (5.75) and (5.78) are two conditions for three unknowns; the third is a choice of convenience, but a crucial one. Constant up and down factors already give recombination because $ud = du$. The stronger CRR symmetry choice is

$$ud = 1, \quad (5.79)$$

meaning that an up move followed by a down move returns the price exactly to its starting value. The payoff of this choice is combinatorial: without recombination the tree has 2^N distinct nodes after N steps — computationally hopeless — whereas with it the branches merge (Figure 5.8) and only $\sim N^2$ nodes survive.

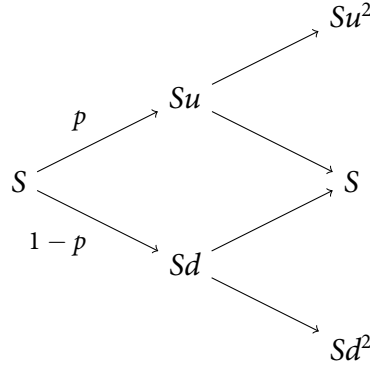


Figure 5.8: The first two steps of a recombining binomial tree ($ud = 1$): the up-down and down-up paths land on the same node, keeping the number of nodes quadratic rather than exponential in the number of steps.

Solving the system: from (5.75),

$$p = \frac{e^{r\Delta t} - d}{u - d}, \quad (5.80)$$

⁷ and the jump sizes are

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}}. \quad (5.81)$$

These are approximate solutions, correct to leading order: expanding $e^{r\Delta t}(u + d) - ud - e^{2r\Delta t}$ in powers of Δt reproduces $\sigma^2\Delta t$, as the variance condition requires. Note the $\sqrt{\Delta t}$ in the exponents — the diffusive scaling of Chapter 3, resurfacing in discrete time.

The bookkeeping of the tree is handled by two indices: the time index $i = 0, \dots, N$ and the up-move count $j = 0, \dots, i$, in terms of which the stock price at node (i, j) is

$$S_{i,j} = S u^j d^{i-j}. \quad (5.82)$$

5.16.2 Option pricing on the binomial tree

To price an option on the tree, we replay the Black–Scholes hedging argument in discrete time. The trader's wealth is $W = \Pi + \Delta S = C$, where $C = C(0) = C(S(0))$ and $\Pi = \Pi(0)$; a price move dS induces dC and $d\Pi$, with $\Delta = \partial C / \partial S$ playing its usual role. Over one step the world splits in two: if S goes up, $C \rightarrow C_u$ and $\Pi \rightarrow \Pi_u$; if down, $C \rightarrow C_d$ and $\Pi \rightarrow \Pi_d$. Risk neutrality of the hedged portfolio demands $\Delta\Pi = \Pi_u - \Pi_d = 0$, i.e.

$$C_u - \Delta S_u = C_d - \Delta S_d \implies \Delta_{\text{CRR}} = \frac{C_u - C_d}{S_u - S_d}, \quad (5.83)$$

the discrete difference quotient replacing the derivative $\partial C / \partial S$ — delta hedging without calculus. The riskless portfolio must then earn the risk-free rate, $\Pi e^{r\Delta t} = (C - \Delta S) e^{r\Delta t}$, which holds precisely when $\Delta = \Delta_{\text{CRR}}$; substituting and simplifying,

$$C_{\text{CRR}} = e^{-r\Delta t} \{p C_u + (1 - p) C_d\}. \quad (5.84)$$

⁷Require $d < e^{r\Delta t} < u$ so that $0 < p < 1$; otherwise this binomial market admits arbitrage or is degenerate.

Pause on the structure of this result: the option price today is the *discounted expectation of tomorrow's prices under the probability p* — and p , from (5.80), is not the real-world probability of an up move but the risk-neutral one, built from r . Equation (5.84) is the Feynman–Kac formula (5.39) in its simplest possible incarnation: one step, two states.

5.16.3 The backward algorithm

Iterating (5.84) from maturity back to the present prices the option; the recipe is called *backward induction*. For a European plain vanilla call it reads:

a) **Initialize:** given X and S , set $T = 1$, N , and $\Delta t = T/N$. Given r and σ_{implied} , compute p , $1 - p$, u , d from (5.80) and (5.81).

b) **Forward sweep** (build the tree): for $i = 1, \dots, N$ and $j = 0, \dots, i$, compute

$$S_{i,j} = S u^j d^{i-j}. \quad (5.85)$$

c) **Terminal payoff:** at maturity ($i = N$), for $j = 0, 1, \dots, N$, set

$$C_{N,j} = \max\{S_{N,j} - X, 0\}. \quad (5.86)$$

d) **Backward sweep** (price the option): for $i = N - 1, \dots, 0$ and $j = 0, \dots, i$, apply

$$C_{i,j} = e^{-r\Delta t} \{p C_{i+1,j+1} + (1 - p) C_{i+1,j}\}. \quad (5.87)$$

The option price is $C_{0,0}$.

A miniature example makes the mechanics concrete. With $N = 3$, the terminal layer $i = 3$ has four payoffs $C_{3,0}, \dots, C_{3,3}$; the layer $i = 2$ combines them pairwise into three values $C_{2,j} = e^{-r\Delta t} \{p C_{3,j+1} + (1 - p) C_{3,j}\}$; the layer $i = 1$ gives two; and the root $i = 0$ delivers the fair price $C_{0,0}$.⁸

5.16.4 Extensions

The real payoff of the binomial framework is how little needs changing to price non-standard contracts; two examples close the chapter.

Barrier options A *barrier option* (of knock-out type) dies if the stock price ever crosses a barrier U during the life of the contract. Set $C_{N,j} = \max(S_{N,j} - X, 0)$ below the barrier and zero at or above it. Then modify the backward sweep at every earlier node,

$$\begin{aligned} i = N - 1, \dots, 0, \quad j = 0, \dots, i, \\ \begin{cases} S_{i,j} \geq U & \implies C_{i,j} = 0, \\ S_{i,j} < U & \implies C_{i,j} = e^{-r\Delta t} \{p C_{i+1,j+1} + (1 - p) C_{i+1,j}\}. \end{cases} \end{aligned} \quad (5.88)$$

This tree checks the barrier only at its monitoring dates; continuous monitoring requires an additional crossing correction or a convergence analysis.⁹

⁸The convergence $C_{\text{CRR}} \rightarrow C_{\text{BS}}$ as $N \rightarrow \infty$ is guaranteed by the central limit theorem: the binomial distribution of the terminal node converges to the Gaussian, and the discrete-time model recovers the continuous-time GBM.

⁹For knock-in barrier options the logic is reversed: the option only comes alive once the barrier is crossed.

American options For an *American* option, the holder may exercise at any time $t \leq T$, and the tree handles this with equal ease. At each node (i, j) the holder faces a choice — exercise now, or hold on — and rationality dictates taking the better of the two:

$$C_{i,j}^A = \max\{S_{i,j} - X, e^{-r\Delta t}[p C_{i+1,j+1}^A + (1-p) C_{i+1,j}^A]\}, \quad (5.89)$$

the maximum of the *exercise value* and the *continuation value* (the discounted no-arbitrage value of keeping the option alive). When $S_{i,j} < X$ the exercise value is nil and immediate exercise is not advantageous, although future exercise opportunities can still affect its value; when $S_{i,j} > X$ early exercise may become optimal, capturing $S_{i,j} - X$ immediately rather than waiting.¹⁰

With options priced — analytically when we can, numerically when we must — we turn in the next chapter to the other great family of financial instruments: bonds, and the stochastic dynamics of interest rates that governs them.

¹⁰For American *calls* on non-dividend-paying stocks with $r \geq 0$, one can show that the continuation value always exceeds the exercise value, so early exercise is never optimal and the American call equals the European one. For American *puts*, by contrast, early exercise can be genuinely optimal.

Interest Rates and Bond Models

Throughout the previous chapter the risk-free interest rate r was a constant — a convenient fiction, and we knew it. In reality interest rates fluctuate, driven by central banks and by the market itself, and their dynamics governs the pricing of *bonds*, the most fundamental fixed-income securities and by volume one of the largest markets in the world. This chapter develops the theory of bond pricing under stochastic interest rates. The plot will feel familiar: as with options, a hedging argument plus no-arbitrage produces a pricing PDE; but there is a new twist — the underlying “asset”, the rate r , is not itself tradable — and resolving it introduces a genuinely new object, the market price of risk. The chapter culminates in the Vasiček model and its complete solution.

6.1 Bond fundamentals

A *bond* is a debt instrument issued by a state or corporation (the *writer*) to an investor (the *holder*). The holder lends capital, and in exchange acquires the right (a) to the reimbursement of the capital lent, (b) with an interest, (c) at a future date T , the *maturity*. As with options, the price of the contract must be *fair*, and the task of the chapter is to say what that means quantitatively.

Constant interest rate. We warm up by assuming $r = \text{const.}$, even though r is in general stochastic. Let $B(t)$ be the value of the bond at time t . Over an infinitesimal step the bond simply accrues interest,

$$B(t)(1 + r dt) = B(t + dt), \quad (6.1)$$

which is nothing but the cost of money. In differential form,

$$\frac{B(t + dt) - B(t)}{B(t)} = \frac{dB(t)}{B(t)} = r dt \implies B(T) = B(t) e^{r(T-t)}, \quad (6.2)$$

the compound-interest law of Section 5.6 again.

In the market, bonds are classified by their maturity: *short-term* (≤ 1 year), *medium-term* (1–5 years), *long-term* (5–10 years), and *very long-term* (10–50 years); holder and writer agree on the maturity at signing. Depending on the issuer one speaks of *government bonds* — by far the most widespread and traded — or *corporate bonds*. For the theory, however, maturity, issuer, and face value are inessential bookkeeping, and it is convenient to normalize them away: for any T , any issuer, and any face value we set

$$B(T) = 1, \quad (6.3)$$

i.e. every bond pays one unit of currency at maturity. With constant r the problem is then completely solved,

$$B(t) = e^{-r(T-t)} ; \quad (6.4)$$

the bond today is worth the discounted unit payment. Everything that follows is about what survives of this formula when r starts to move.

6.2 Stochastic interest rates

The assumption $r = \text{const.}$ is not really tenable. The rate $r(t)$ is a *spot rate* — the rate for instantaneous borrowing, quoted now — and its future values are unknown; by the EMH of the previous chapter, it should itself follow an SDE, $dr(t)$.

Why do rates fluctuate? The causes are of two kinds, and it is worth keeping them apart.

The first kind is *exogenous* to the market but endogenous to the economy: rates are steered at the table of the central banks (FED, BCE, ...) as their principal policy instrument. Two standard scenarios: when inflation is high, the bank *raises* r to encourage saving over spending; when the economy stagnates, it *lowers* r to stimulate consumption and investment. The direction of the next intervention may be guessable, but its timing is not — and every surprise decision instantly modifies the value of all outstanding bonds.

The second kind is *endogenous* to the market, and works in two stages. Bonds are born on the *primary market*, where the issuer places a certain volume with a face value but no fixed price; only a restricted circle of institutions can participate, through a competitive auction mechanism.¹ They then move to the *secondary market*, where small banks and individual investors trade them freely — and there $B(t)$ fluctuates like any other price.

The fluctuations are anything but small. Figure 6.1 shows the US 10-Year Treasury constant-maturity rate — the world's reference “risk-free” rate — which rose from about 4% in the early 1960s to above 14% in the inflationary early 1980s, then declined for four decades toward 1–2% by 2020. A model with constant r cannot even begin to describe such a history.

6.3 The yield to maturity and the yield curve

If the growth rate of a bond is no longer constant, the exponent in $B(T) = B(t) e^{r(T-t)}$ must be replaced by an effective, maturity-dependent rate. One writes

$$B(T) = B(t) e^{R(t,T)(T-t)}, \quad (6.5)$$

where $R(t, T)$ is the *yield to maturity* (rendimento a scadenza): the constant rate that, applied over $[t, T]$, would reproduce the actual growth of the bond. It depends on both the present time t and the maturity T . Using the normalization $B(T) = 1$ and solving for R ,

$$B(t) = e^{-R(t,T)(T-t)} \implies \ln B(t) = -R(t,T)(T-t) \implies R(t,T) = -\frac{1}{T-t} \ln B(t). \quad (6.6)$$

So the market prices of bonds determine $R(t, T)$, and plotting it against the maturity T at fixed t produces the *yield curve* (curva dei rendimenti) — the single most watched graph in fixed income, to which we return in the final section.

¹In Italy, the BOTs (Buoni Ordinari del Tesoro) are placed exactly this way.

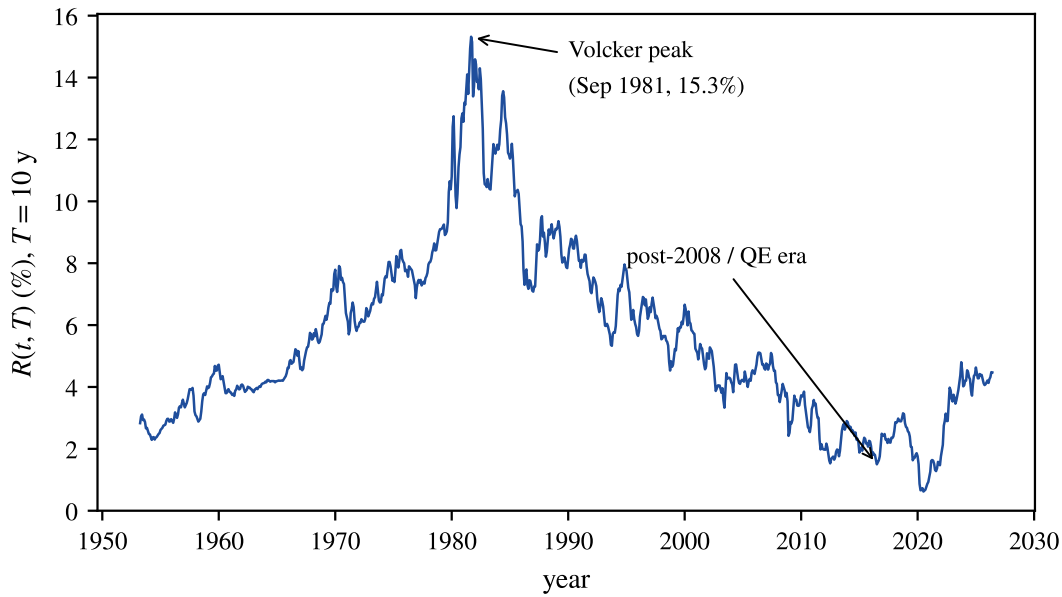


Figure 6.1: US 10-Year Treasury constant-maturity yield, monthly averages, 1953–2026. Data: FRED, series GS10 (Federal Reserve Bank of St. Louis). Generated by `images/code_for_images/treasury_10y.py`.

From the yield to maturity one recovers the instantaneous rate as the short-maturity limit,

$$r(t) := \lim_{(t-T) \rightarrow 0} R(t, T) = \lim_{(t-T) \rightarrow 0} \left(-\frac{1}{T-t} \ln B(t) \right), \quad (6.7)$$

which is formally indeterminate — $B(T) = 1$ makes the numerator vanish together with the denominator — but converges, exactly as a derivative does. The *spot rate* $r(t)$ is the object our stochastic models will describe.

6.4 The Vasiček model for bond pricing

The bond pricing model of Vasiček[42] stands to bonds as Black–Scholes stands to options, and we will construct it along the same lines: hypotheses, hedged portfolio, no-arbitrage, PDE.

Hypotheses.

- (i) There is market risk, but no credit risk.
- (ii) The short rate is assumed to be a Markov Itô diffusion.
- (iii) The no-arbitrage principle holds.
- (iv) The interest rate follows the SDE

$$dr(t) = f(r, t) dt + S(r, t) dW_t, \quad (6.8)$$

with drift and diffusion functions f and S to be specified later.

The parallel with the stock case, $dS(t) = \mu S dt + \sigma S dW_t$, is deliberate, but the analogy must be handled with care: there, the underlying S was a *tradable* asset; here, one cannot buy “one unit of interest rate”. The chain of dependence is $r \rightarrow dr \rightarrow B(r, t; T)$, in parallel with $S \rightarrow dS \rightarrow O(S, t)$.

Let us compute how a bond price responds to the rate. Given $B = B(r, t; T)$, Itô’s formula applied along (6.8) gives

$$dB = \left(\frac{\partial B}{\partial t} + f(r, t) \frac{\partial B}{\partial r} + \frac{1}{2} S^2(r, t) \frac{\partial^2 B}{\partial r^2} \right) dt + S(r, t) \frac{\partial B}{\partial r} dW_t. \quad (6.9)$$

It is convenient to give names to the relative drift and volatility of the bond.² We define

$$\mu(t, T) = \frac{1}{B(r, t; T)} \left(\frac{\partial B}{\partial t} + f(r, t) \frac{\partial B}{\partial r} + \frac{1}{2} S^2(r, t) \frac{\partial^2 B}{\partial r^2} \right), \quad (6.10)$$

$$\sigma(t, T) = -\frac{1}{B(r, t; T)} S(r, t) \frac{\partial B}{\partial r} > 0, \quad (6.11)$$

so that $\mu(t, T)$ is the instantaneous return of the bond and $\sigma(t, T)$ its volatility. The minus sign in the definition of σ is a sign convention chosen to make σ positive, and it encodes a genuine piece of economics: the rate and the bond price are *anticorrelated*. If $dr > 0$, i.e. $r(t + dt) > r(t)$, then $B(r + dr) < B(r)$, so $dB < 0$. A concrete example: let $T = 1$ year and suppose the BCE raises the annual rate from 2% to 4%. Newly issued bonds now yield 4%; the old ones, still promising 2%, are suddenly less attractive, and their market price drops until their effective yield matches.

6.4.1 The hedging portfolio

We now look for the analogue of the delta-hedged portfolio. There is an immediate obstruction: in Black–Scholes we hedged the option *with the underlying stock*, but here the underlying is r , which cannot be held. The resolution is to hedge *one bond with another*: bonds of different maturities respond to the same r , so a suitable combination can be made riskless.

Let $B_i = B(r, t; T_i)$, $\mu_i = \mu(t, T_i)$ and $\sigma_i = \sigma(t, T_i)$. Choose holdings h_1, h_2 and position values $V_i = h_i B_i$ in a self-financing portfolio of value $\Pi = V_2 - V_1$. Then

$$d\Pi = h_2 dB_2 - h_1 dB_1 = (V_2 \mu_2 - V_1 \mu_1) dt - (V_2 \sigma_2 - V_1 \sigma_1) dW_t.$$

The holdings are adjusted so that $V_2 \sigma_2 = V_1 \sigma_1$. For $\sigma_1 \neq \sigma_2$, the value and hedge conditions give

$$V_1 = \frac{\Pi \sigma_2}{\sigma_1 - \sigma_2}, \quad V_2 = \frac{\Pi \sigma_1}{\sigma_1 - \sigma_2}.$$

Consequently the riskless portfolio satisfies

$$d\Pi = \Pi \frac{\mu_2 \sigma_1 - \mu_1 \sigma_2}{\sigma_1 - \sigma_2} dt = r \Pi dt,$$

and hence

$$\sigma_1(\mu_2 - r) = \sigma_2(\mu_1 - r).$$

The adjustable quantities are the holdings, not the quoted bond prices. This is the same self-financing condition used in Chapter 5.

²The notation follows the notes: μ and σ here refer to the bond’s return and volatility, not the stock’s.

Rearranging so that each maturity keeps to its own side,

$$\frac{\mu(t, T_1) - r}{\sigma(t, T_1)} = \frac{\mu(t, T_2) - r}{\sigma(t, T_2)}, \quad (6.12)$$

valid $\forall t \leq \min(T_1, T_2)$ and, crucially, $\forall T_1, T_2$. This is the **Vasiček identity**, a dynamic identity of the model: the excess return per unit of volatility, $\frac{\mu-r}{\sigma}$, is the *same for all bonds*, whatever their maturity — a kind of separation of variables, with the maturity dropping out of the ratio.

The market price of risk. Since the ratio in (6.12) does not depend on the maturity, it defines a function of r and t alone. We give it a name:

$$q(r, t) := \frac{\mu(r, t; T) - r}{\sigma(r, t; T)}, \quad (6.13)$$

the *market price of risk*. It measures the compensation — excess return per unit of volatility — that the market demands for bearing interest-rate risk, i.e. the aggregate risk aversion of bond investors. In Black–Scholes no such object appeared, because hedging with the tradable underlying eliminated the risk premium entirely; here, where r is not tradable, a residue of the market's attitude toward risk survives in the pricing equation.

To obtain that equation, write $q\sigma = \mu - r$ and substitute the definitions of μ and σ ; the terms reassemble into

$$\boxed{\frac{\partial B}{\partial t} + (f + Sq) \frac{\partial B}{\partial r} + \frac{1}{2} S^2 \frac{\partial^2 B}{\partial r^2} = rB} \quad (6.14)$$

with boundary condition $B(r, t = T; T) = 1$. This is the **term structure equation**, the exact analogue of the Black–Scholes equation $\partial_t O(S, t) + \dots = rO$ for bonds.

Several remarks are in order before we solve it. First, note the structural difference from Black–Scholes: there the discounting rate r on the right-hand side was a fixed constant, while here the r multiplying B is the fluctuating state variable itself — the same symbol, two very different roles. Second, the equation is not yet specified: the functional forms of f and S must be modelled, and GBM will *not* do — an interest rate does not grow exponentially (if it did, no state could ever afford to issue bonds). Third, taking q constant is an additional modelling assumption, not a consequence of the ratio defining it. With the bond convention $dB/B = \mu dt - \sigma dW$, the Girsanov kernel of Section 5.11 is $\varphi = q$. Thus $dW = q dt + dW^Q$ and the short-rate drift under Q is $f + Sq$. Finally, the Feynman–Kac formula of Chapter 5 offers an alternative representation of the solution,

$$B(r, t; T) = E^Q \left[\exp \left\{ - \int_t^T r(s) ds \right\} \mid r(t) = r \right], \quad (6.15)$$

a discounted expectation as before — except that now $r(s)$ is stochastic, so the discount factor sits *inside* the expectation, integrated along each trajectory. One could evaluate this average directly, but we will follow the PDE route instead.

6.5 The Vasiček solution

6.5.1 The Ornstein–Uhlenbeck model for the interest rate

To make the term structure equation concrete we must choose the model

$$dr(t) = f(r, t) dt + S(r, t) dW_t, \quad (6.16)$$

i.e. commit to functional forms for f and S . What do we require of a sensible rate model? That for long times r neither explodes nor collapses, but hovers around a stable level with small fluctuations — exactly the behaviour of the mean-reverting process we solved in Chapter 4. Vasiček's choice is indeed the Ornstein–Uhlenbeck process[41], shifted to revert to a nonzero level: the (essentially unique, by Doob's theorem) Markov, Gaussian, stationary process. Its Itô SDE is

$$dr(t) = \alpha[\theta - r(t)]dt + \sigma dW_t, \quad (6.17)$$

with $\alpha, \theta, \sigma > 0$ and $[\alpha] = [t]^{-1} = \tau^{-1}$. Here θ is the *long-run mean* toward which the rate is pulled, and we note already that θ is *not* simply the historical mean of the observed time series.³

Although the solution parallels Chapter 4, we spell it out, both because the nonzero mean changes the formulas and because the notes' derivation is compact. Apply Itô's lemma to the function $g(r, t) = r e^{\alpha t}$, for which $\partial_r g = e^{\alpha t}$ and $\partial_t g = \alpha r e^{\alpha t}$:

$$dg(r, t) = [\alpha r e^{\alpha t} + \alpha(\theta - r) e^{\alpha t}]dt + \sigma e^{\alpha t} dW_t = \alpha\theta e^{\alpha t} dt + \sigma e^{\alpha t} dW_t; \quad (6.18)$$

the mean-reverting drift has been absorbed, by the integrating factor, and only an inhomogeneous source survives. Integrating from t to T (where T is a sufficiently long time),

$$r(T) e^{\alpha T} - r(t) e^{\alpha t} = \alpha\theta \int_t^T e^{\alpha s} ds + \sigma \int_t^T e^{\alpha s} dW_s, \quad (6.19)$$

and solving for $r(T)$,

$$r(T) = \theta(1 - e^{-\alpha(T-t)}) + r(t) e^{-\alpha(T-t)} + \sigma e^{-\alpha T} \int_t^T e^{\alpha s} dW_s. \quad (6.20)$$

The terms have a direct interpretation: the initial rate decays away exponentially, the long-run mean θ fades in to replace it, and the noise accumulates with exponentially fading memory.

The moments follow at sight. The Itô integral averages to zero, so

$$\mathbb{E}[r(T) | r(t)] = \theta(1 - e^{-\alpha(T-t)}) + r(t) e^{-\alpha(T-t)} \xrightarrow{T-t \gg \tau} \theta, \quad (6.21)$$

and the mean converges to θ for long times — the long-run economic average. By the Itô isometry,

$$\text{Var}[r(T) | r(t)] = \sigma^2 e^{-2\alpha T} \int_t^T e^{2\alpha s} ds = \frac{\sigma^2}{2\alpha} (1 - e^{-2\alpha(T-t)}) \xrightarrow{T-t \gg \tau} \frac{\sigma^2}{2\alpha}, \quad (6.22)$$

so the fluctuations saturate rather than growing without bound. The process being Gaussian at all times,

$$r(T) | r(t) \sim \mathcal{N}\left(\theta + (r(t) - \theta)e^{-\alpha(T-t)}, \frac{\sigma^2}{2\alpha}(1 - e^{-2\alpha(T-t)})\right), \quad (6.23)$$

$$r(T) | r(t) \xrightarrow[T-t \rightarrow \infty]{\text{law}} \mathcal{N}\left(\theta, \frac{\sigma^2}{2\alpha}\right).$$

while for short times the mean expands as $\mathbb{E}[r(T) | r(t) = r] = r + \alpha(\theta - r)(T - t) + O((T - t)^2)$

The qualitative behaviour has a name that finance uses constantly: *mean reversion*. Reading the SDE over an infinitesimal step,

$$dr(t) = r(t + dt) - r(t) = \alpha[\theta - r(t)]dt + \sigma dW_t, \quad (6.24)$$

³Under the historical measure, θ is the long-run mean and can be estimated from sufficiently long rate data. The pricing parameter is the risk-adjusted mean θ^* defined below. See Section 6.6.

the deterministic part acts like a restoring force: whenever the rate sits above θ it is pushed down, whenever below, up, and the trajectory keeps oscillating around the mean (Figure 6.2). Taking the ensemble average of the increment,

$$\mathbb{E}[r(t+dt) - r(t)] = \alpha[\theta - \mathbb{E}[r(t)]] dt, \quad (6.25)$$

which identifies α as the *speed of mean reversion*: it quantifies how fast the rate returns to θ after a fluctuation. In simulations one averages finitely many trajectories; this estimates the ensemble relaxation with sampling error.

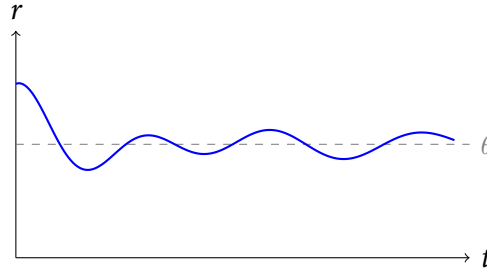


Figure 6.2: Mean reversion in the Vasicek model: the rate $r(t)$ oscillates around the long-run mean θ , pulled back by the restoring drift at speed α .

The model has one well-known blemish. Being Gaussian with unbounded support, the rate can fluctuate to *negative* values, especially at long times if θ is small. In principle this should not happen; in practice, remarkably, bonds with negative yields *have* existed — marginally around 2008, then systematically in Europe and Japan — so the blemish is partly a feature.⁴

An alternative is the nonnegative CIR model[6]. The construction in the handwritten notes starts from a Gaussian OU variable x , whose square is $r = x^2$:

$$dx = -\gamma x dt + \delta dW_t. \quad (6.26)$$

The variable x is signed; it cannot be identified with the nonnegative square root $\sqrt{r}$. Itô's formula gives

$$dr = (\delta^2 - 2\gamma r) dt + 2\delta x dW_t = \alpha(\theta - r) dt + \sigma\sqrt{r} d\tilde{W}_t,$$

where $\alpha = 2\gamma$, $\theta = \delta^2/(2\gamma)$, $\sigma = 2\delta$, and $d\tilde{W}_t = \text{sgn}(x_t)dW_t$ (take the sign at zero to be 1). The process $\tilde{W}$ is Brownian, since its quadratic variation is t . This construction gives a special CIR parameter choice, not the general model.⁵

Yet another route to positivity is the *exponential Vasicek* model, obtained by modelling the *logarithm* of the rate as an OU process:

$$Y(t) = \ln r(t) \implies r(t) = \exp\{Y(t)\} > 0, \quad (6.27)$$

so that r is log-normal — approximately what one sees in the data.

⁴Negative yields can be read as an insurance premium: investors accept a small guaranteed loss — a contained loss, compared with other assets — in exchange for safety; central-bank policy did the rest. For example, the Swiss National Bank set its sight-deposit rate at -0.75% in January 2015; longer-dated Swiss government bonds also traded at negative yields. See the SNB's October 2016 policy discussion[40].

⁵The general CIR diffusion permits independent positive α , θ , σ . Starting from $r > 0$, the Feller condition $2\alpha\theta \geq \sigma^2$ ensures that zero is not reached; otherwise zero can be reached, although the process remains nonnegative. The squared-OU construction above does not satisfy this condition. The finite-time transition law is a scaled noncentral chi-squared distribution, and the stationary law is Gamma, with shape $2\alpha\theta/\sigma^2$ and rate $2\alpha/\sigma^2$ [6, 2].

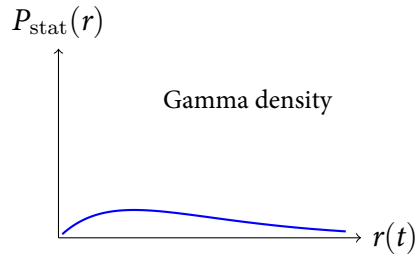


Figure 6.3: An illustrative stationary CIR Gamma density, with shape 2 and rate 1 in scaled rate units. These parameters describe a general CIR model, not the special squared-OU construction above.

6.5.2 Solution of the term structure equation

We now return to the term structure equation (6.14) and solve it with the OU specification. Assuming constant q and substituting $f = \alpha(\theta - r)$ and $S = \sigma$, the drift correction combines with the market price of risk into

$$f + S q = \alpha(\theta^* - r), \quad \theta^* = \theta + \frac{\sigma q}{\alpha}, \quad (6.28)$$

so the whole effect of risk aversion is to shift the long-run mean from θ to the *risk-adjusted* value θ^* — an economical piece of bookkeeping. The problem is then

$$\frac{\partial B}{\partial t} + \alpha(\theta^* - r) \frac{\partial B}{\partial r} + \frac{1}{2} \sigma^2 \frac{\partial^2 B}{\partial r^2} = r B, \quad B(T) = 1. \quad (6.29)$$

The equation is linear in r in its coefficients, which suggests an exponential-affine *ansatz*:

$$P(r, t; T) = A(t, T) e^{-r B_0(t, T)}, \quad (6.30)$$

with terminal conditions $A(T, T) = 1$ and $B_0(T, T) = 0$ (so that $P(r, T; T) = 1$ as required). The derivatives of the *ansatz* are

$$\partial_t P = \dot{A} e^{-r B_0} - r \dot{B}_0 A e^{-r B_0}, \quad \partial_r P = -A B_0 e^{-r B_0}, \quad \partial_r^2 P = A B_0^2 e^{-r B_0}, \quad (6.31)$$

and substituting into the PDE and dividing through by $A e^{-r B_0}$,

$$\frac{\dot{A}}{A} - r \dot{B}_0 - \alpha(\theta^* - r) B_0 + \frac{1}{2} \sigma^2 B_0^2 = r. \quad (6.32)$$

Now comes the decisive observation: this must hold *for every value of r* , and r appears only linearly. The coefficient of r and the r -independent part must therefore vanish *separately* — the PDE splits into two ordinary differential equations.

Collecting the terms linear in r (including the right-hand side),

$$-\dot{B}_0 + \alpha B_0 - 1 = 0 \implies \dot{B}_0 = \alpha B_0 - 1, \quad (a)$$

and collecting the constant terms,

$$\frac{\dot{A}}{A} - \alpha \theta^* B_0 + \frac{1}{2} \sigma^2 B_0^2 = 0. \quad (b)$$

Equation (a) is a linear first-order ODE. Its general solution is $B_0(t, T) = C e^{\alpha t} + \frac{1}{\alpha}$, and imposing $B_0(T, T) = 0$ fixes $C = -\frac{1}{\alpha} e^{-\alpha T}$, whence

$$B_0(t, T) = \frac{1}{\alpha} (1 - e^{-\alpha(T-t)}). \quad (6.33)$$

Note the two limits: $B_0 \approx T - t$ for short maturities (the bond behaves like $e^{-r(T-t)}$, as it must), while $B_0 \rightarrow 1/\alpha$ for long maturities — mean reversion caps the sensitivity of long bonds to the current rate.

With B_0 known, equation (b) determines A by separation of variables:

$$\frac{dA}{A} = \left[\alpha \theta^* B_0 - \frac{1}{2} \sigma^2 B_0^2 \right] dt, \quad (6.34)$$

with $A(T, T) = 1$, i.e. $\ln A(T, T) = 0$. Integrating from t to T ,

$$\ln A(t, T) = \int_t^T \left[-\alpha \theta^* B_0(s, T) + \frac{1}{2} \sigma^2 B_0^2(s, T) \right] ds, \quad (6.35)$$

and carrying out the (elementary if tedious) integrals yields

$$\ln A(t, T) = \left(\theta^* - \frac{\sigma^2}{2\alpha^2} \right) [B_0(t, T) - (T - t)] - \frac{\sigma^2}{4\alpha} B_0^2(t, T), \quad (6.36)$$

which passes the sanity check $\ln A(T, T) = 0$, since both $B_0(T, T)$ and $T - T$ vanish.

Assembling the pieces, the Vasiček bond price is

$$P(r, t; T) = A_{\text{Vas}}(t, T) e^{-r B_0(t, T)} =: P_{\text{Vas}}(r, t; T; \alpha, \theta^*, \sigma), \quad (6.37)$$

an explicit closed formula — the bond-market counterpart of the Black–Scholes formula.

6.6 Calibration

A pricing formula is only useful once its parameters are pinned to the market, and the Vasiček model has three: α , σ , and θ^* . The yield to maturity gives us the handle. On the empirical side,

$$R(t, T) = -\frac{1}{T-t} \ln P(r, t; T) \quad \longrightarrow \quad \text{measured from quoted bond prices}, \quad (6.38)$$

while on the theoretical side the model predicts

$$R_v(r, t; T; \alpha, \theta^*, \sigma) = -\frac{1}{T-t} \ln P_v(r, t; T; \alpha, \theta^*, \sigma). \quad (6.39)$$

The pricing mean θ^* must be inferred from market prices as well as historical dynamics.⁶

The calibration proceeds in two steps, each using the data best suited to it. First, one fits α and σ — the dynamical parameters — from the historical time series of the spot rate $r(t)$, by a best fit of the observed relaxation and fluctuations. Second, one fits θ^* — the risk-adjusted level, invisible in the history because it contains the market price of risk — from today's *yield curve*: at the observed maturities (1, 2, 3, 4, 5, 10, 20 years, ...) one imposes

$$R_{\text{emp}}(t, T) - R_{\text{the}}(r, t; T; \alpha, \sigma, \theta^*) = 0 \quad (6.40)$$

and fits θ^* , generally by minimizing the discrepancies over maturities; one parameter need not match every quoted yield exactly. Figure 6.4 illustrates an upward-sloping model curve with synthetic observations; it is not an empirical calibration.

⁶The phrase “tirato fuori dal mercato” is a paraphrase in the handwritten lecture notes, not a verified quotation from Björk. For the distinction between historical dynamics and pricing dynamics, see Björk[2], the chapters on short-rate models.

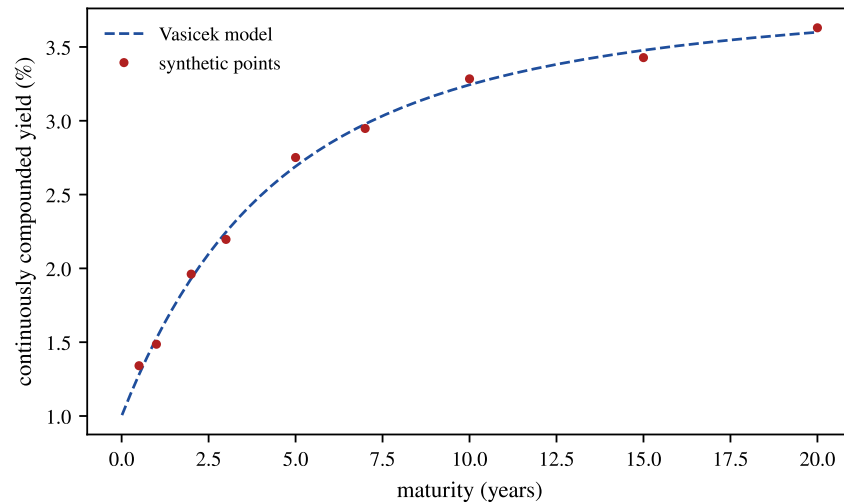


Figure 6.4: Illustrative Vasicek yield curve ($r = 1\%$, $\alpha = 0.4$, $\theta^* = 4\%$, $\sigma = 1\%$ in annual units), with synthetic perturbed points. No market data were fitted.

With all parameters fixed, the model prices not only the bonds themselves but the derivatives written on them: in the market one actively trades *bond options*, whose valuation depends on exactly the parameters just calibrated. The interplay between an analytically solvable model and a market calibration is the same we met for Black–Scholes — and, as there, the model’s Gaussian simplicity is both its strength and its limitation. The next chapter confronts those limitations head-on.

Advanced Financial Models

The Black–Scholes framework rests on two statistical pillars: log-returns are Gaussian, and volatility is constant. Chapter 5 already hinted, through the volatility smile, that the market itself does not believe either. In this chapter we confront the model with the data and repair it in two directions. First, we introduce the *stable distributions* of Lévy, whose fat tails describe the empirical statistics of returns far better than the Gaussian; second, we promote the volatility to a stochastic process of its own, obtaining the *stochastic volatility* models — Stein–Stein, Heston, and their relatives — that are the workhorses of modern quantitative finance. The chapter closes by redoing Black–Scholes in the Heston world, where a new and instructive obstacle appears: volatility cannot be hedged with the stock alone.

Before any statistics, Figure 7.1 explains both why the GBM conquered finance and why this chapter is necessary. The upper panel shows ten years of the real S&P 500; the lower panel shows a geometric Brownian motion simulated with the *same* drift and the *same* volatility, estimated from those very data. At a glance the two are remarkably alike — same trend, same jitter, same look at every scale — which is precisely the mimicry that made the GBM the default model of Chapter 5. Look longer, though, and the real series gives itself away: the cliff of February–March 2020, when the index lost a third of its value in four weeks with single days beyond -12% , has no counterpart in the simulated panel in this realization; in a Gaussian model such a large daily loss has extremely small probability under the fitted Gaussian model. Everything that follows in this chapter is, in one way or another, the statistics of that difference.

7.1 Empirical properties of log-returns

Let us first recall what the GBM predicts, so that we know exactly what to test. For $dS = \mu S dt + \sigma S dW_t$, the log-returns satisfy

$$\mathbb{E}\left[\ln \frac{S(t + \Delta t)}{S(t)}\right] = \mu^* \Delta t, \quad \mu^* = \mu - \frac{\sigma^2}{2}, \quad \text{Var}\left[\ln \frac{S(t + \Delta t)}{S(t)}\right] = \sigma^2 \Delta t, \quad (7.1)$$

and shifting the time window, $t \rightarrow t + \tau$, $t + \Delta t \rightarrow t + \Delta t + \tau$, changes nothing: the returns are stationary, consistent with $t_j - t_i = (t_j + \tau) - (t_i + \tau)$. For the price itself the model gives

$$\mathbb{E}[S(t)] = S_0 e^{\mu t}, \quad \text{Var}[S(t)] = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1), \quad (7.2)$$

although in reality μ itself changes over the years — a first warning sign.

The decisive test is the histogram. Take a real time series, compute the log-returns, extract the empirical mean and standard deviation $\mu_e \leftrightarrow m$, $\sigma_e \leftrightarrow s$, and superimpose on the histogram

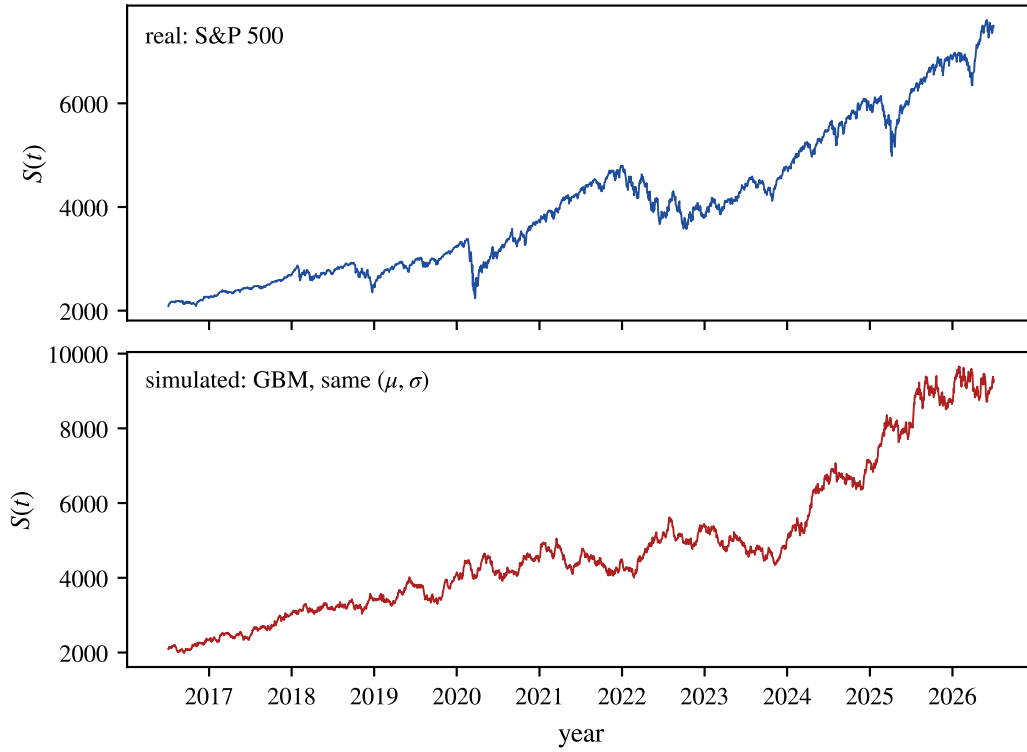


Figure 7.1: Top: the S&P 500, daily closes 2016–2026 (FRED series SP500). Bottom: a geometric Brownian motion simulated with the same starting value and the same daily drift and volatility estimated from the data. The two are nearly indistinguishable by eye — except for the 2020 crash in the real series, which is much more extreme than typical moves in the fitted Gaussian model. Generated by `images/code_for_images/sp500_returns.py`.

the Gaussian that the model predicts,

$$\text{PDF}\{\ln r_{\Delta t}(t)\} \approx \mathcal{N}(\mu_e, \sigma_e^2) = \frac{1}{\sqrt{2\pi\sigma_e^2}} \exp\left\{-\frac{(\ln r_{\Delta t}(t) - \mu_e)^2}{2\sigma_e^2}\right\}. \quad (7.3)$$

The fit fails, and it fails in a characteristic way, shown on real data in Figure 7.2: the empirical distribution is far more peaked at the centre and, above all, far heavier in the tails. Extreme returns — crashes and rallies — occur orders of magnitude more often than any Gaussian allows. The returns are *not* normally distributed, and we need a family of distributions built to accommodate this.

7.2 Stable distributions

The appropriate family was identified by Lévy, and imported into finance by Mandelbrot: the *stable distributions*, whose theory is also known as the *generalized Central Limit Theorem* (generalized CLT), with contributions by Gnedenko. Why “generalized”? The ordinary CLT says that sums of i.i.d. variables with *finite variance* converge to the Gaussian; the generalized version asks the bolder question — what are *all* the possible limits of such sums, finite variance or not? — and the answer is precisely the stable family.

A stable distribution is characterized by four interlocking properties, which we examine in turn: stability under convolution, a closed-form characteristic function, power-law asymptotic behaviour, and the corresponding cumulative distribution function (CDF).

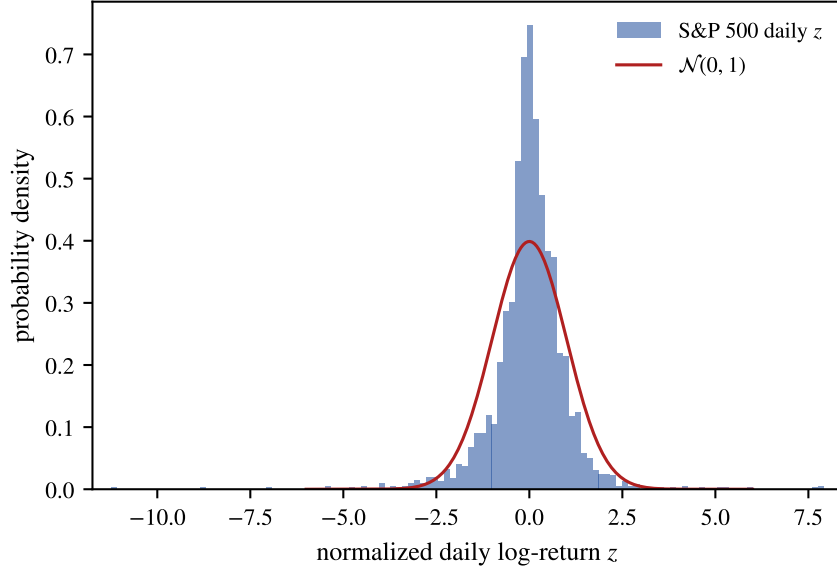


Figure 7.2: The histogram test on real data: normalized daily log-returns of the S&P 500 (2016–2026, FRED series SP500) against the Gaussian with the same mean and variance. The empirical distribution has a sharper central body and far heavier tails — returns beyond $\pm 4\sigma$, rare under the Gaussian, are clearly populated. Generated by `images/code_for_images/sp500_stylized.py`.

7.2.1 Stability

Let X_1, X_2 be i.i.d. random variables with PDF $P(x)$, so that $P_2(x_1, x_2) = P(x_1)P(x_2)$, and consider their sum $S_2 = X_1 + X_2$ (so that $X_2 = S_2 - X_1$). The PDF of the sum is the convolution

$$P_2(S_2) = \int dx P(x) P(S_2 - x) = P * P. \quad (7.4)$$

Convolutions are unwieldy; Fourier transforms turn them into products. Defining the *characteristic function* $\tilde{\phi}(k) = \mathcal{F}[P]$,

$$\mathcal{F}[P_2(S_2)] = \tilde{\phi}_2(k) = \int dS_2 e^{ikS_2} P_2(S_2) = \mathcal{F}[P * P] = \mathcal{F}[P] \cdot \mathcal{F}[P] = \tilde{\phi}^2(k), \quad (7.5)$$

and for N i.i.d. variables $X_1, \dots, X_N$ with $S_N = \sum_{i=1}^N X_i$,

$$\mathcal{F}[P_N(S_N)] = \tilde{\phi}^N(k), \quad (7.6)$$

with no general implication that the distribution is heavy-tailed. A distribution is called *stable* when this operation does not change its shape: the sum has the same distribution as the summands, up to a rescaling and a location shift (the centred symmetric cases below need no shift).

The Gaussian is the prototype. For $P_0(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/(2\sigma^2)}$, with scale parameter σ , the characteristic function is

$$\tilde{\Phi}(k) = \int dx e^{ikx} P_0(x) = \exp\left\{-\frac{\sigma^2 k^2}{2}\right\}, \quad (7.7)$$

so that

$$\tilde{\phi}_2(k) = \mathcal{F}[P * P] = e^{-\sigma^2 k^2} \implies \mathcal{F}^{-1}[\tilde{\phi}_2(k)] = P_0(x, N=2) : \quad (7.8)$$

the sum of two Gaussians is again a Gaussian — the Gaussian is stable. How does the scale grow? Rewriting the $N = 2$ result,

$$P_0(x, N = 2) = \frac{1}{\sqrt{2\pi(2\sigma^2)}} \exp\left\{-\frac{x^2}{2 \cdot 2\sigma^2}\right\} = \frac{1}{\sqrt{2}} P_0\left(\frac{x}{\sqrt{2}}\right), \quad (7.9)$$

and in general

$$P_N(x, N) = \frac{1}{a_N} P_0\left(\frac{x}{a_N}\right), \quad a_N = \sqrt{N} = N^{1/\alpha}, \quad \alpha = 2, \quad (7.10)$$

where the a_N are rescaling factors. The exponent α appearing in $a_N = N^{1/\alpha}$ is the *characteristic exponent* (or *Lévy index*) of the distribution; for the Gaussian, $\alpha = 2$, which is just the familiar $\sqrt{N}$ growth of a random walk. As we now see, other values of α are possible — and each governs a different, self-consistent statistical world.

The Cauchy distribution. The most instructive non-Gaussian member of the family is the *Cauchy* distribution (the Lorentzian of spectroscopy), which realizes $\alpha = 1$:

$$P_c(x) = \frac{\delta}{\pi} \frac{1}{\delta^2 + x^2}, \quad \delta = \text{scale parameter}. \quad (7.11)$$

Its characteristic function is the two-sided exponential

$$\tilde{\phi}(k) = \mathcal{F}[P_c(x)] = e^{-\delta|k|}, \quad (7.12)$$

and stability is verified at sight:

$$\tilde{\phi}_2(k) = e^{-2\delta|k|} \implies P_c(x, N = 2) = \frac{2\delta}{\pi} \frac{1}{4\delta^2 + x^2} = \frac{1}{a_2} P_c\left(\frac{x}{a_2}\right), \quad (7.13)$$

a Cauchy again, with rescaling $a_N = N^{1/\alpha}$, $\alpha = 1$: the scale of the sum grows *linearly* in N , much faster than the Gaussian $\sqrt{N}$. Asymptotically $P_c(x) \sim 1/x^{1+\alpha} = 1/x^2$ for $|x| \rightarrow \infty$, and this slow power-law decay has a startling consequence: the Cauchy distribution has *no finite mean and no finite variance*. It is a pathological but deeply instructive object — a perfectly legitimate, normalizable distribution for which the standard toolkit of means and variances simply does not apply. The family of stable laws interpolates between these behaviours, guided by the index α .

General α -stable distributions. For a general symmetric α -stable distribution the characteristic function is

$$\tilde{\phi}_\alpha(k) = \exp\{-\delta|k|^\alpha\}, \quad \forall \alpha \text{ admissible and } \delta \text{ (scale)}, \quad (7.14)$$

an exponential of $-|k|^\alpha$ for $\alpha \in (0, 2)$. The two cases already met are recovered as $\delta = \frac{\sigma^2}{2}$, $\alpha = 2$ (Gaussian) and $\delta > 0$, $\alpha = 1$ (Cauchy). The admissible range is $0 < \alpha \leq 2$, and within it the moments deteriorate progressively: for $1 < \alpha < 2$ the variance does not exist; for $0 < \alpha \leq 1$ even the mean is gone. The distributions retain the scale $\delta^{1/\alpha}$; it is their asymptotic power-law tails that are scale-free. The density itself is recovered by inverse Fourier transform,

$$P_\alpha(x) = \frac{1}{2\pi} \int dk e^{-ikx} e^{-\delta|k|^\alpha}, \quad \int dx P_\alpha(x) = 1, \quad (7.15)$$

which has closed form only in special cases ($\alpha = 2$, $\alpha = 1$), but whose properties are known in general.

7.2.2 Asymptotic behaviour and CDF

The third property is the one that matters most for data analysis: for every $\alpha < 2$, the tails of the density decay as a *power law*,

$$P_\alpha(x) \propto \frac{C_\alpha}{|x|^{1+\alpha}}, \quad |x| \rightarrow \infty. \quad (7.16)$$

In practice one works with the fourth property, the cumulative distribution,

$$P_{<}(x) = \int_{-\infty}^x P(x') dx', \quad P_{>}(x) = \int_x^{+\infty} P(x') dx' = 1 - P_{<}(x), \quad (7.17)$$

because integrating the histogram tames the statistical noise in the tail. Writing $\alpha' = \alpha + 1$ for the density exponent (so that $1 < \alpha' < 3$ describes the non-Gaussian stable tails), the tail of the CDF follows by direct integration:

$$P_{>}^*(x) = \int_x^{+\infty} dx' P_\alpha(x') \xrightarrow{|x| \rightarrow \infty} C_\alpha \int_x^\infty dx' x'^{-\alpha'} = \frac{C_\alpha}{\alpha' - 1} x^{-(\alpha' - 1)} = \frac{C_\alpha}{\alpha} x^{-\alpha}. \quad (7.18)$$

Taking logarithms of both sides,

$$\ln P_{>}^*(x) = \ln \frac{C_\alpha}{\alpha} - \alpha \ln x, \quad (7.19)$$

so in a *log-log plot* the tail is a straight line of slope $-\alpha$: this is the standard experimental signature, and the standard way of measuring α from data (Figure 7.3). The contrast with the Gaussian could not be sharper: $\ln P_0(x) \sim -x^2$ plunges quadratically, while $\ln P_\alpha(x) \sim -\alpha \ln x$ descends along a gentle straight line — the fat tail.

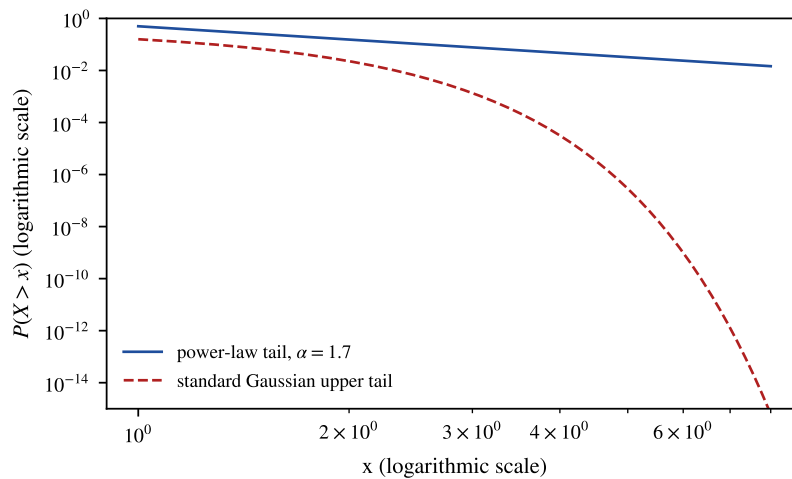


Figure 7.3: Tail of the cumulative distribution in a log-log plot: a stable law gives a straight line of slope $-\alpha$, while the Gaussian bends down much faster (its leading log-tail is quadratic in x , not in $\ln x$). The straight-line tail is the experimental fingerprint of fat tails.

7.2.3 Empirical evidence

How do real markets score on this test? The first measurement is due to Mandelbrot, who studied the daily log-returns of cotton prices[26] and found a beautiful power-law tail with $\alpha \approx 1.7$;

Fama repeated the analysis on stock returns with similar results[13] (empirical analyses of asset prices for which Fama would receive the 2013 Nobel Prize in Economics). But a value $\alpha = 1.7 < 2$ is not acceptable at face value: it implies *infinite variance*, hence in particular an undefined volatility — awkward, for the quantity on which all of Chapter 5 was built.

The modern reference measurement is that of Mantegna and Stanley [28], who analysed the S&P 500 at $\Delta t = 1$ minute over 1984–1989, using the standardized variable

$$z = \frac{\frac{\Delta S_{\text{SP}}(t)}{S(t)} - \mu}{\sigma}, \quad (7.20)$$

and found, once again, a stable-law behaviour with $\alpha \approx 1.4$.

The test is easy to repeat on modern data, and Figure 7.4 does so on ten years of daily S&P 500 closes. The empirical tail lies well above the fitted Gaussian in the extreme-return range. The line of slope -3 is a reference guide, not an exponent fitted to this dataset. A slope of -3 in the CCDF would correspond to a density tail proportional to $|z|^{-4}$, outside the non-Gaussian stable range $\alpha < 2$.

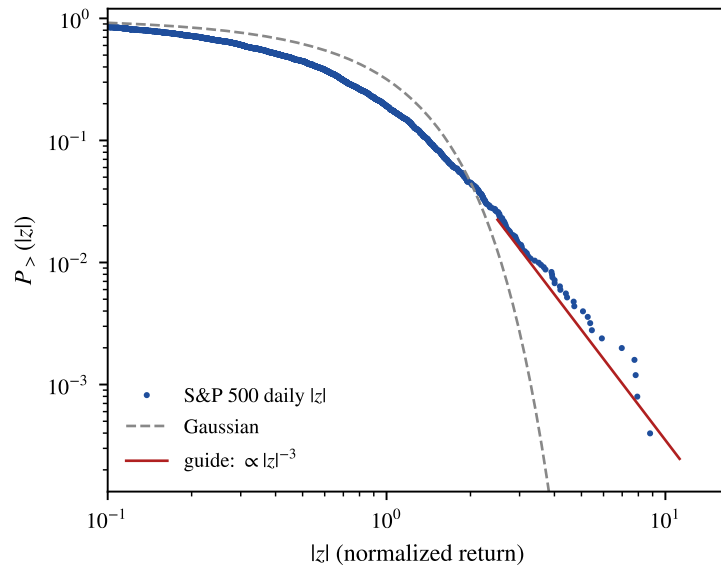


Figure 7.4: Complementary CDF of the absolute normalized daily log-returns $|z|$ of the S&P 500 (2016–2026, FRED series SP500), in log-log scale, against the Gaussian prediction (dashed) and the inverse-cubic guide line $\propto |z|^{-3}$ (red). The fat tail — the straight-line regime — is the empirical refutation of the GBM hypothesis. Generated by `images/code_for_images/sp500_returns.py`.

Two observations enrich the picture. First, the S&P 500 aggregates 500 assets, and its distribution is non-Gaussian because the components are *not independent* — when they fall, they fall together, so the standard CLT does not apply to the aggregation; disentangling such cross-correlations requires *random matrix theory*. Second, the scaling is impressive: approximate scaling can hold across a limited range of time horizons Δt , with the PDF collapsing as $P_{\Delta t}(x) = a(\Delta t)^{-1} P_*(x/a(\Delta t))$ over the fitted central range — the self-similarity of stable laws at work in real data.

Truncated stable distributions. The resolution of the infinite-variance puzzle is that the true tail is *not* a pure stable law all the way to infinity. The far tail is better modelled by a power law *truncated by an exponential*,

$$P_{\alpha,\lambda}(x) \sim |x|^{-(1+\alpha)} e^{-\lambda|x|}, \quad (7.21)$$

which preserves the stable behaviour over the observable range while restoring a finite variance. An equally popular parametrization is the *Student-t* distribution,

$$P_S(t) = C_S \left(1 + \frac{x^2}{\nu \hat{\sigma}^2} \right)^{-\frac{\nu+1}{2}}, \quad (7.22)$$

where $\hat{\sigma}$ is a scale parameter and ν a shape parameter (the number of degrees of freedom). For $\nu = 3$ the asymptotic tail decays as x^{-4} — the *inverse quartic law* — which is compatible with a large body of empirical observations.¹

7.3 Volatility

The second pillar to re-examine is the volatility. In reality σ is not constant — and worse, it is a *hidden* quantity: no market quotes it directly, and extracting it is a delicate statistical task. Throughout this section the basic observable is the log-return

$$dX_t = \ln \frac{S(t + \Delta t)}{S(t)}. \quad (7.23)$$

Three estimators of volatility are in common use, in increasing order of temporal resolution.

The first is the *historical volatility*, extracted from the sample variance of the time series, exactly as described in Chapter 5 (Section 5.15).

The second is the *high-frequency volatility*, built from the mean absolute return rather than the squared one,

$$\sigma_{\text{hf}} = \frac{1}{N} \sum_{i=1}^N |dX_i|, \quad (7.24)$$

for N returns at the same sampling interval. This is a scale for returns over that interval; a per-unit-time volatility also requires division by $\sqrt{\Delta t}$. To relate it to σ , assume for a moment Gaussian returns, $P(x) = \mathcal{N}(0, \sigma^2)$; then

$$\begin{aligned} \sigma_{\text{hf}} = \mathbb{E}[|x|] &= \int dx |x| P(x) = 2 \int_0^\infty dx x P(x) \\ &= \frac{2}{\sqrt{2\pi} \sigma} \int_0^\infty dx x e^{-x^2/(2\sigma^2)} = \sqrt{\frac{2}{\pi}} \sigma \approx 0.8 \sigma, \end{aligned} \quad (7.25)$$

so the mean absolute return is a proxy that *underestimates* σ by a known factor $\sqrt{2/\pi}$ — easily corrected, and statistically more robust against outliers than the sample variance.

The third quantity in the notes is the return-by-return proxy

$$\sigma_{\text{ist}} = \frac{|dX_t|}{\sqrt{\Delta t}}.$$

It does not converge to the instantaneous volatility as $\Delta t \rightarrow 0$: even for constant σ , its limiting distribution is $\sigma|Z|$ with $Z \sim \mathcal{N}(0, 1)$. The differential rule $(dW)^2 = dt$ concerns quadratic variation, not the square of each individual increment. For a continuous Itô log-price with variance

¹The Student- t arises naturally as a mixture of Gaussians whose variance is itself random (inverse-Gamma distributed; equivalently, the precision is scaled χ^2 -distributed) — a first hint that random volatility and fat tails are two faces of the same phenomenon. This is the parametric approach to fitting the PDF of returns.

process v , the appropriate bridge is the realised quadratic variation,

$$\sum_i (\ln S_{t_{i+1}} - \ln S_{t_i})^2 \xrightarrow[|\pi| \rightarrow 0]{P} \int_0^T v_s ds.$$

Estimating an instantaneous value requires local averaging and regularity assumptions, rather than one return.²

7.4 Volatility clustering

With return magnitudes as a volatility proxy, one can finally look at volatility *as a time series* — and what one sees is striking. Volatility is *intermittent*: long quiet stretches alternate with sudden bursts of turbulence, in a pattern strongly reminiscent of the velocity field of a turbulent fluid. Figure 7.5 shows the phenomenon on real data — ten years of daily S&P 500 log-returns: the calm of 2017 and the fury of March 2020 are not different draws from one distribution, they are different *regimes*, and large moves visibly arrive in clusters. The statistics makes the analogy quantitative.

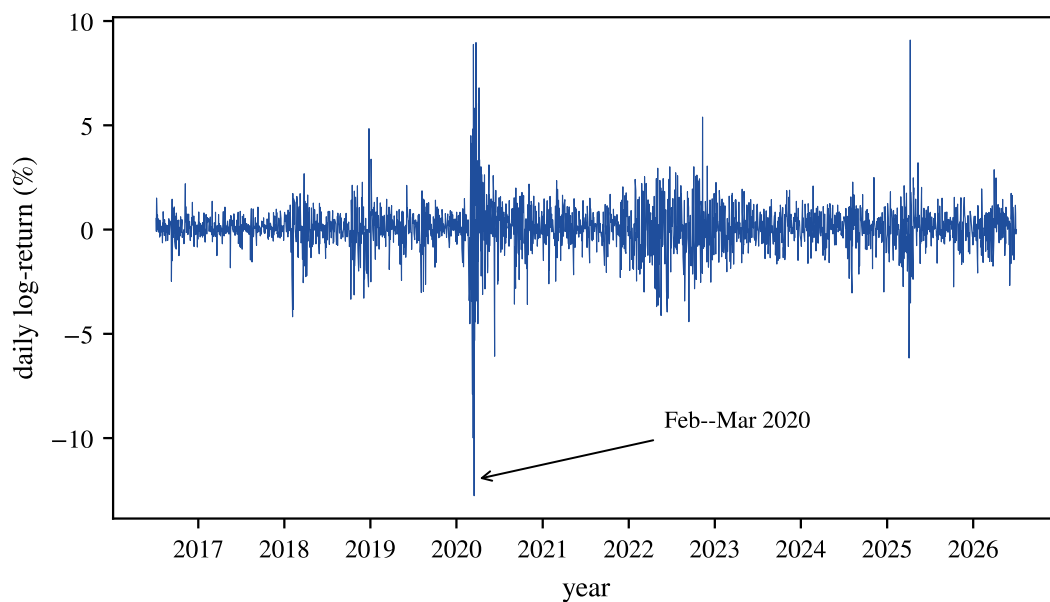


Figure 7.5: Daily log-returns of the S&P 500, 2016–2026 (FRED series SP500). Quiet stretches alternate with bursts of turbulence — most dramatically the COVID crash of February–March 2020 — and large returns of either sign cluster together: this is volatility clustering, incompatible with a constant σ . Generated by `images/code_for_images/sp500_returns.py`.

The log-returns themselves are essentially uncorrelated,

$$\text{Corr}(dX_i, dX_{i+\tau}) \approx 0 \quad (\text{decaying within minutes}), \quad (7.26)$$

as an empirical approximation for many liquid-market samples, with important dependence on sampling frequency and market frictions. But their *absolute values* are strongly and persistently

²See Björk[2], Chapter 4, for quadratic variation and the Itô rules. This corrects the identification of a squared increment with its mean in the handwritten heuristic.

correlated, with a slow power-law decay of the autocorrelation,

$$\text{Corr}(|dX_i|, |dX_{i+\tau}|) \sim \tau^{-\beta}, \quad \beta \approx 0.3, \quad (7.27)$$

so a large move today makes large moves likely for days and weeks to come, in either direction[19]. Figure 7.6 shows both facts at once on our daily data: the return autocorrelations are weaker than those of their magnitudes, although some lags fall outside the reference band; the $\tau^{-0.3}$ curve is an illustrative guide, not a fitted decay.

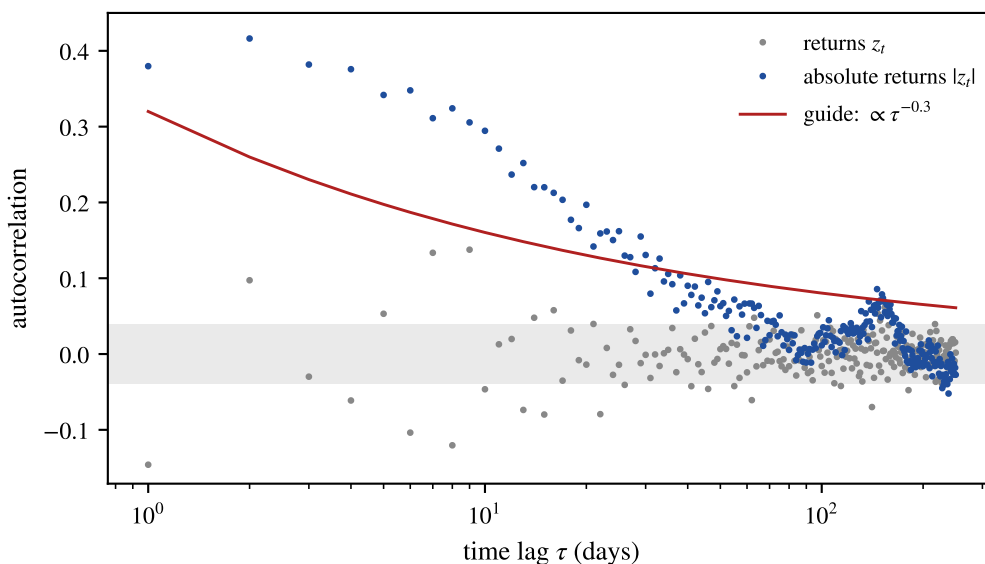


Figure 7.6: Sample autocorrelation of the daily S&P 500 log-returns (grey) and of their absolute values (blue), against the time lag τ ; the shaded band is the 95% noise level. Magnitude correlations are more persistent than linear return correlations. The red $\tau^{-0.3}$ curve is a guide, not a fit; the band is the approximate pointwise iid reference $\pm 1.96/\sqrt{N}$, not a simultaneous confidence band. Generated by `images/code_for_images/sp500_stylized.py`.

Nonzero correlations of return magnitudes show that the returns are not independent, even when their linear correlations are small. This conclusion does not follow from comparing two differently normalized volatility estimators.

This is *volatility clustering*, and it kills the constant- σ hypothesis for good; even the distribution of the volatility itself turns out to have heavy tails, and is well fitted empirically by a log-normal or an inverse-Gamma law[4]. The natural response is to give σ its own dynamics.

7.5 Fractional Brownian motion*

Fractional Brownian motion (fBM) is a Gaussian process with correlated stationary increments.³ In the notation of the first reference, write $B_H(t)$, with $B_H(0) = 0$, zero mean and covariance

$$E[B_H(t)B_H(s)] = \frac{1}{2} (t^{2H} + s^{2H} - |t-s|^{2H}), \quad H \in (0, 1).$$

³This section supplements the handwritten notes, but fractional Brownian motion is listed in the course program. Its sources are Mandelbrot and Van Ness[27] and Gatheral, Jaisson and Rosenbaum[16], rather than Björk.

For $H = \frac{1}{2}$ this is ordinary Brownian motion. In general $\text{Var}(B_H(t)) = t^{2H}$, so changing H changes the variance scaling as well as the dependence of increments. The correlation of successive unit increments is $2^{2H-1} - 1$: positive for $H > \frac{1}{2}$ and negative for $H < \frac{1}{2}$. More generally, the unit-increment covariance at lag n is

$$\frac{1}{2} ((n+1)^{2H} - 2n^{2H} + (n-1)^{2H}) \sim H(2H-1)n^{2H-2}.$$

Long-range dependence of these increments occurs for $H > \frac{1}{2}$; for $H < \frac{1}{2}$ their negative correlations are summable. For $H \neq \frac{1}{2}$, fBM is neither Markov nor a semimartingale in its natural filtration, so the ordinary Itô calculus cannot simply be applied to it.

The rough-volatility literature finds local log-volatility regularity consistent with small H , around 0.1 in the cited study[16]. Roughness at short lags and persistence over long lags are distinct properties; small H does not mean long-range dependence of fBM increments. A Markov model is not generally prohibited from having slowly decaying correlations. Nor does market efficiency alone fix a Hurst exponent: arbitrage conclusions depend on the price model and the allowed trading strategies. A common construction retains Brownian driving noise for the traded price while introducing roughness into its volatility.

7.6 Stochastic volatility models

To model a time-varying σ , one couples the price to a second stochastic process $Y(t)$ that drives the volatility. The general *stochastic volatility* model is the two-dimensional SDE system

$$(1) \quad dS(t) = \mu S(t) dt + f[Y(t)] S(t) dW_1(t), \quad (7.28)$$

$$(2) \quad dY(t) = \alpha(m - Y(t))dt + g[Y(t)] dW_2(t), \quad (7.29)$$

whose ingredients deserve individual introductions. The function $f[Y(t)]$ converts the auxiliary process into the actual volatility of the stock, $\sigma(t) = f[Y(t)]$; its choice distinguishes the models below. The drift of Y is of Ornstein–Uhlenbeck type, so the volatility is *mean-reverting* — pushed up or down toward its long-run level m , in agreement with the observed alternation of calm and turbulent regimes. The function $g[Y(t)]$ is the noise amplitude of the volatility itself, universally known as the *vol of vol*. Finally — and this is the empirically crucial ingredient — the two Wiener processes are *correlated*, with coefficient $\rho \in [-1, 1]$:

$$dW_2(t) = \rho dW_1(t) + \sqrt{1 - \rho^2} dZ(t), \quad (7.30)$$

where Z is a Wiener process independent of W_1 . The linear combination is Gaussian and has independent increments because W_1 and Z are independent Wiener processes. Its mean and variance complete the verification. Its increments have zero mean,

$$\mathbb{E}[dW_2] = \rho \mathbb{E}[dW_1] + \sqrt{1 - \rho^2} \mathbb{E}[dZ] = 0, \quad (7.31)$$

and the correct variance,

$$\mathbb{E}[dW_2^2] = \rho^2 \mathbb{E}[dW_1^2] + (1 - \rho^2) \mathbb{E}[dZ^2] = \rho^2 dt + (1 - \rho^2) dt = dt, \quad (7.32)$$

while the cross-correlation with W_1 is, by construction,

$$\langle dW_1(t) dW_2(t) \rangle = \rho dt. \quad (7.33)$$

Empirically $\rho < 0$ for equities (fits give $\rho \approx -0.7$ for stock indices): when the price falls, the volatility rises — bad news makes the market panic. This asymmetry is the *leverage effect*, so named from its first interpretation in terms of firms' debt leverage, and its fingerprints were all over the crises of 2007–2008.

Specific models. Different choices of the pair (f, g) define the models found in the literature, of which three are canonical.

1. **Stein–Stein**[39]: $f[Y] = Y = \sigma$ and $g[Y] = \kappa > 0$, so the volatility itself follows a classical Gaussian OU process,

$$d\sigma = \alpha(m - \sigma) dt + \kappa dW_2, \quad (7.34)$$

with stationary law $\mathcal{N}(m, \kappa^2/(2\alpha))$ on $\mathbb{R}$ when initialized in equilibrium. The Gaussian diffusion coefficient can be signed; its square determines the instantaneous variance. Replacing it by its absolute value in a correlated model requires care because it changes the noise coupling.

2. **Heston**[21]: $f[Y] = \sqrt{Y}$ and $g[Y] = \kappa\sqrt{Y}$, so that the *variance* $v = \sigma^2$ follows

$$d\sigma^2 = \alpha(m - \sigma^2) dt + \kappa\sqrt{\sigma^2} dW_2(t), \quad (7.35)$$

which is exactly a CIR process (Chapter 6) for σ^2 : its transition law is scaled noncentral chi-squared, and its values are nonnegative; strict positivity requires the Feller condition. This built-in positivity, together with a semi-analytical option-pricing formula, has made Heston the industry standard.

3. **Exponential OU** (Scott[37]): $f[Y] = e^Y$, so $\sigma = e^Y > 0$, with $g[Y] = \kappa$ and

$$dY = \alpha(m - Y) dt + \kappa dW_2(t), \quad \text{PDF}\{\sigma\} = \text{LN}(\sigma), \quad (7.36)$$

a log-normal volatility: positive as well, but difficult to treat analytically.

The frontier has since moved on: recent models replace the driving noise with *fractional* Brownian motion, rougher and more irregular than the standard one, in the “rough volatility” program. A sobering general remark closes the survey: elementary density formulas are generally unavailable, although characteristic functions and integral representations are known in important cases. What is known is the phenomenology — leptokurtic, fat-tailed distributions at high frequency, slowly re-Gaussianizing over long horizons (the *aggregational Gaussianity* of the stylized-facts literature[5]) — which is precisely what the data show, and what the GBM could not produce.

7.7 Option pricing under Heston’s model

Does the Black–Scholes program survive stochastic volatility? The Heston model (1993)[21] is the natural laboratory, being the direct generalization of Black–Scholes:

$$dS = \mu S dt + \sqrt{v} S dW_1, \quad (7.37)$$

$$dv = \kappa(\theta - v) dt + \sigma\sqrt{v} dW_2, \quad (7.38)$$

where v is the variance and $\langle dW_1 dW_2 \rangle = \rho dt$, the correlation being a specified parameter of the joint driving noise. Figure 7.7 shows what the model produces: price paths of realistic roughness, driven by a volatility that is itself a fluctuating, mean-reverting process.

In Black–Scholes only the price $S(t)$ moves, and one protects oneself by holding stock. In Heston one still hedges dW_1 by buying S , but for dW_2 there is nothing to buy: volatility is not exchanged. One therefore needs another instrument, sensitive to volatility *and* tradable — another option.

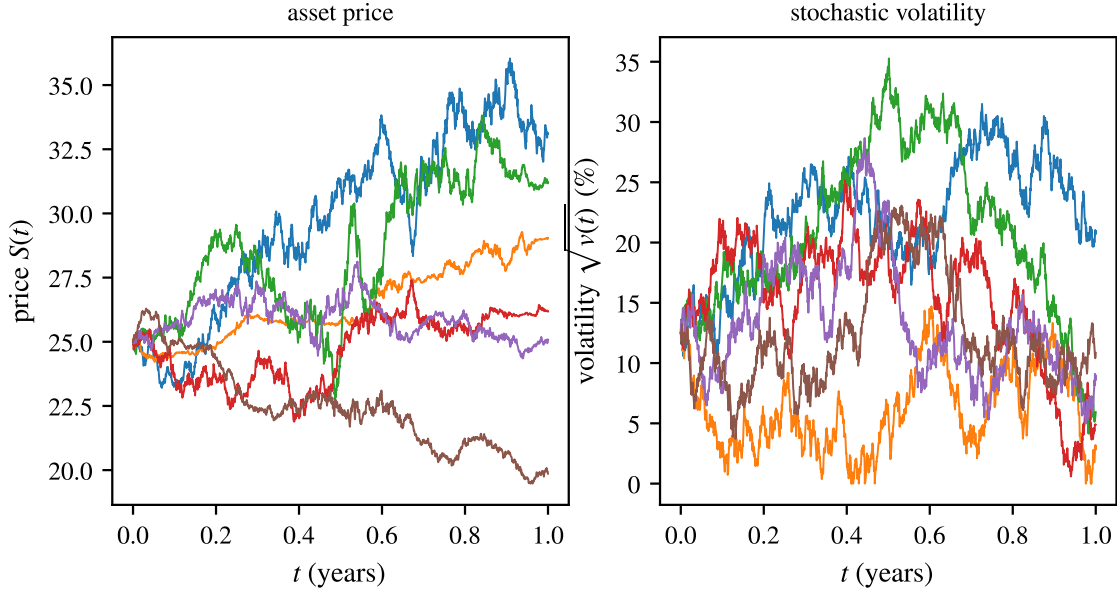


Figure 7.7: Monte Carlo simulation of the Heston model (six sample paths, full-truncation Euler scheme): the asset price $S(t)$ (left) and its stochastic volatility $\sqrt{v(t)}$ (right), with negative price–volatility correlation $\rho = -0.125$ to account for the leverage effect. Simulations of this kind are routinely used in finance for the pricing of plain vanilla and exotic options under stochastic volatility. Generated by `images/code_for_images/heston_sim.py`.

7.7.1 Hedging with two options

Consider then two options O_1 and O_2 on the same stock, with maturities T_1 and T_2 : O_2 is the *target* option, the one the writer has sold, and O_1 is the instrument used to cover dW_2 , with

$$T_1 > T_2 \geq t \quad (7.39)$$

(the hedging option must outlive the target). The writer's wealth is a portfolio of cash, a fraction Δ of stock (which takes care of dW_1), and a fraction Δ_1 of the hedging option (which takes care of dW_2),

$$W(S, v, t) = \Pi(S, v, t) + \underbrace{\Delta(S, v, t) S}_{dW_1} + \underbrace{\Delta_1(S, v, t) O_1(S, v, t)}_{dW_2} = O_2(S, v, t), \quad \Delta, \Delta_1 \in \mathbb{R}, \quad (7.40)$$

where the last equality is replication: the portfolio must reproduce the sold option. For the self-financing replication, write dG for the gain on the current hedged positions, as in Chapter 5. Then

$$\Pi(S, v, t) = O_2 - \Delta S - \Delta_1 O_1 \implies dG = dO_2 - \Delta dS - \Delta_1 dO_1. \quad (7.41)$$

To expand dO_1 and dO_2 we need the *two-dimensional Itô formula*, the natural extension of (4.39): for $f[X(t), Y(t), t]$ with $dX = a dt + b dW_1$ and $dY = c dt + d dW_2$,

$$df = \left(\frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + c \frac{\partial f}{\partial y} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} + \frac{1}{2} d^2 \frac{\partial^2 f}{\partial y^2} + \rho b d \frac{\partial^2 f}{\partial x \partial y} \right) dt + b \frac{\partial f}{\partial x} dW_1 + d \frac{\partial f}{\partial y} dW_2, \quad (7.42)$$

the only novelty being the mixed term, generated by $dW_1 dW_2 = \rho dt$. Applied to the two option prices, and collecting the second-order operators into

$$\mathcal{D} = \frac{1}{2} v S^2 \frac{\partial^2}{\partial S^2} + \frac{1}{2} \sigma^2 v \frac{\partial^2}{\partial v^2} + \rho \sigma v S \frac{\partial^2}{\partial S \partial v}, \quad (7.43)$$

this gives, for $i = 1, 2$,

$$dO_i = \left\{ \frac{\partial O_i}{\partial t} + \mu S \frac{\partial O_i}{\partial S} + \kappa(\theta - v) \frac{\partial O_i}{\partial v} + \mathcal{D} O_i \right\} dt + \sqrt{v} S \frac{\partial O_i}{\partial S} dW_1 + \sigma \sqrt{v} \frac{\partial O_i}{\partial v} dW_2. \quad (7.44)$$

Now rearrange the portfolio differential as $dG + \Delta dS + \Delta_1 dO_1 = dO_2$ and compare the noise terms on the two sides. Collecting dW_1 on the left gives $\sqrt{v} S(\Delta + \Delta_1 \partial_S O_1) dW_1$, while the right carries $\sqrt{v} S \partial_S O_2 dW_1$; collecting dW_2 gives $\Delta_1 \sigma \sqrt{v} \partial_v O_1 dW_2$ against $\sigma \sqrt{v} \partial_v O_2 dW_2$. The noise dW_2 cancels between the two members if

$$\sigma \sqrt{v} \frac{\partial O_2}{\partial v} = \Delta_1 \sigma \sqrt{v} \frac{\partial O_1}{\partial v} \implies \Delta_1 = \frac{\partial O_2 / \partial v}{\partial O_1 / \partial v} \quad (\text{the vega ratio}), \quad (7.45)$$

the option fraction, provided $\partial_v O_1 \neq 0$ and $t < T_2$: the two options are held in proportion to their volatility sensitivities, and the volatility noise is gone. Likewise for dW_1 ,

$$\sqrt{v} S \frac{\partial O_2}{\partial S} = \sqrt{v} S \left(\Delta + \Delta_1 \frac{\partial O_1}{\partial S} \right) \implies \Delta = \frac{\partial O_2}{\partial S} - \Delta_1 \frac{\partial O_1}{\partial S}, \quad (7.46)$$

the optimal stock fraction: the familiar delta hedge, with a *correction* for the delta carried along by the hedging option.

7.7.2 The Heston PDE

With both noises cancelled, dG is now riskless, and by the no-arbitrage principle it must earn the risk-free rate,

$$dG = r \Pi dt = r \{O_2 - \Delta S - \Delta_1 O_1\} dt, \quad (7.47)$$

with the optimal Δ and Δ_1 substituted in. On the other hand, dG can be computed from the Itô expansions of dO_1 and dO_2 above; when the two expressions are equated, every term proportional to μS cancels identically between the O_2 terms, the stock term, and the O_1 terms — the same disappearance of the historical drift that blessed the Black–Scholes derivation. What survives can be separated so that everything referring to O_1 stands on one side and everything referring to O_2 on the other:

$$\begin{aligned} & \left\{ \frac{\partial O_1}{\partial t} + rS \frac{\partial O_1}{\partial S} + \kappa(\theta - v) \frac{\partial O_1}{\partial v} + \mathcal{D} O_1 - rO_1 \right\} \left(\frac{\partial O_1}{\partial v} \right)^{-1} \\ &= \left\{ \frac{\partial O_2}{\partial t} + rS \frac{\partial O_2}{\partial S} + \kappa(\theta - v) \frac{\partial O_2}{\partial v} + \mathcal{D} O_2 - rO_2 \right\} \left(\frac{\partial O_2}{\partial v} \right)^{-1}, \end{aligned} \quad (7.48)$$

With $\mathcal{A}_r O = O_t + rSO_S + \kappa(\theta - v)O_v + \mathcal{D}O$, this is

$$\frac{\mathcal{A}_r O_1 - rO_1}{\text{vega}(O_1)} = \frac{\mathcal{A}_r O_2 - rO_2}{\text{vega}(O_2)} = \lambda(S, v, t), \quad (7.49)$$

a *universal* function, the same for every option on this stock whatever its maturity — the same separation-of-variables manoeuvre that produced the Vasiček identity in Chapter 6. The function λ is the *price of volatility risk*: the extra return demanded per unit of volatility exposure, a variance-drift adjustment (not the Brownian kernel itself), related to the market price of risk q of Chapter 6, and it appears here for the same reason as there — because the risk source (v , like r) is not tradable.

Undoing the ratio, every option price must satisfy the **Heston PDE**:

$$\underbrace{\frac{\partial O}{\partial t} + rS \frac{\partial O}{\partial S} + \frac{1}{2} \nu S^2 \frac{\partial^2 O}{\partial S^2}}_{\text{Black-Scholes}} + \rho \sigma \nu S \frac{\partial^2 O}{\partial S \partial \nu} + \kappa(\theta - \nu) \frac{\partial O}{\partial \nu} + \frac{1}{2} \sigma^2 \nu \frac{\partial^2 O}{\partial \nu^2} - \lambda \frac{\partial O}{\partial \nu} = r O \quad (7.50)$$

The first three terms are exactly Black–Scholes; the new ones are the volatility drift and diffusion, the mixed derivative, and the $-\lambda \partial_\nu O$ term carrying the price of volatility risk (which may equivalently be folded into the drift as $[\kappa(\theta - \nu) - \lambda] \partial_\nu O$; under Q the variance drift is $\kappa(\theta - \nu) - \lambda$; setting $\lambda = 0$ while keeping the historical parameters is an additional assumption). The equation duly collapses onto Black–Scholes when the volatility freezes ($\sigma = 0$, $\nu = \text{const}$ and $\kappa(\theta - \nu) = 0$).

Is it solvable? For Heston's affine specification $\lambda(S, \nu, t) = \lambda_0 \nu$, the risk-neutral variance drift remains affine and the solution is available through Fourier integrals. For a European call the solution retains the Black–Scholes skeleton,

$$C(S, \nu, t) = S P_1 - X e^{-r(T-t)} P_2 \neq C_{\text{BS}}, \quad (7.51)$$

where $P_2 = Q(S_T > X \mid \mathcal{F}_t)$ and $P_1 = Q^S(S_T > X \mid \mathcal{F}_t)$. Here Q^S is the stock-numeraire measure; the two probabilities are not under the same measure. If $f_j(u) = E^{Q_j}[e^{iu \ln S_T} \mid \mathcal{F}_t]$, with $Q_1 = Q^S$ and $Q_2 = Q$, Fourier inversion gives

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \text{Re} \left[\frac{e^{-iu \ln X} f_j(u)}{iu} \right] du.$$

The affine characteristic functions f_j are given in Heston's original paper[21]; the remaining integral is evaluated numerically. This is a representation of the distribution through its characteristic function, not a claim that the density is unknown.

The model earns its keep on the data as well: Drăgulescu and Yakovenko showed that the Heston return distribution fits the Dow Jones index across time lags from one day to a year, reproducing the fat-tailed short-horizon distributions and their slow re-Gaussianization in one stroke[9].

The lesson of the chapter in one line: the market's deviations from Black–Scholes are systematic and measurable, and the models that capture them — stable laws for the tails, stochastic volatility for the clustering — remain tractable enough to price with.

7.8 Summary: the stylized facts

It is useful to close the chapter by collecting the empirical regularities met along the way — the *stylized facts* of financial returns, so called because they hold, with remarkable universality, across markets, epochs, and asset classes[5].

1. The geometric Brownian motion with constant volatility is not a realistic description of asset price dynamics.
2. In many liquid-market samples, linear return correlations are small, but they are *not independent*: higher-order correlations survive through the volatility.
3. In the high-frequency limit the distribution of returns is leptokurtic and non-Gaussian, with heavy tails, sometimes approximated by $P(r) \propto |r|^{-4}$ (the inverse quartic law), and it converges only slowly to a Gaussian over long horizons (*aggregational Gaussianity*).

4. The volatility is itself a stochastic process, with intermittent fluctuations (volatility clustering), long-range power-law autocorrelation, and a fat-tailed distribution (log-normal or inverse Gamma).
5. Stochastic volatility models — Stein–Stein, Heston, exponential OU — provide a realistic description of these facts, give rise dynamically to non-Gaussian return distributions, and remain tractable enough for pricing, volatility forecasting, and simulation.

One last application of the Fokker–Planck machinery awaits, and it takes us out of the market altogether: the distribution of income in a society.

Econophysics, Equilibrium, and Pareto

This final chapter applies the Fokker–Planck formalism to a problem outside the financial markets: the *distribution of income* in a population. It is, in a sense, the purest piece of econophysics in the book — no contracts, no hedging, just a statistical-mechanics question asked of society: if incomes diffuse randomly under simple rules, what stationary distribution do they reach? The answer turns out to reproduce, from one and the same equation, the two regimes observed in a range of income datasets: an exponential (Boltzmann) distribution for the lower and middle class, and a power-law (Pareto) tail for the rich.

8.1 Power laws in economics

Many phenomena, in economics and beyond, follow a *power law*. The classic example is *Zipf’s law*: rank the words of a text by frequency, and the frequency falls off as a power of the rank. The same pattern appears in the sizes of cities, the sizes of firms, and — the case that concerns us — the distribution of wealth.

Wealth and income are distinct quantities; here one studies the *gross income* r (reddito), and studies the complementary CDF — the fraction of individuals with income at least r — rescaled by the mean income $r/\bar{r}$ to compare across countries and years. When this is done, the data speak clearly: two groups emerge. The bulk of the population — the lower and middle class — follows an *exponential* law; the wealthiest few percent follow a *power-law* (Pareto) tail. Figure 8.1 shows the two regimes in the form in which they appear in the empirical literature: in a log-linear plot the exponential bulk is a straight line from which the tail lifts off; in a log-log plot the Pareto tail is a straight line into which the bulk bends. The power-law tail was discovered by Pareto in 1897[32]; empirical evidence and proposed mechanisms are reviewed by Yakovenko and Rosser[43], and its remarkable stability across countries and decades cries out for a mechanism.

8.2 A diffusion model for income

The mechanism we propose is diffusion: incomes perform a stochastic process in “income space”. The physical distinction that will generate the two regimes is the *source* of the income. For the lower and middle class, income comes from labour — wages, roughly deterministic, with additive fluctuations. For the upper class, income comes from *assets* — capital gains, intrinsically

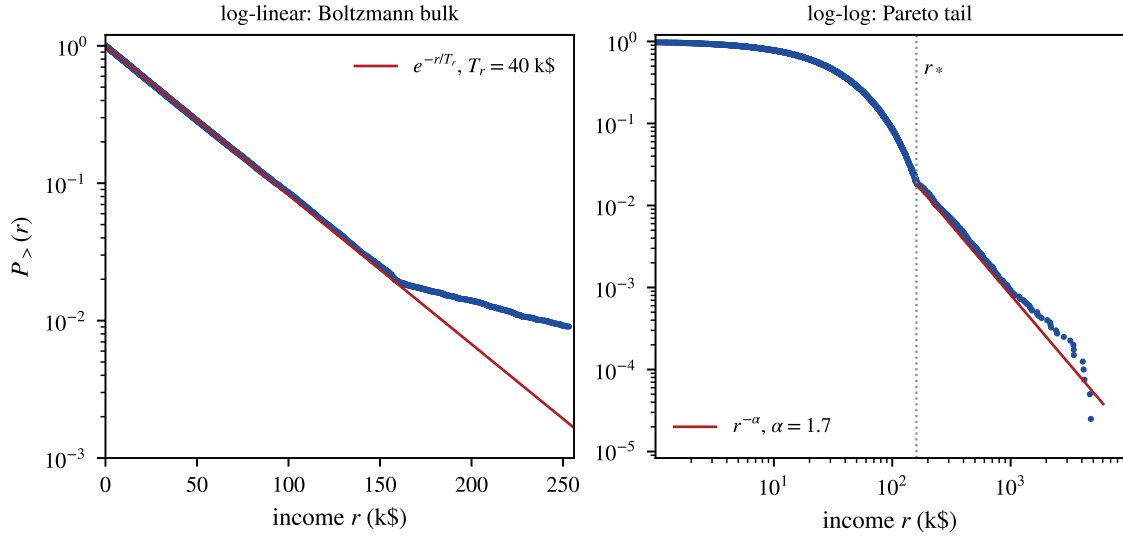


Figure 8.1: Simulated illustration of a two-regime income distribution, not empirical US observations or a fit. The chosen exponential scale is $T_r = 40$ in thousands of dollars, with Pareto exponent $\alpha = 1.7$ and crossover $r_* = 4T_r$; continuity of the CCDF gives tail mass $e^{-4} \simeq 1.83\%$. Left: log-linear axes. Right: log-log axes. The qualitative comparison is motivated by Yakovenko and Rosser[43].

stochastic, with fluctuations proportional to the capital itself. One model, two regimes of its coefficients.

General hypotheses. The framework rests on four hypotheses.

1. Incomes are divisible into an continuous set of income levels $r \in \mathbb{R}_+$, with $r \geq r_{\min}$.
2. $P(r, t)$ is the fraction of individuals with income r at time t , and N is the total number of income earners, so that $NP(r, t) dr = dm(r, t) dr$ counts the earners in $[r, r + dr]$. We take N constant, since there is approximately a one-to-one map between individuals and income positions (people change salary, change job, inherit — but the slots remain).
3. The distribution is normalized, $\int P(r', t) dr' = 1$.
4. The diffusion has *no memory*: the process is Markov.

8.2.1 The master equation

For a Markov diffusion with continuous paths, the forward equation for $P(r, t)$ is a Fokker-Planck equation, with drift and diffusion coefficients $A(r)$ and $B(r)$:

$$\frac{\partial}{\partial t} P(r, t) = \frac{\partial}{\partial r} [A(r) P(r, t)] + \frac{\partial^2}{\partial r^2} [B(r) P(r, t)], \quad (8.1)$$

written here with the sign convention of the notes. For the diffusion away from a reflecting boundary, its local conditional moments are

$$E[dr | r] = -A(r) dt, \quad E[(dr)^2 | r] = 2B(r) dt + o(dt) : \quad (8.2)$$

a positive A describes an average *loss* of income, and B the intensity of its fluctuations.

Equilibrium (stationary) solution. Now the economic input: empirically, the *shape* of the income distribution is approximately constant over time — individuals move up and down, but the histogram barely changes from year to year. This is an equilibrium statement, and we implement it by the *stationary* Fokker–Planck equation,

$$\frac{\partial}{\partial t} P(r, t) = 0. \quad (8.3)$$

The right-hand side of (8.1) then collapses into a total derivative,

$$0 = \frac{d}{dr} [A(r) P(r)] + \frac{d^2}{dr^2} [B(r) P(r)] = \frac{d}{dr} \left\{ A(r) P(r) + \frac{d}{dr} [B(r) P(r)] \right\}, \quad (8.4)$$

and the negative of the object in braces is the probability flux $J(r)$ of Chapter 3,

$$J(r) = -A(r) P(r) - \frac{d}{dr} [B(r) P(r)], \quad (8.5)$$

constant in r by the equation above. On $r \geq 0$, impose a reflecting boundary $J(0) = 0$ and a normalizable density with no probability supplied from infinity. The constant flux then vanishes, so $J = 0$ everywhere:

$$A(r) P(r) + \frac{d}{dr} [B(r) P(r)] = 0. \quad (8.6)$$

Equilibrium with zero flux — detailed balance, a statistical mechanician would say — is a first-order ODE for $P(r)$: given any choice of A and B , the stationary distribution follows by one integration. It remains to feed in the economics.

8.3 Lower and middle class

For the lower and middle class both coefficients are constants,

$$A(r) = A_0, \quad B(r) = B_0 > 0. \quad (8.7)$$

The reading is concrete. A constant, positive A_0 gives local drift $E[dr | r] = -A_0 dt$, while the local fluctuation intensity is $E[(dr)^2 | r] = 2B_0 dt + o(dt)$. These are infinitesimal increment moments, not equations asserting a falling population mean at equilibrium. Reflection at zero contributes to the global moment balance.

This additivity is the statistical signature of *labour* income.

With constant coefficients the stationary equation (8.6) becomes

$$A_0 P(r) + B_0 \frac{dP}{dr} = 0 \implies \frac{dP}{P} = -\frac{A_0}{B_0} dr, \quad (8.8)$$

and integrating,

$$\int \frac{dP(r')}{P(r')} = -\frac{A_0}{B_0} r' \implies P(r) \propto C e^{-\frac{A_0}{B_0} r} = C e^{-r/T_r}, \quad (8.9)$$

an *exponential* distribution (positive and normalizable provided $A_0 > 0$), with scale

$$T_r = \frac{B_0}{A_0}. \quad (8.10)$$

The form is unmistakable to a physicist: it is a Boltzmann distribution, $e^{-E/k_B T}$ with income in place of energy, and T_r plays the role of an *effective temperature of income* — large when fluctuations (B_0) are strong relative to erosion (A_0), as follows directly from the zero-flux equation above. At equilibrium the distribution is stable even though every individual income keeps fluctuating; and if the mean income $\langle r \rangle$ falls, the “temperature” falls with it, compressing the whole distribution. The exponential form resembles an equilibrium energy distribution, but this income diffusion model does not impose conservation of total money.

8.4 Upper class

For the upper class the phenomenology is different in kind, not just in degree: income is driven by *multiplicative feedback*, because capital generates returns proportional to itself. The coefficients that encode this are

$$A(r) = a r, \quad B(r) = b r^2 \quad (a, b > 0), \quad (8.11)$$

giving

$$E[dr | r] = -a r dt, \quad E[(dr)^2 | r] = 2b r^2 dt + o(dt) : \quad (8.12)$$

the average relative erosion is the same as before, but the fluctuations now *grow with the square of the income* — positive feedback, the rich man’s gains and losses both scaling with his fortune. This multiplicativity is the statistical signature of *asset* income, and the reader will recognize in $A \propto r, B \propto r^2$ the drift-diffusion structure of the geometric Brownian motion of Chapter 4.

We solve the stationary equation (8.6) with these coefficients. Expanding the derivative,

$$A(r) P(r) + B(r) \frac{dP}{dr} + P(r) \frac{dB}{dr} = 0 \implies P'(r) + \frac{A(r) + B'(r)}{B(r)} P(r) = 0, \quad (8.13)$$

and substituting $A(r) = ar, B(r) = br^2, B'(r) = 2br$,

$$P'(r) + \frac{ar + 2br}{br^2} P(r) = 0 \implies P'(r) + \frac{a + 2b}{b} \frac{1}{r} P(r) = 0. \quad (8.14)$$

This is again a separable first-order ODE,

$$\frac{dP}{P} = -\frac{a + 2b}{b} \frac{dr}{r}, \quad (8.15)$$

whose integral is worth organizing as in the notes: splitting the coefficient as $\frac{a+2b}{b} = \frac{a}{b} + 2$,

$$\int \frac{dP(r')}{P(r')} = - \int \frac{B'(r')}{B(r')} dr' - \int \frac{A(r')}{B(r')} dr', \quad (8.16)$$

the first integral gives $\ln B(r) = \ln(br^2)$, and the second

$$\int \frac{A(r')}{B(r')} dr' = \frac{a}{b} \int \frac{dr'}{r'} = \frac{a}{b} \ln r, \quad (8.17)$$

so that, for $r \gg r_0$,

$$P(r) \approx \frac{C}{B(r)} e^{-\int \frac{A(r')}{B(r')} dr'} = \frac{C}{b r^2} r^{-a/b}, \quad (8.18)$$

that is,

$$P(r) = C r^{-(2+a/b)} = C r^{-(\alpha+1)}, \quad (8.19)$$

a *power law*: the multiplicative structure has converted the exponential into a Pareto tail. Correspondingly, the complementary CDF decays as $r^{-\alpha}$, with

$$\alpha = 1 + \frac{a}{b}, \quad (8.20)$$

the **Pareto exponent**. Since $a, b > 0$, the model predicts $\alpha > 1$ automatically. Empirically, fits across countries give $\alpha \approx 1.6$ – 1.8 , which through (8.20) implies $b > a$: for the upper class, the multiplicative fluctuations of capital *dominate* its average erosion — and it is precisely this dominance of noise that populates the far tail of great fortunes.¹

The two regimes follow from the same stationary equation. One Fokker–Planck equation, one zero-flux equilibrium condition; additive coefficients give Boltzmann’s exponential for the many, multiplicative coefficients give Pareto’s power law for the few, and the crossover between the two regimes can be compared with income data. The tools were exactly those of the rest of the book — the same stationary Fokker–Planck equation that priced no options and hedged no portfolios here draws, with two lines of algebra, the statistical portrait of a society. This is econophysics at its best: not finance dressed up in physics notation, but the genuine transfer of a physical way of thinking — equilibrium, diffusion, universality — to economic matter.

8.5 The minority game*

Note: This section is a placeholder, to be written. It will present the *minority game* — the simplest agent-based model of a market of adaptive, heterogeneous, competing agents — and its phase transition between an efficient and an inefficient phase, solved exactly with the statistical mechanics of disordered systems.

¹A Pareto exponent $\alpha < 2$ means that the *variance* of the income distribution diverges: inequality in the tail is so extreme that no “typical spread” exists. That α falls in the same narrow window for very different countries suggests that the mechanism — multiplicative dynamics of capital — is universal, largely indifferent to institutional detail.

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